ACADEMIC MATERIAL DISCLAIMER

CHM101 — FUOYE COURSE MATERIAL

This document is identified as academic material associated with the Federal University Oye-Ekiti (FUOYE). The original content, notes, diagrams, and other intellectual property remain with FUOYE and/or the respective lecturers, authors, or rights holders, as applicable.

For educational use only. This copy is provided for personal study and reference. It is not an official publication unless confirmed by FUOYE, and it should not be presented as an official university-issued version.

No ownership of the original material is claimed by the person sharing or using this copy. Please respect the university's and the original creators' rights. Do not redistribute, sell, modify, or publicly upload the material without permission from the appropriate rights holder.

If you are the rights holder and believe this copy has been shared improperly, please contact the person or platform hosting it and request appropriate action.

DISCLAIMER NOTICE • This page does not itself grant permission to reproduce or distribute the underlying material.



Introductory INORGANIC & PHYSICAL Chemistry

FOR UNIVERSITY STUDENTS





FIRST EDITION

OS Ayanda
KO Sodeinde
EG Olumayede
PO Okolo
SO Afolabi

TABLE OF CONTENTS

Brief Contents iii

Table of Contents iv

Preface v

Physical Quantities and Error Treatments 1

Structure of the Atom 8

States of Matter 31

The Mole Concept and Avogadro's Number 43

Determination of Relative and Average Atomic Masses 48

Stoichiometry and Chemical Equations 51

Acids, Bases and Salts 73

Chemical Bonding and Intermolecular Forces 90

Chemical Equilibrium and Thermodynamics 96

Introductory Chemical Kinetics and Nuclear Chemistry 111

Introduction to Colligative Properties of Liquids 125

Reduction-Oxidation Reactions 134

Practice Questions and Answers 150

PREFACE

This book grew out of many years of teaching inorganic and physical chemistry and searching for a textbook that would be accessible to students and demonstrate that chemistry is a modern, vital and evolving field. This book covers two main branches of chemistry – inorganic and physical chemistry – in a broadly equal proportion. By covering two branches in a single volume, the book is able to show the connections between the two, reflecting the way that modern chemical research is carried out.

Introductory inorganic and physical chemistry is a course in the undergraduate chemistry program that introduces, or develops many distinct topics. The authors include specialists in physical chemistry (OS Ayanda, KO Sodeinde and SO Afolabi) and inorganic chemistry (OS Lawal, PO Okolo and EG Olumayede). This book comprises of twelve (12) sections. Each section is treated in such a way that the knowledge of inorganic and physical chemistry is advanced. The topics provide a short, interesting and elementary introduction to inorganic and physical chemistry for the first-year students of chemical, biological and physical sciences as well as the engineering and medical students.

The authors understand the parts of chemistry that students find difficult at the start of a University degree and have taken care to build on what the students know from their secondary school level. The authors have drawn on experience to guide students through difficult concepts. They believe that readers may want to dip in to find the information they are looking for, rather than read the whole sections, therefore, the authors

have ensured that each section of the book is presented as short and self-contained sections.

One of the most important ways to master this course is to work on problems. Conceptional questions also ensure that students learn to express their ideas in the language of science. Therefore, many examples are worked through in each section. The practice questions and answers at the end of the chapters cover a range of difficulties suitable for students.

We welcome the suggestions/comments of both the students and instructors on how best the material may be used and how the book can be improved. Please contact us at chemistry.science@fuoye.edu.ng

Olushola S Ayanda

PHYSICAL QUANTITIES AND ERROR TREATMENTS

K.O. Sodeinde

Physical Quantities and Units

A physical quantity is defined as an attribute of a physical body (system, phenomenon, or process) that may be common for many physical bodies in a qualitative sense, In a quantitative sense, it is unique for each of them and may be inherent in different material bodies, processes, or phenomena. A physical quantity determined quantitatively is an individual property of a concrete object. For example, one may distinguish length, width, and height of an object. With the knowledge of length, width, and height, one can form an opinion concerning the distinctions between objects. The concept "dimension" of a physical quantity is taken to mean quantitative definiteness of the quantity, being peculiar to any concrete material object, system, phenomenon, or process. The value of a physical quantity is an expression of the dimension of the physical quantity in terms of a certain number of units accepted for it.

System of Units

A unit is an established standard for a physical quantity against which particular examples of that physical quantity can be compared. The act of comparing a physical quantity to a unit is called **measurement** and the measure of a particular physical quantity is the ratio of that physical quantity to the unit.

A set of fundamental units that is sufficient for measuring the properties of the class of phenomena under consideration is called a **system of units**.

The International System of Units

The International System of Units, universally abbreviated SI (from the French *Le Système International d'Unités*), is the modern metric system of measurement. The SI unit is rapidly becoming the dominant measurement system used in science, technology and international commerce. The SI is founded on seven base units for seven base quantities assumed to be mutually independent (Table 1).

Table 1: SI Base Units

Physical Quantity	Symbol	Unit
Length	l	metre
Mass	m	kilogram
Time	t	second
Electric current	I	ampere
Thermodynamic temperature	T	kelvin
Luminous intensity	I_v	candela
Amount of substance	n	mole

Base and Derived Quantities and Units

Physical quantities are related to one another by mathematical equations, for example:

$$Velocity = \frac{Displacement}{Time}$$

Some physical quantities are chosen as base quantities. Other physical quantities are obtained from the base quantities using the appropriate algebraic relationships and these are called the **Derived Quantities**. Base quantities are said to have fundamental units and derived quantities have

derived units. For example, if displacement and time are chosen to be base quantities, then velocity is a derived quantity.

SI Derived Units

Derived quantities are defined in terms of the seven base quantities via a system of quantity equations. The SI derived units for these derived quantities are obtained from these equations and the seven SI base units. Examples of SI derived units are listed in Table 2.

Table 2: SI Derived Units

Derived quantity	Unit
Area	m^2
Volume	m^3
Speed, Velocity	m/s
Acceleration	m/s^2
Density	kg/m^3
Current density	A/m^2

Base and Derived Quantities and Units

Significant figures

Whenever a measurement is made, the number of meaningful digits that is written down implies the error in the measurement. For example, if you say that the length of an object is 0.528 m, you imply an uncertainty of about 0.001 m. To record this measurement as either 0.5 or 0.52819667 would imply that you only know it to 0.1 m in the first case or to 0.00000001 m in the second. You should only report as many significant figures as are consistent with the estimated error. The quantity 0.528 m is said to have three significant figures, that is, three

digits that make sense in terms of the measurement. Notice that this has nothing to do with the "number of decimal places". The same measurement in centimetres would be 52.8 cm and still be a three-significant figure number. The accepted convention is that only one uncertain digit is to be reported for a measurement. In the example, if the estimated error is 0.02 m you would report a result of 0.53 ± 0.02 m, not 0.528 ± 0.02 m. Students frequently are confused about when to count a zero as a significant figure. The rule is: If the zero has a non-zero digit anywhere to its left, then the zero is significant, otherwise it is not. For example, 5.00 has 3 significant figures; the number 0.0005 has only one significant figure, and 1.0005 has 5 significant figures. A number like 6000 is not well defined. Rather one should write 6×10^3, one significant figure, or 6.00×10^3, three significant figures

Precision

Precision refers to the degree or extent of reproducibility among replicate measurements of the same quantity. Consider three determinations of the percentage energy loss in a conversion process, determined by one scientist to be 22.60, 22.75 and 22.98 per cent, and three results obtained for the same energy loss, by a second scientist, to be 22.63, 22.62, and 22.62 per cent. The results of the second scientist exhibit much less variation among themselves than do those of the first, so the precision of the set of second results is better than that of the first.

Accuracy

Not all measurements of physical quantities can be entirely accurate. It is important to know, therefore, just how much the measured value is likely to deviate from the unknown true value of the quantity. Accuracy thus refers to the closeness of a measurement, or the mean of multiple measurements to its true or accepted value. The distinction between

precision and accuracy may be likened to the result of shooting a series of arrows at an archery target. Precision refers to how close together the several arrows hit and accuracy refers to how close to the bull's-eye each lands. It is possible for a replicate series of measurements or determinations to be very precise and yet highly inaccurate. It is, however, quite meaningless to consider the accuracy of a series of values unless the precision is reasonably good. A chemist desires to achieve acceptable precision and accuracy in all of his work and to assess how accurate and precise his work and methods are.

Error

The scientist is continually interested in the cause and the magnitude of errors in his measurements. He examines the quantitative data he obtains not with the question as to whether error is present but rather with the question as to how much error and uncertainty exist. He recognizes that error is always present and that he will not completely eliminate error even though he does continually strive to recognize, to minimize, and to evaluate error in his measurements.

Absolute and Relative Errors

The **absolute error** in a measured quantity is the uncertainty in the quantity and has the same units as the quantity itself. For example, if you know a length is 0.428 m ± 0.002 m, the 0.002 m is an absolute error. The **relative error** (also called the **fractional error**) is obtained by dividing the absolute error in the quantity by the quantity itself. The relative error is usually more significant than the absolute error. For example, a 1 mm error in the diameter of a skate wheel is probably more serious than a 1 mm error in a truck tire. Note that relative errors are dimensionless. When reporting relative errors, it is usual to multiply the

fractional error by 100 and report it as a percentage. Error may be arbitrarily divided into two categories; systematic (determinate) and random (indeterminate) errors.

Systematic Error

Systematic errors are those one-sided errors which can be traced to a specific source, either in the strategic scheme of the experiment or in the apparatus used to perform it. Such errors can often be minimized by a modified plan. Even when the errors cannot be completely suppressed in this way, an understanding of their origins often makes it possible to deduce a correction factor that can be applied to the final result, or at least to estimate the probable residual error in that result. When one or more large errors appear to be present, it is frequently possible to discover their origins by a series of carefully controlled experiments in which the experimental conditions and quantities are varied widely in a systematic way.

The resultant error must follow one of three courses:

a. the error may remain relatively constant and independent of the experimental conditions,
b. the magnitude of the error may vary systematically with one or more of the experimental conditions, or
c. the error may persist as a random error.

If the error in a measurement proves to be constant in magnitude, such possibilities as instrument calibration must be considered. If a systematic variation of the error is evident, the parameter linked to this variation frequently indicates the cause. When an apparently random error is encountered, it may be a systematic error linked to some experimental condition not yet investigated or controlled. For example, an apparently

random error could ultimately prove to be associated with variations in atmospheric humidity, perhaps indicating that a chemical or material is absorbing water during the experiment. Systematic errors tend to make the observed or calculated values consistently too high or too low. This means that systematic errors can make results highly inaccurate without affecting the precision of replicate results. Varying at least some experimental factors in replicate experiments can minimize the danger of retaining one-sided errors without recognizing their presence. In critical analyses, duplicate sets of samples should be analyzed by entirely different methods since it is unlikely that the same systematic errors would appear to the same extent in entirely different analytical procedures.

Random Error

The cause of a random error may or may not be known. Some personal judgment is required in all measurements, such as in reading instrument dials or meters, noting just when a container is filled to a predetermined calibration mark, etc. Some random errors arise within the method itself, such as impurity of a supposedly pure material, variations with stirring and with speed of mixing reagent. Random variations in room temperature and other environmental factors may introduce random error into analytical results.

STRUCTURE OF THE ATOM

E.G. Olumayede

Early Ideas of Atoms

The early Greek philosophers tried to understand the nature of the world through reason and logic, but not through experiment and observation. Democritus is one of the earlier Greek philosophers that believed the universe is made of matters and the matters are made up of atoms. He suggested that atom is a tiny, indivisible, solid object.

Dalton's Atomic Theory

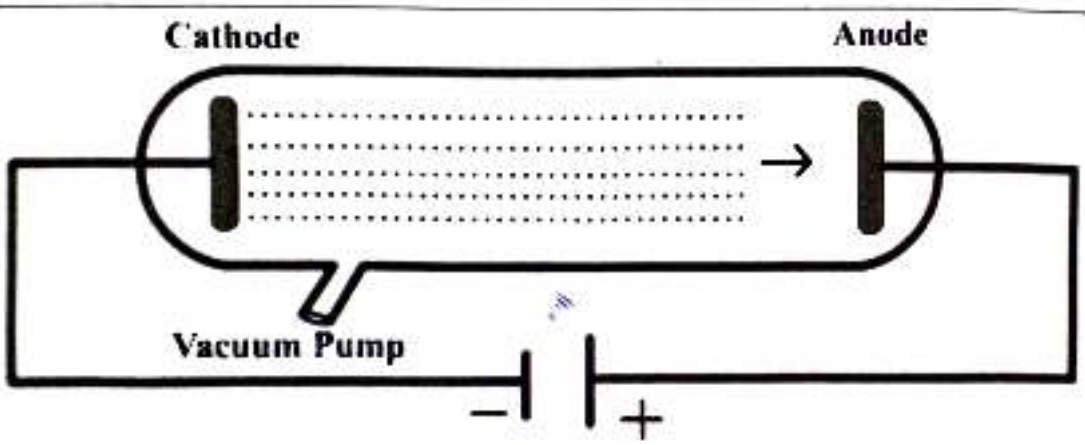
John Dalton believed in both logical thinking and experimentation. He proposed the first scientific theory that relates chemical changes to the structure, properties, and behaviour of the atom.

Dalton's Atomic Theory states that:

1. Matter is made of tiny particles called atoms.
2. Atoms are indivisible. During a chemical reaction, atoms are rearranged, but they do not break apart, nor are they created or destroyed.
3. All atoms of a given element are identical in mass and other properties.
4. The atoms of different elements differ in mass and other properties.
5. Atoms of one element can combine with atoms of another element to form "compounds" – new complex particles. In a given compound, however, the different types of atoms are always present in the same relative numbers.

Michael Faraday

For several years, the Dalton's Atomic Theory was believed to be true and different chemical experiments were performed to test it. However, Michael Faraday gave the first evidence to show that electric charges were not continuous but exist in form of discrete particles.



This experiment was performed in the latter half of 19th century. The experiment was carried out by scientist like Crookes, Goldestein. When a high voltage is applied to gas at low pressure, stream of particles moved from cathode to the anode. These are called cathode rays. At low pressure "glow" was observed

Figure 1: Faraday's Experiment

From Faraday's experiment (Fig. 1), we see that the amount of electricity needed to deposit N atoms (1 mole) during electrolysis of mono-valent atoms is exactly the same. It can be concluded that a certain amount of electricity must be associated with one atom of univalent element. It was also discovered that only multiple of this quantity of electricity are necessary to deposit one atom of divalent or trivalent element.

J. J. Thomson

In 1897, a scientist named J. J. Thomson conducted some research that suggested that Dalton's Atomic Theory was not entirely true. J. J. Thomson used a cathode ray tube which is a small glass tube with a cathode (a negatively charged metal plate) and an anode (a positively charged metal plate) at opposite ends (Fig. 2).

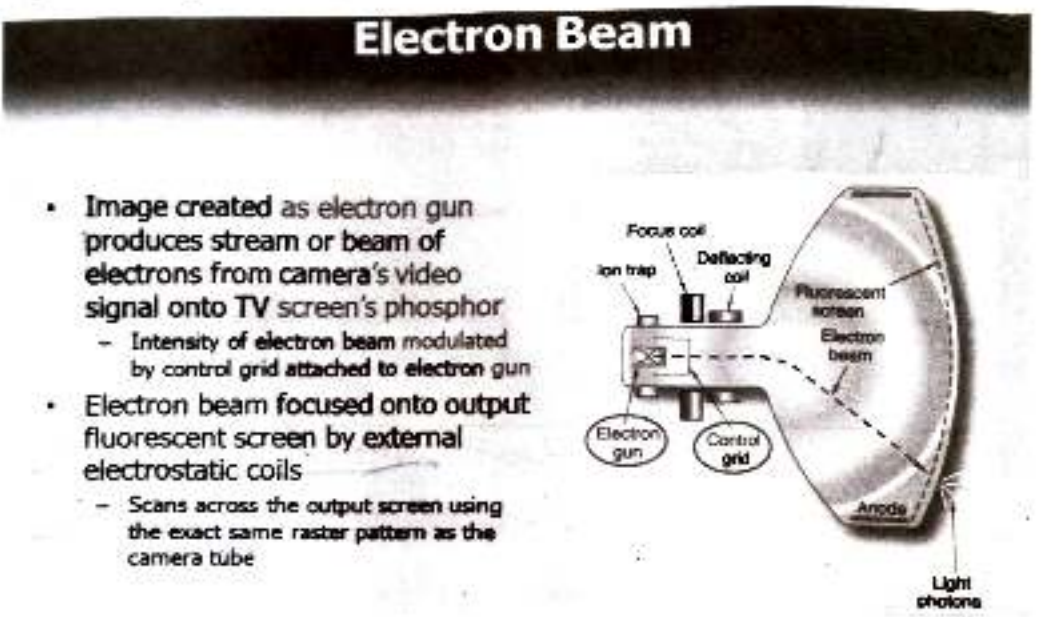


Figure 2: Thomson Cathode Ray Tube Experiment

The steps are

i. to allow the beam to go through a straight line and make a spot on the screen
ii. deflect to magnetic field
iii. adjust it back to electric field

From the above observations, Thompson was able to determine e/m ratio (charge to mass ratio of electrons).

In the magnetic field, magnetic force acting on each electron causes it to accelerate in the direction of the force.

$$HeV = \frac{MV^2}{r} = \frac{e}{m} = \frac{V}{Hr}$$

Similarly, in the electric field

$$Ee = HeV$$

$$V = \frac{E}{H}$$

$$\frac{e}{m} = \frac{E}{rH^2}$$

From this experiment, it was found that irrespective of the material of the cathode, the charge to mass ratio is always the same value i.e. it was constant and equal to 1.76×10^8 c/g. The particles were called "electron".

Robert Millikan Oil Drop Experiment

With his oil drop experiment (Fig. 3), Robert Millikan was able to give final evident that electrons are discrete particles. What Millikan did was to spray a cloud of oil droplet between two charge plates. Due to gravity, the tiny droplets of oil experienced a gravitational force (mg), where m is the mass and g is the acceleration due to gravity.

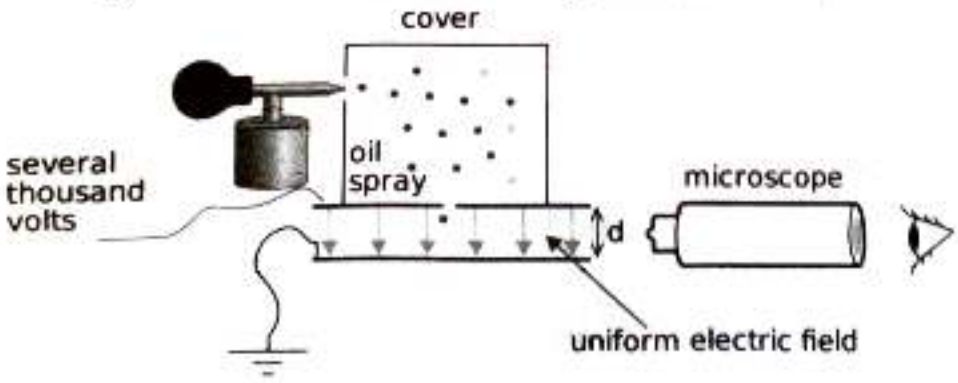


Figure 3: Robert Millikan Oil Drop Experiment

The oil chamber was irradiated with x-rays, causing the oil to become negatively charged. The oil droplet then experiences upward force due to

electric field, the magnitude is equal to Eq, where E is the electric field strength and q is the total charge on the drop.

Therefore, when $Eq = mg$, we can calculate q.

It was observed that the charge on different drops varied, the values obtained were always small whole numbers, in multiples of 1.6×10^{-19} C, the charge on a single electron. So far, we have $m = 9.1 \times 10^{-28}$ g from Thompson experiment and he modelled an atom as a sphere of positive electricity, with negative electrons embedded in it. This mean that mass is associated with positive electricity and uniformly spread through atom.

Rutherford's Nuclear Model

In attempt to disprove the J.J Thompson's idea, Rutherford bombarded metal foils with alpha rays and found that most of the particles pass through the foil (Fig. 4). However, few were slightly deflected and occasionally one underwent a deflection 90^0 or more and bounced back. Rutherford postulated that atoms had a centre or nucleus, which was small compare with the volume of the rest of the atom.

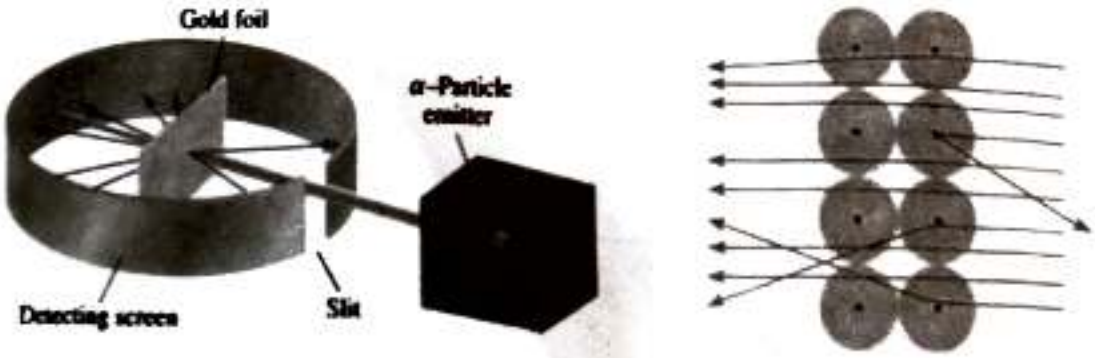


Figure 4: Rutherford's Model

If Thomson's model was accurate, there was nothing hard enough for these small particles to hit that would cause any change in their motion.

Rutherford described firing alpha particles at gold foil like shooting a high-powered rifle at tissue paper. While Rutherford's discovery of the positively charged atomic nucleus offered insight into the structure of the atom. Rutherford's model, though, suggested that the positive charge wasn't distributed like batter, but rather, was concentrated into a tiny particle at the centre of the atom, while most of the rest of the atom was empty space.

H.G.J Moseley Experiment (Discovery of Atomic Number)

Moseley used different element as the cathode with high voltage (Fig. 5). This allows the x-rays to pass through slit of a narrow beam and then diffracted. When the diffracted rays were allowed to fall on photographic plate, an x-ray spectrum was produced. This was analysed and it was observed that this characterized the target material.

Moseley then measured the mass of the emitted rays and found that the $\sqrt{v} \propto$ Atomic mass, where v = frequency of X-rays emitted. Atomic number gives in order of appearance of the elements in the periodic table. Moseley was able to deduce the expression:

$$V = \frac{1}{\lambda} = C(z - b)$$

where, λ is the wave length of the x-rays, c and b are constant and Z is equal to relative ½ of the atomic number.

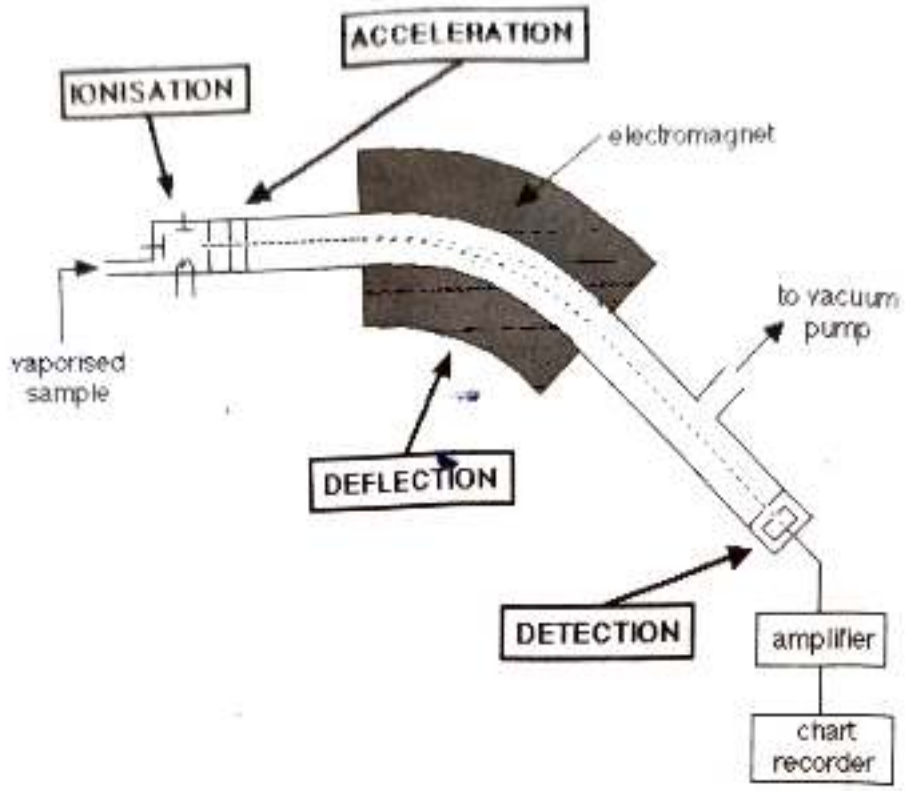


Figure 5: Moseley Experiment

Properties of electrons, protons, and neutrons

The properties of sub-atoms particles (electrons, protons, and neutrons) are as shown in the table below.

Particles	Charge	Mass (atomic mass units)
Protons	+	1
Neutrons	Nil	0
Electrons	-	1

The third column shows the masses of the three subatomic particles in grams. The second column, however, shows the masses of the three subatomic particles in "atomic mass units". An **Atomic Mass Unit (a.m.u.)** is defined as one-twelfth the mass of a carbon-12 atom. Atomic mass units are useful, because, as you can see, the mass of a proton and the mass of a neutron are almost exactly 1.0 in this unit system

Atomic Spectrum

As the various experiments on the discovery of atoms were going on, other scientists were carrying out experiments on the light given out when various elements were heated. Atoms of different elements heated in a flame produced different colours. The different colours emitted were examined by passing the light through a glass prism.

It was observed that a line spectrum was obtained for each element. A line spectrum consists of a series of colour lines (Fig. 6). The pattern differs from element to element but the spectrum of a particular element is so characteristic of the element and it can be used to identify the element. Simple line spectrum is that of hydrogen atom.

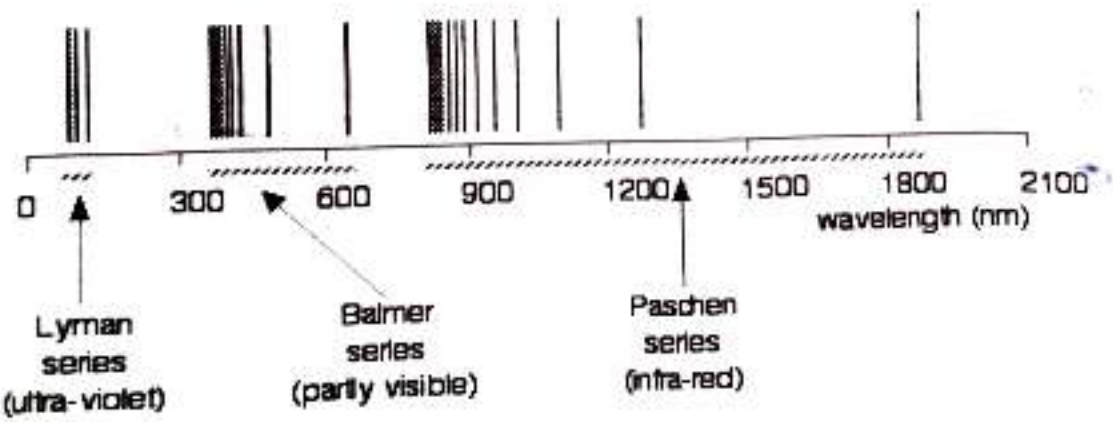


Figure 6: A Typical Atomic Line Spectrum

Rydberg suggested that the lines of a particular wavelength fit the equation:

$$\frac{1}{\lambda} = R\left\{\frac{1}{{n_1}^2} + \frac{1}{{n_2}^2}\right\}$$

where, R is the Rydberg's' constant, n_1 and n_2 = integers

However, Plank examined radiation emitted by hot bodies and found that energy was not being emitted continuously but in "packet" which he called **quanta.**

Bohr model

It was put forward to explain (i) the line spectra (ii) the apparent contradiction between the classical thermodynamics and its application to electrons in atoms (iii) the periodic law – arrangement of the elements in order of atomic number, elements with the same chemical properties occur periodically.

The postulate of Bohr theory

1. The electrons move in an orbit around the central nucleus
2. Only certain orbits are allowed i.e. those whose angular momentum of an electron can only take on values which are whole multiple of $h/2\pi$ i.e. $mvr = nh/2\pi$ where, n = 1, 2..... This condition is called "quantum condition".
3. The energy of an electron in a given orbit is fixed. As long as the electrons stay in that orbit, it neither absorbs nor radiates energy. The energy of an electron in an orbit is given as:

$$E_n = \frac{-Rch}{n^2}$$

where, E_n = Energy of electron in the nth orbit, R = Rydberg's' constant, h = planks constant and c = speed or velocity of the light

Types of Energy

1. **Electronic energy**: this is the energy due to the motion of electron about the atomic nucleus i.e. from charge interaction between electrons and the nucleus.
2. **Translational, rotational and vibrational energies**: These are subjected to the limitations of quantum theory.

When an electron changes from one orbit to another, it must absorb or emit an amount of energy sufficient to bring it to another state. The

quantity of the energy equal to the difference in energy between the two orbits.

$$\Delta E = E_{n_2} - E_{n_1} = \frac{-Rch}{{n_2}^2} - \frac{-Rch}{{n_1}^2}$$
$$= -Rch \left\{\frac{1}{{n_2}^2} - \frac{1}{{n_1}^2}\right\}$$

But we know that $\lambda = \frac{1}{v}$, where v is the frequency of light

Therefore, $\frac{1}{\lambda} = R \left\{\frac{1}{{n_2}^2} - \frac{1}{{n_1}^2}\right\}$

Example

Show that $\Delta E_{n=3\rightarrow n=1} = \Delta E_{n=3\rightarrow n=2} - \Delta E_{n=2\rightarrow n=1}$

Solution

$$\Delta E_{n=3\rightarrow n=1} = K\left\{\frac{1}{1^2} - \frac{1}{3^2}\right\}$$

$$\Delta E_{n=3\rightarrow n=2} = K\left\{\frac{1}{2^2} - \frac{1}{3^2}\right\}$$

$$\Delta E_{n=2\rightarrow n=1} = K\left\{\frac{1}{1^2} - \frac{1}{2^2}\right\}$$

Add up Equations 2 and 3

$$\Delta E_{n=3\rightarrow n=1} = K\left\{\frac{1}{4} - \frac{1}{9} + \frac{1}{1} - \frac{1}{4}\right\}$$

$$\Delta E_{n=3\rightarrow n=1} = K\left\{\frac{1}{1} - \frac{1}{9}\right\}$$

Bohr found that the maximum number of electrons in any level is $2n^2$. The energy levels were called K, L, M, N shells i.e. n = 1, 2, 3 ..... Thus, we expect K, L, M, N shells to have maximum electrons of 2, 8, 18, 32 electrons respectively.

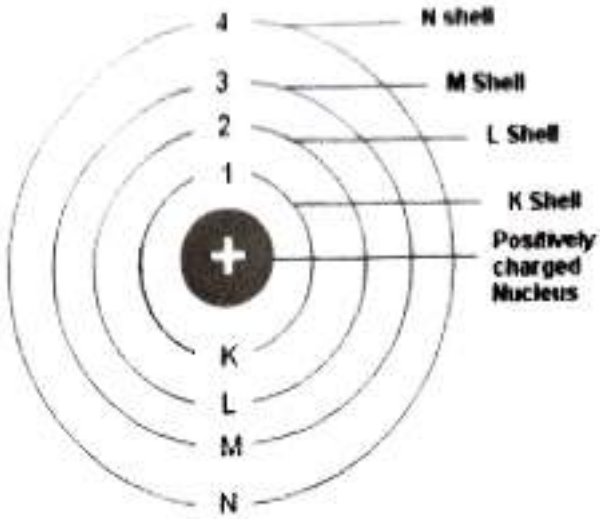


Figure 7: Bohr Model

Wave Mechanical Model

The electrons do not move around the nucleus in circular orbits like planets orbiting the sun. Electrons move randomly; however, there is more chance to find them close to nucleus. Hence, they act as a wave. The waves are characterized by wavelength (λ), frequency (ν) and speed of propagation (c). The relationship between these three variables is given as:

$$\lambda v = c$$

c can be expressed as h/m. where, m = the mass of the body and h = the Planck's constant, with the value 6.63×10^{-34} J.sec or 3.99×10^{-13}kJ mol^{-1}. Meanwhile, the value of c is 3.00×10^{8}ms^{-1}

Example

A body of mass 0.23mg moves at a wave length of 1.45×10^{-25}m. Calculate the frequency of the body movement

Solution

Using the relationship $\lambda v = c$

$$\lambda = \frac{c}{v} = \frac{h}{mv}$$

From this, make frequency (v) the subject of the formula and substitute

$$v = \frac{h}{m\lambda}$$

$$= \frac{6.63 \times 10^{-34} Jsec}{2.3 \times 10^{-7} kg \times 1.45 \times 10^{-25} m}$$

$$v = 1.988 \times 10^{-2}$$

Exercise

a. Calculate the frequency of a wave having its wavelength of 631nm.
b. Calculate the wavelength of the first two lines of the Balmer series.
c. Calculate the wavelength of the first two lines of the layman series

Principal Energy Levels

As stated above, the atoms have certain discrete energy levels. The energy levels contain orbitals of the same number (E1, E2, E3 and so on). The lowest energy level is (E1) and is called the Ground state. Furthermore, each energy level has subshells which are denoted as s, p, d, f. The maximum numbers of electron in each subshell are: s =2, p =6, d =10 and f= 14. Some of these subshells however overlapped.

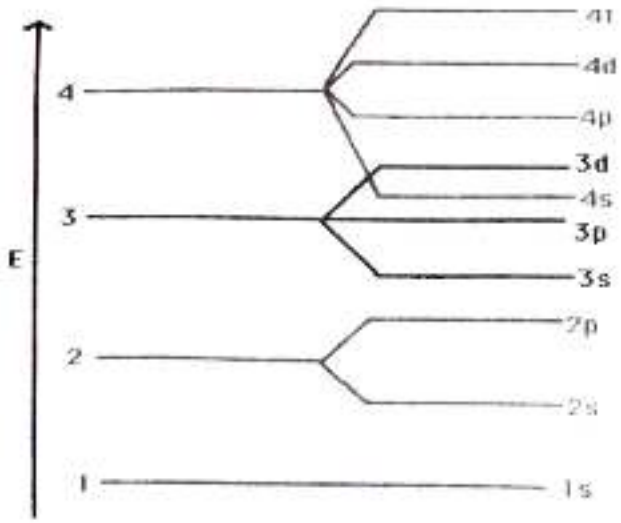


Figure 8: Energy Level Diagram

The energy levels arrangement helps us in writing the electron configuration of an element

E.g. H (1) = $1s^1$

He (2) = $1s^2$

Li (3) = $1s^2\ 2s^1$

Ar (18) = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

K (19) = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

Exercise

1. Write the electronic configuration of an element M, whose atomic mass is 17
2. What would be the electronic configurations of the following positively charged ions: K^-, Ca^{2-}, Al^{3-}?
3. What would be the electron configurations of the following negatively charged ions: F^-, Cl^-, Br^-, O^{2-}?

Quantum Number

In an atom, any electron can be described by a set of four quantum numbers.

Name	Symbol	Allowed value	Probable principle determined
Principal	N	1,2,3.....	Size and energy of orbit
Azimuthal	L	0,1, 2,.....n-1	Shape of orbital
Magnetic	M	-1 ... 0... +1	Orientation of orbital
Spin	ms	+1/2, -1/2	Spinning direction

Shape of Orbitals

The "s" sublevels are spherical. However, sublevels "p" consists of two lobes. The label x, y, or z on a given p-orbital tells along which axis the lobes of that orbital are directed.

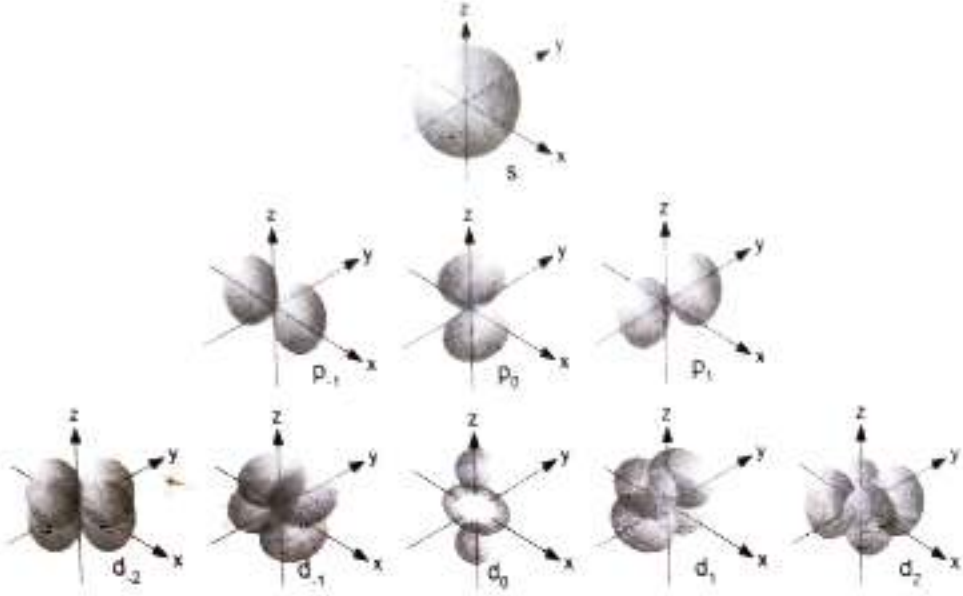
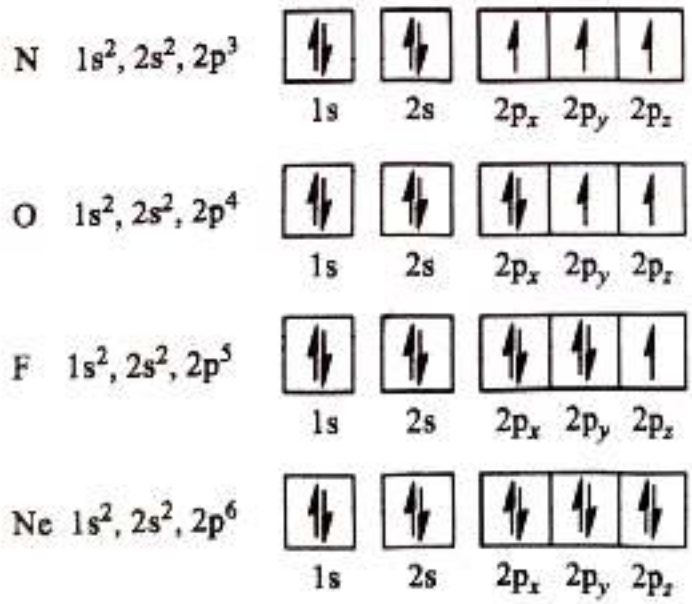


Figure 9: Shape of Orbitals

Methods of Representing Electron Configuration using Box Notations
In this representation, we use boxes to represent orbitals, with an arrow head up to represent an electron. The pair of arrows with heads in opposite directions to represent two electrons with paired spins).



Electron Configuration: a description of the orbitals of an atom or ion occupied by electrons. E.g. C (6) = $1s^2\ 2s^2\ 2px^1\ 2py^1\ 2pz^0$

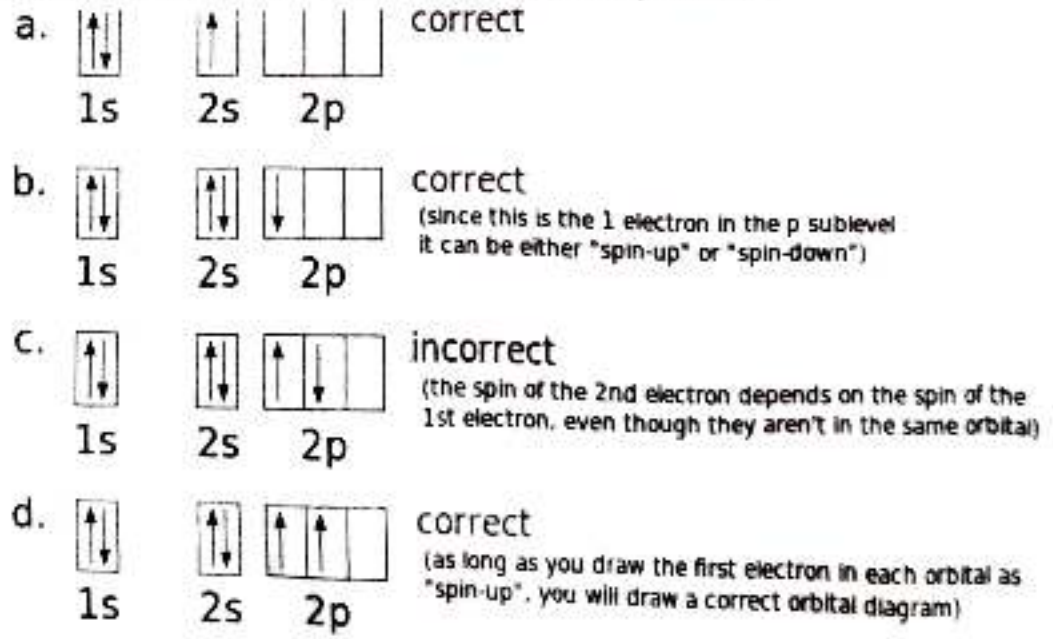
Noble gas notation: we use symbol of the noble gas immediately preceding the particular atom to indicate the electron configuration of all filled levels.

$F\ (9) = 1s^2\ 2s^2\ 2p^5 = [He]\ 2s^2\ 2p^5$

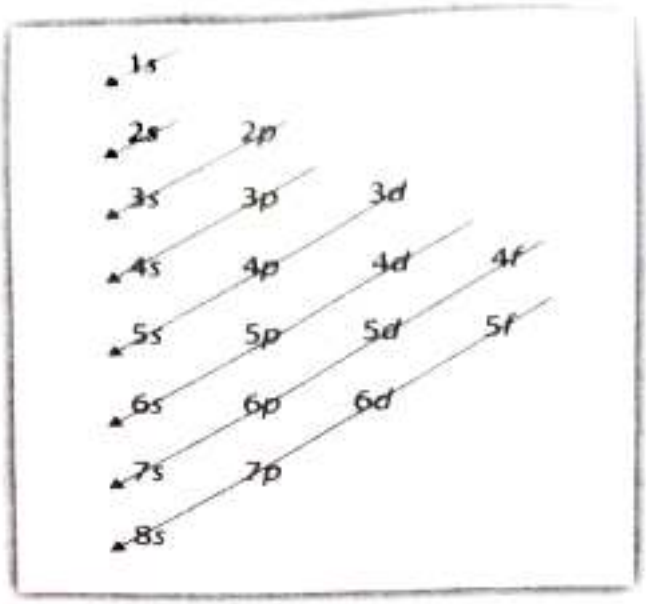
$Mg\ (12) = 1s^2\ 2s^2\ 2p^6\ 3s^2 = [Ne]\ 3s^2$

Order of filling: in filling electrons to orbitals, there are set of rules:

1. *Pauli Exclusion Principle:* This principle states that no two electrons in an atom can have the same set of values for the four quantum numbers.
2. *Heisenberg's uncertainty principle:* According to this principle, it is impossible to determine accurately both the position and momentum of a particle at the same time
3. *Hund's rule:* This state that when there is a set of orbitals of equal energy, electrons in the orbitals are singly paired before pairing in any orbital with opposite spin. That is, each orbital becomes half-filled before any of them are completely filled.



4. *Aufbau's principle*: According to this principle; electrons are added to the various orbitals in order of their increasing energy starting with the orbital of lowest energy.



Valence Shell

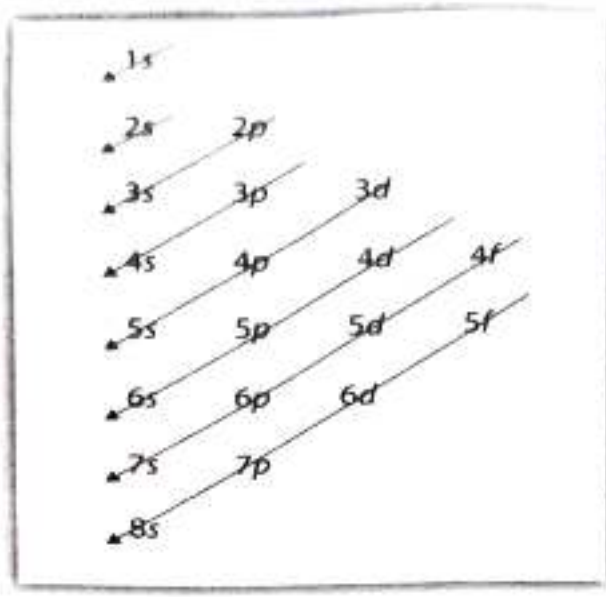
This is the outermost shell of an atom (i.e. its outside level). The electrons in this outermost shell is called the outermost electrons of an atom. The valence electrons determine the combining power of the atom and how the atom would take part in chemical bonding. For example,

- H has electron configuration = $1s^1$ with 1electron in the valence shell
- He has electron configuration = $1s^2$ with 1 electron in the valence shell
- Na has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^1$ with 1electron in the valence shell
- Al has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 2p^1$ with 3 electrons in the valence shell
- Ca has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ with 2 electrons in the valence shell

Development of the Periodic Table

In an attempt to arrange electrons of atoms, Bohr observed that chemical properties depend on the number of electrons in the outermost shell of an atom. With these, Bohr was able to set the periodic table.

Atomic number	Element	K shell	L shell	M shell	N shell
1	H	1			
2	He	2			



Valence Shell

This is the outermost shell of an atom (i.e. its outside level). The electrons in this outermost shell is called the outermost electrons of an atom. The valence electrons determine the combining power of the atom and how the atom would take part in chemical bonding. For example,

- H has electron configuration = $1s^1$ with 1electron in the valence shell
- He has electron configuration = $1s^2$ with 1 electron in the valence shell
- Na has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^1$ with 1electron in the valence shell
- Al has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 2p^1$ with 3 electrons in the valence shell
- Ca has electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ with 2 electrons in the valence shell

Development of the Periodic Table

In an attempt to arrange electrons of atoms, Bohr observed that chemical properties depend on the number of electrons in the outermost shell of an atom. With these, Bohr was able to set the periodic table.

Atomic number	Element	K shell	L shell	M shell	N shell
1	H	1			
2	He	2			

3	Li	2	1		
4	Be	2	2		
5	B	2	3		
6	C	2	4		
7	N	2	5		
8	O	2	6		
9	F	2	7		
10	Ne	2	8		
11	Na	2	8	1	
12	Mg	2	8	2	
13	Al	2	8	3	
14	Si	2	8	4	
15	P	2	8	5	
16	S	2	8	6	
17	Cl	2	8	7	
18	Ar	2	8	8	
19	K	2	8	8	1
20	Ca	2	8	8	2

Mendeleyev realized that after a certain number of elements, similar or related properties appeared again. He ordered these elements vertically underneath each other, but still ordering the elements according to their atomic mass. During the 1800s, most of the elements had been discovered, and attempt was made to classify the elements according to their similarities.

In 1829, Johann Dobereiner noted chemical similarities in several groups of thirteen elements and placed these elements into what he called triads. His groupings included the triads of 1) chlorine, bromine, and iodine, 2) sulphur, selenium, and tellurium, 3) calcium, strontium, and barium, and 4) lithium, sodium, and potassium. In all of the triads, the atomic weight

of the second element was almost exactly the average of the atomic weights of the first and third element.

In 1864, John Newlands proposed a listing of the elements in groups of seven, according to the increasing order of atomic weight. He observed a connection between the chemical properties of elements and their atomic masses. Therefore, he stated that if the known elements, beginning with lithium are arranged in order of increasing mass, the eighth element will have properties similar to the first, the ninth similar to the second, the tenth similar to the third, and so on. Newlands called his relationship the law of octaves, comparing the elements to the notes in a musical scale. Newlands tried to force all the known elements to fit into his octaves but many of the heavier elements, when discovered, did not fit into his patterns.

Mendeleev's Table and Chemical Behaviour

By 1869, a total of 63 elements had been discovered. As the number of known elements grew, scientists began to recognize patterns in the way chemicals reacted and began to devise ways to classify the elements. Dmitri Mendeleev, Russian chemist, was the first scientist to make a periodic table much like the one we use today.

Mendeleev's table listed the elements in order of increasing atomic mass. Then he placed elements underneath other elements with similar chemical behaviour. For example, lithium is a shiny metal, soft enough to be cut with a spoon. It reacts readily with oxygen and reacts violently with water.

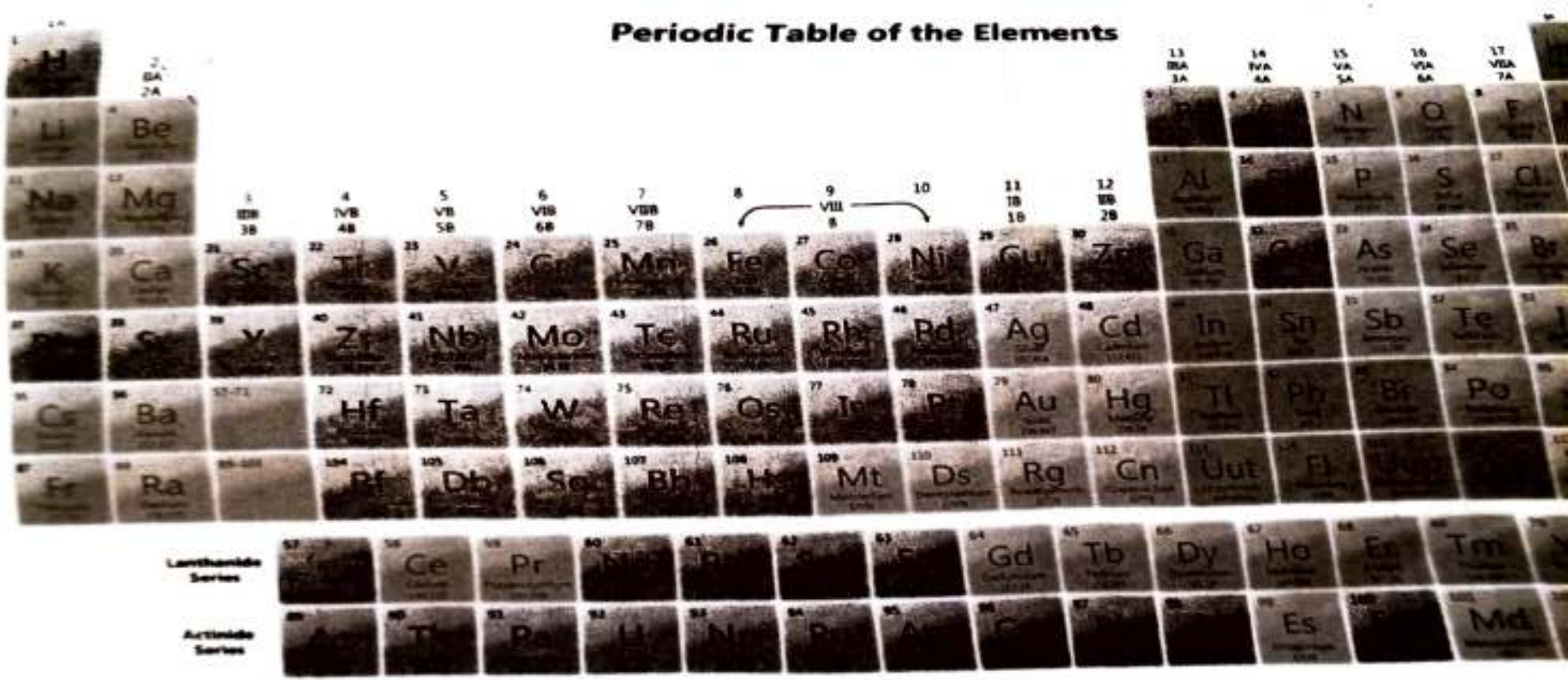


Figure 10: Periodic Table

Families of the Periodic Table

Remember that Mendeleev arranged the periodic table so that elements with the most similar properties were placed in the same group. A **group** is a vertical column of the periodic table. All of the 1A elements have one valence electron. This is what causes these elements to react in the same ways as the other members of the family. The elements in 1A are all very reactive and form compounds in the same ratios with similar properties with other elements. Because of their similarities in their chemical properties, Mendeleev put these elements into the same group. Group 1A is also known as the **alkali metals**. Although most metals tend to be very hard, these metals are actually soft and can be easily cut.

Group 2A is also called the **alkaline earth metals or Beryllium family**. Once again, because of their similarities in electron configurations, these elements have similar properties to each other.

Group 3 (IIIB) – the Boron family
Group 4 (IVB) – the Carbon family
Group 5 (VB) – the Nitrogen family
Group 6 (VIB) - the Nitrogen family
Group 7 (VIIB) – the halogen family
Group 8 (VIIIB) – the noble or inert gas family

Metals, Non-metals, and Metalloids

There is a progression from metals to non-metals across each row of elements in the periodic table. The diagonal line at the right side of the table separates the elements into two groups: the metals and the non-metals. The elements that are on the left of this line tend to be metals, while those to the right tend to be non-metals (with the exception of hydrogen which is a non-metal). The elements that are directly on the diagonal line are metalloids, with some exceptions. Aluminium touches

the line, but is considered a metal. Metallic character generally increases from top to bottom down a group and right to left across a period, meaning that francium (Fr) has the most metallic character of all of the discovered elements. Most of the chemical elements are metals. Most metals have the common properties of being shiny, very dense, and having high melting points. Metals tend to be **ductile** (can be drawn out into thin wires) and **malleable** (can be hammered into thin sheets). Metals are good conductors of heat and electricity. All metals are solids at room temperature except for mercury. In chemical reactions, metals easily lose electrons to form positive ions. Examples of metals are silver, gold, and zinc

Variation of Properties in Periodic Table

1. Atomic Size

Atomic size is the size of an atom and is determined by the size of its outermost occupied orbital.

The simplest way to determine the size of an atom is to determine the distance between atoms in a sample of the element.

- ***Trend along the group***: the atomic size increases from up to down in a column
- ***Trend along the period***: and decreases from right to left across a row.

Ion: an atom with an unequal number of protons and electrons (in the normal conditions, an atom is neutral).

$$Na + Energy \longrightarrow Li^{+}\ (ion) + e^{-}$$

2. Ionization energy

This is the energy that is necessary to remove an electron from an atom or a molecule i.e. the energy required to remove the most loosely held electron from an atom in the gas phase. The Ionisation Energy is

measure for the force by which an electron is bound in the atom. The Ionisation Energy is a function of the radius and the charge of an atom

- ***Trend along the groups***: ionization energy increases from down to up in a column and from left to right across a row.
- ***Trend along the period***: the ionization energy increases from left to right along a period due to decrease in size of the atoms i.e. electrons are closer to the nucleus.

In general, metals have relatively low ionization energies. Therefore, a relatively small amount of energy is needed to remove an electron from a typical metal.

Note: the valence electrons can be removed easily than other electrons (because they are further apart from the nucleus).

3. Ionic radius

The radius of an atom is defined as half distance between the centres of two bonded atoms. For bonded ions, the distance is referred to as ionic radii.

- ***Trend along the groups***: similarly charged ions increase in size from top of the table to the bottom.
- ***Trend along the period***: (i) for positive ions, within a period ionic radius decrease from left to right due to increasing nuclear charge. (ii) for negative ions, within a period ionic radius decrease from left to right like positive ions.

4. Electron Affinity (EA)

The electron affinity is the energy that accompany the process of adding an electron to an atom in its ground state, that is, the energy accompanying the acceptance or addition of an electron to an atom shell.

$$M + e^- \longrightarrow M^- + \text{Electron Affinity (EA)}$$

It should be noted that the smaller an atom is, the greater its affinity to attract or add an electron to its shell. Hence,

- *Trend along the groups*: electron affinity decreases from down to up in a group
- *Trend along the period*: electron affinity increases from left to right along a period due to decrease in size of the atoms i.e. electrons are closer to the nucleus.

5. Electronegativity

This is the relative ability of an atom to attract shared electrons to it. It is used to predict whether bonds between atoms will be predominantly ionic or covalent in character.

- *Trend along the groups*: Electronegativity decreases from bottom to the top in any group due to decrease in size.

- *Trend along the period*: Electronegativity increases from left to right across any period due to increase in the nuclear charge.

Thus, the most electronegative elements are found in the upper corner of the periodic table

Examples

1. Arrange the following in order of increasing atomic radii:
   a) Li, K, Ti b) Be, Mg, Zn
2. Arrange the following in order of decreasing order of first ionization potential:
   a) Be, N, O b) Mg, Na^+, Al

Solution

1a. $Li \rightarrow K \rightarrow Ti$ b. $Be \rightarrow Mg \rightarrow Zn$

2a. O, N, Be b. $Na^+ \leftarrow Al \leftarrow Mg$

STATES OF MATTER

K.O. Sodeinde

All the objects around us whether living or non-living are matter. Water we drink, food we eat, air we breathe, cars and furniture are all examples of matter. Matter is anything that has mass and occupies space. Matter appears in a huge variety of forms such as rocks, trees, computer, clouds, people, etc. Each pure kind of matter is called substance. A state of matter is one of the distinct forms that matter takes on. Four states of matter are observable in everyday life: **solid, liquid**, gas, and **plasma**.

Matter in the solid state maintains a fixed volume and shape, with component particles (atoms, molecules or ions) close together and fixed into place. Matter in the liquid state maintains a fixed volume, but has a variable shape that adapts to fit its container. Its particles are still close together but move freely. The component particles of gases are loosely packed and can move around freely. Matter in the plasma state has variable volume and shape, but as well as neutral atoms, it contains a significant number of ions and electrons, both of which can move around freely. Plasma is the most common form of visible matter in the universe.

The term **phase** is sometimes used as a synonym for state of matter, but a system can contain several immiscible phases of the same state of matter.

Gaseous State

The gaseous state is arguably the most widely studied state which has led to the development of many important chemical theories and principles.

The basic physical properties of gases include expansion, relatively low density, compression, diffusion, etc.

Measurable properties of gases are as follows:

i. Mass: It is expressed in gram or kg.

ii. Volume: It is equal to the volume of the container and is expressed in terms of litre (L), millilitre (mL), cubic centimetre (cm^3), cubic metre (m^3) or cubic decimetre (dm^3).
$1\ L = 1000\ mL = 1000\ cm^3 = 1\ dm^3$
$1\ m^3 = 10^3\ dm^3 = 10^6\ cm^3 = 10^6\ mL = 10^3 L$

iii. Pressure: Gas pressure is measured with manometer and atmospheric pressure is measured by barometer.
1 atm = 760 mm of Hg = 760 torr, 1 atm = 101.325 kPa = 101325 Pa = 101.325 Nm^{-2} = 1.01325 bar, 1 bar = 10^5 Pa.
Measurement of pressure of gas: (a) Open end manometer, $p_{gas} = p_{atom} - h$ (b) Closed end manometer, $p_{gas} = h$ where h is difference in the mercury levels in the two columns of density (d) (of a gas)

iv. Temperature: It is measured in Celsius scale (°C) or in Kelvin scale (K). SI unit of temperature is kelvin (K), $T\,(K) = t^0C + 273$. Standard temperature and pressure (STP) stand for 273.15 K (0°C) temperature and 1 bar (i.e. exactly 105 Pascal) pressure. At STP, molar volume of an ideal gas is 22.71098 L /mol.

A more detailed behaviour of gases is described by the kinetic theory of gases. The kinetic theory of gases is based on the following assumptions:

i. A gas consists of large number of finites, discrete particles, called molecules.

ii. Volume occupied by gas molecules is negligible as compared to the total volume of the gas.

iii. There is continuous rapid random motion of gas molecules. The molecules collide with each other and with the walls of container.
iv. The molecules are perfect elastic bodies and there is no loss of kinetic energy during collisions.
v. There are no attractive forces between the gaseous molecules.
vi. The pressure exerted by a gas is due to the bombardment of gas molecules against the walls of the container.
vii. The average kinetic energy (KE) is directly proportional to absolute temperature.

$$KE = \frac{3}{2}RT$$

Therefore,

the average kinetic energy per molecule = $KE = \frac{3}{2}KT$.

k is Boltzmann constant, it is gas constant per molecule.

$$K = \frac{R}{NA} = 1.38 \times 10^{-23} J/K/mol$$

From the above postulates, the kinetic gas equation derived is

$$pv = \left(\frac{1}{3}\right) MnU^2;$$

where $U = root\ mean\ square\ velocity$

$$U = \sqrt{\frac{3RT}{M\ Velocities\ of\ gas\ molecules}}$$

The different velocities possessed by gas molecules are:

i. Most probable velocity (α): It is the velocity possessed by maximum fraction of gas molecules at a particular temperature.

$$\propto = \sqrt{\frac{2RT}{M}}$$

ii. Average velocity (v): This is the average of the different velocities of all the molecules.

$$v = \sqrt{\frac{8RT}{\pi M}}$$

iii. Root mean square velocity (U_{rms}): It is the square root of the mean of the square of the different velocities of the molecules.

Gas Laws

Boyle's Law: Robert Boyle in 1661 postulated that the volume of a fixed mass of gas at constant temperature is inversely proportional to the pressure. i.e.

$$V \propto \frac{1}{P}, \quad V = \frac{k}{P}, \quad k = PV$$

(Temperature, T, and number of moles, n, are constant)

If P_1V_1 are the initial pressure and volume of a given gas sample and P_2V_2 are final pressure and volume; then we can say:

$$P_1V_1 = k = P_2V_2 \text{ or } P_1V_1 = P_2V_2$$

Boyle's law strictly only applies to ideal gases at very low pressures, when there are very few molecular collisions and very few interactions between the molecules.

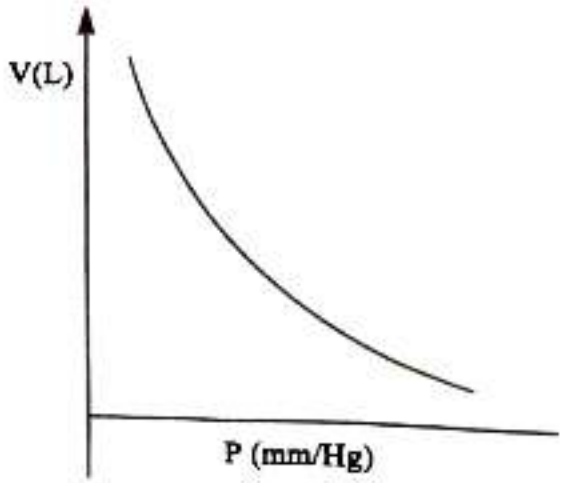


Figure 11: Graphical Representation of Boyle's Law

Charles' Law: The volume of a fixed mass of gas at constant pressure is directly proportional to the absolute temperature i.e. $V \alpha T$ (P & n are constant).

$$V = KT. \quad V/T = K$$

$$\frac{V_1}{T_1} = K = \frac{V_2}{T_2}$$

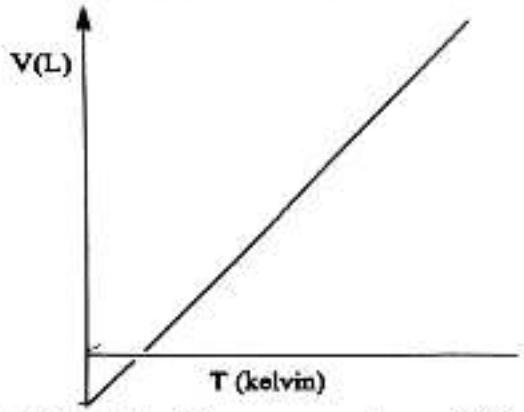


Figure 12: Graphical Representation of Charles' Law

The Combined Gas Law (General Gas Law Equation)

The combination of Boyle's law; $V \alpha 1/P$ (T, n constant) and Charles' law: $V \alpha T$ yield: $V \alpha T/P$ (n constant), $V = KT/P$ where K is the proportional constant (n constant).

$$PV = KT,$$

Therefore, $K = \frac{PV}{T}$

If the pressure, volume and temperature change from initial level (1) to final level (2), then

$$\frac{P_1V_1}{T_1} = K,$$

$$\frac{P_2V_2}{T_2} = K,$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \text{ .... General Gas Law Equation}$$

Examples

1. 10.0 L of a gas has a pressure of 0.7atm and temperature of 23°C. What will be the volume if the pressure is increased to 1.4atm and the temperature to 380K?

Solution

Initial Conditions	Final Condition
$V_1 = 10L$	$V_2 = ?\ L$
$P_1 = 0.7atm$	$P_2 = 1.4atm$
$T_1 = 23 + 273 = 296K$	

$$\frac{p_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Substituting the values in the above equation, we have

$$\frac{0.7 \times 10}{296} = \frac{X \times 1.4}{380}$$

$$\frac{0.7 \times 10 \times 380}{296 \times 1.4} = X = 6.4189L$$

Gay-Lussac's Law: At constant volume, the pressure of a fixed mass of gas is directly proportional to the absolute temperature. i.e. $P \alpha T$ (V, n constant).

$$P = KT, K = \frac{P}{T}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Avogadro's Law: Consider a ball containing a fixed mass of gas. If we add to its mass at constant T and P, the volume of gas (V) will increase. Experimentally, it has been found that the amount of gas in moles is

proportional to the volume. i.e. $V \propto n$ (T and P constant). The law is also expressed as "equal volume of all gases at the same temperature and pressure contain the same number of molecules".

$V = Kn$ where K is the proportionality constant.

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} = K$$

If $V_1 = V_2$ then $n_1 = n_2$ (T and P constant)

Diffusion and Effusion of Gases

Diffusion is the migration of molecules of different substances as a result of the random motion of molecules. Although gas molecules are consistently having collisions resulting in frequent changes in direction, the net rate at which a gas moves in a particular direction depends on the average speed. Diffusion always proceeds from a region of higher concentration to one of lower concentration. **Effusion** is a process related to diffusion, it is the escape of gas molecules from their container through an orifice (pinhole).

Graham's Law of Diffusion: Under constant temperature and pressure, the rates of diffusion of gases are inversely proportional to the square root of their densities.

Mathematically, $\left(\frac{r_1}{r_2}\right) = \left(\frac{\sqrt{d_2}}{\sqrt{d_1}}\right) = \left(\frac{\sqrt{M_2}}{\sqrt{M_1}}\right)$

Dalton's Law of Partial Pressure: At constant temperature, the total pressure exerted by a mixture of non-reacting gases is the sum of partial pressures of different gases present in the mixture.

$$P_T = p_1 + p_2 + p_3 + \cdots$$

Partial pressure of a gas = mole fraction of the gas × total pressure. If n_1, n_2 and n_3 are moles of non-reacting gases filled in a vessel of volume V at temperature T, the total pressure, p is given by

$$PV = (n_1 + n_2 + n_3)RT$$

This is the equation of state of a gaseous mixture.

Example

The volume of hydrogen collected over water is 453 mL at 18° C and 780 mm Hg. What is its volume dry at STP?

Solution

$V_1 = 453$ mL, $V_2 = ?$

$T_1 = 18° C + 273 = 291$ K, $T_2 = 0° C + 273 = 273$ K

$$\frac{p_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$V_2 = \frac{P_1V_1T_2}{(T_1 \times P_2)}$$

$$V_2 = \frac{780 \times 453\,mL \times 273k}{(291 \times 760)} = 436mL\ H_2$$

Derivation of Ideal Gas Law

Recall Boyle's law $V \propto \frac{1}{p}$ (T, n constant)

Charles' law V α T (P, n constant)

Avogadro's law V α n (P, T constant)

Combination of the three laws yields: $V \propto \frac{nT}{P}$

If R is the proportionality constant, the equation becomes

$$V = \frac{RnT}{P}$$

$PV = nRT$ where constant R = gas constant

For one mole (1) of a gas, the equation becomes $PV = RT$

$$R = \frac{PV}{nT}$$

For one mole of a gas at standard temperature and pressure (STP), the volume = 22.4 L

Substituting this,

$$R = \frac{1atm \times 22.4\,L}{1mole \times 273} = 0.082\,L\,atm/mol/k$$

Ideal gases obey the ideal gas law

Ideal Gas Equation, Density and Molar Mass Relationship

$$PV = nRT$$

$$No.of\ moles = \frac{mass\ (m)}{molar\ mass\ (M)}$$

No. of moles = mass (m)/molar mass (M)

$$PV = \frac{m}{M} \times RT$$

$$P = \frac{m}{(V \times M)} \times RT$$

Recall $density = \frac{mass}{volume}$

We have $P = \frac{density}{molar\ mass} \times RT$

Example

The density of a gaseous organic compound is 340 g/L at 45°C and 1.7 atm. What is its molar mass?

Solution

T (K) = 45+273 = 318K

P = 1.74 atm

Density = 340 g/L

From equation (v), $molar\ mass = \frac{density}{PRT}$

$$= \frac{340}{1.74} \times 0.082 \times 318 = 7.49 g/mol \approx 5215.2 g/mol$$

Exercise: A compound X with density 8.9 g/L was synthesized in the laboratory at 60°C and 2.73atm. Calculate the molar mass of X.

Deviation from Ideal Behaviour (Real Gases)

- At high pressure and low temperature, the gases deviate considerably from the ideal behaviour. The actual volume of molecules in an ideal gas is negligible while in a real gas it is appreciable.
- No attractive forces between molecules in an ideal gas while these exist in a real gas
- Collisions in an ideal gas is perfectly elastic whereas it is not so in a real gas.

Deviation can be expressed in terms of compressibility factor (Z), expressed as $Z = \frac{pV}{nRT}$

In case of ideal gas, $pV = nRT$, Z = 1, In real gas, $pV \neq nRT$, $Z \neq 1$

Negative deviation: Here, Z < 1 (gas is more compressible).

Positive deviation where Z > 1 (gas is less compressible).

The factors affecting the deviation are:

i. Nature of the gas: Generally, the most easily liquefiable and highly soluble gases show larger deviation.

ii. Pressure: The deviation is more at high pressure. CO_2 and N_2 show negative deviation at low pressure and positive derivation at high pressure.

iii. Temperature: The deviation is more at low temperature and H_2 and He always show positive deviations at 0°C

Van der Waal's Equation

The volume of a gas is the free space in the container in which molecules move about. The volume of an ideal gas is the same as the volume of the container; v. Van der Waal assumed that molecules of a real gas are frigid spherical particles which possess a certain, definite volume called extended volume not available for the molecules to move in.

Real gas volume = ideal gas – volume occupied by gas moles. For n moles of the gas, the corrected volume is (v – nb), where b is a constant and is characteristics of each gas.

Pressure

A molecule in the interior of a gas is attracted by other moles on all sides. These attractive forces cancel out. But a molecule about to strike the wall of the vessel is attracted by molecules on one side only. Thus, it experiences an inward pull. Therefore, it strikes the wall with reduced velocity and the actual pressure of the gas P, will be less than the ideal pressure.

$P_{actual} = P_{ideal} - P$ (P = factor by which the ideal pressure is reduced)

P is determined by the force of attraction between molecules (A) striking the wall of container and mols (B) pulling them inward. The net force of

attraction is therefore proportional to the concentration of (A) typ molecules and also of (B) type molecules i.e.

$$PC_A \times C_B$$

$$\frac{Pn}{v} \times \frac{n}{v}$$

$$p = \frac{an^2}{v^2}$$

Where n is total number of gas molecules in volume V and a proportionality constant of the gas. Thus, the pressure P in the ideal g equation is corrected as:

Recall, $PV = nRT$

$$\left(P + \frac{an^2}{v^2}\right)(v - nb) = nRT \quad \text{Van Der Waal Equation}$$

For one mole of a gas (n = 1), the equation becomes;

$$\left(P + \frac{a^2}{v^2}\right)(v - b) = RT$$

Constants a and b are called Van Der Waal constants and ar characteristic of each gas.

References

Gurnett D. A., Bhattacharjee A, 2005. Introduction to Plasma Physic With Space and Laboratory Applications. Cambridge, UK Cambridge University Press. p. 2. ISBN 0-521-36483-3.

THE MOLE CONCEPT AND AVOGADRO'S NUMBER

P. O. Okolo

If we want to buy rice, we may measure it in cups. This is because it is not practicable to count the individual grain of rice; a cup of rice may contain 2.0×10^{12} grains of rice. Similarly, scientists quantify the elementary entities (electrons, atoms, ions, molecules, etc.) in moles (mols) – a mole (mol) of any particle contains exactly 6.022×10^{23} of such particles; this is the Avogadro's number (N).

$$\mathbf{1\ mole} = 6.022 \times 10^{23} = \boldsymbol{Avogadro's\ number\ (N)}$$

The mole is the reacting units of chemical species, and therefore it is the basis upon which the amounts of reactants and products could be measured and monitored.

Mole Concept and the Periodic Table

The Periodic Table presents the average molar masses of atoms of elements. Each molar mass is actually one mole of the atoms of that element. For example, copper (Cu) is enlisted as 63.55 on the Periodic Table. This means that one mole 6.022×10^{23} of Cu atoms weighs $63.55g$.

$$63.55g\ Cu = 1\ mole\ of\ Cu = 6.022 \times 10^{23}\ atoms\ of\ Cu$$

Example 1

Calculate:

a. The mass of 0.0892 mol of Cu.

b. The number of moles of Cu in 72.5g sample.

c. The mass of a sample of Cu containing 2.00×10^{16} of Cu atoms, and

d. The mass of a single atom of Cu. (Given: Cu = 63.55 g/mol)

Solution

a. Since 63.55g/mol Cu means 1 mol Cu weighs 63.55g Cu,

$\therefore$ 0.0892 mol Cu will weigh

$$\frac{0.0892\ mol\ Cu}{1\ mol\ Cu} \times 63.55g\ Cu$$

$$= 5.6687g\ Cu$$

b. Since $63.55g\ Cu = 1\ mol\ Cu$

$$\therefore 72.5g \qquad Cu = \frac{72.5gCu}{63.55gCu} \times 1\ mol\ Cu = 1.1408\ mol\ Cu$$

c. From Avogadro's number and Periodic Table,

$$6.022 \times 10^{23}\ atoms\ of\ Cu = 1\ mol\ Cu = 63.55g\ Cu$$

$$\therefore 2.00 \times 10^{16}\ atoms\ of\ Cu = \frac{2.00 \times 10^{16}\ atoms\ Cu}{6.022 \times 10^{23}\ atoms\ Cu} \times 63.55g\ Cu$$

$$= 21.1059g\ Cu$$

d. From Avogadro's number and Periodic Table,

$$6.022 \times 10^{23} atoms\ Cu = 1\ mol\ Cu = 63.55g\ Cu$$

$$\therefore \quad 1\ atom\ Cu = \frac{1\ atom\ Cu}{6.022 \times 10^{23}\ atom\ Cu} \times 63.55g\ Cu$$

$$= 1.055297243 \times 10^{-22} g\ Cu$$

Mole Concept, Molar Mass and Molecular Mass

A formula unit of a molecule represents one mole of such compound.

For example:

a. H_2SO_4 =1 mol of H_2SO_4 molecules = $1 \times 6.022 \times 10^{23}$ molecules of H_2SO_4

b. $2MgSO_4$ = 2mols of $MgSO_4$ molecules = $2 \times 6.022 \times 10^{23}$ molecules of $MgSO_4$.

c. $^1/_2\,O_2$ mol of O_2 molecules = $\frac{1}{2} \times 6.022$ molecules of O_2

The molar mass of a compound is the mass in grams per mole, while the molecular mass of the same compound is its mass in atomic mass units (a.m.u, symbol u)

Example 2

Calculate the (i) molecular mass and (ii) molar mass of $(NH_4)_2SO_4$ (N=14, H=1, S=32, O=16).

Solution

$$\begin{aligned}(NH_4)_2SO_4 &= 2N + 8H + S + 4O \\ &= 2 \times 14 + 8 \times 1 + 1 \times 32 + 4 \times 16 \\ &= 28 + 8 + 32 + 64 \\ &= 132\end{aligned}$$

∴ Molecular mass = 132 a.m.u or 132u, and molar mass = 132g/mol

Example 3

Calculate:

a. the number of SO_4^{2-} ions in 150 g sample of H_2SO_4, and

b. the volume of $HCl(d = 1.18g/mL)$ that contains 2.00 mol Cl atoms.

$(S = 32, H = 1, Cl = 35.5)$

Solution

a. 1 mol H_2SO_4 ⟶ 1 mol SO_4^{2-} ion

∴ $98g$ $H_2SO_4 \equiv 6.022 \times 10^{23}$ SO_4^{2-} ions

$$\therefore 150g\ H_2SO_4 \equiv \frac{150g\ H_2SO_4}{98g\ H_2SO_4} \times 6.022 \times 10^{23}\ SO_4^{2-}\ \text{ions}$$

$$\equiv 9.2173 \times 10^{23}\ SO_4^{2-}\ \text{ions}$$

b. 1 mol HCl ⟶ 1 mol H atoms + 1 mol Cl atoms

i.e 1 mol Cl atoms ⟶ 1 mol HCl molecules

$\therefore\ 2mol\ Cl\ atoms \equiv 2\ mol\ HCl\ molecules$

But $2\ mol\ Cl = 2 \times 35.5Cl = 71.0g\ Cl$

Now, from density of HCl

$1.18g\ HCl = 1ml$

$$\therefore 71g\ HCl = \frac{71g\ HCl}{1.18g\ HCl} \times 1ml\ HCl$$

$$= 60.1695\ mL\ HCl$$

Exercises

1. Calculate the total number of Cl atoms in 150 mL of liquid $CCl_4(d = 1.589g/mL)$.
2. Find the number of grams of carbon in 300 g of sucrose $(C_{12}H_{22}O_{11})$ $(C = 12, H = 1, O = 16)$.
3. Calculate the volume (mL) of aqueous 40% sucrose having a density of 0.911 g/mL, required to furnish 1.5 mol C atoms (sucrose = $C_{12}H_{22}O_{11}; C = 12, H = 1, O = 16$).
4. Silver (Ag) is an ornamental metal. Calculate the mass (g) of Ag in 3 atoms of Ag. (Ag=107.9).
5. How many grams of Zn are there in 0.54 mol of Zn (Zn=65.39)
6. Compute the mass (g) of americium in 12 atoms of Am (Am= 243).
7. Calculate (i) the number of moles of atoms and (ii) the number of atoms in a 20 g sample of Ce (Ce=140.1).

8. How many silicon (Si) atoms are there in a silicon chip that has a mass of 6.78mg? ($Si = 28.09$).
9. Calculate (1) the number of moles of a sample of cobalt (Co) containing 3.00×10^{20} atoms and (ii) the mass of the sample ($Co = 58.93$).
10. Determine the mass percent of sulphur (S) in penicillin ($C_{14}H_{20}N_2SO_4$) ($C = 12, H = 1, N = 14, S = 32, 0 = 16$).

References

1. Ndiokwere CL, Ukhun ME, 2006. First year University inorganic and physical chemistry. Mindex Publishing, Benin City.
2. Hill JW, Petrucci RH, McCreary TW, Perry SS, 2005. General Chemistry, 4th edn. Pearson prentice Hall, New Jersey.
3. Chang R, 1991. Chemistry, 4th edn. McGraw-Hill Inc. New Jersey.

DETERMINATION OF RELATIVE AND AVERAGE ATOMIC MASSES

P. O. Okello

If we know the combining ratios of atoms in compounds, it becomes easy to measure atomic masses. By international convention the modern atomic mass standard is the pristine isotope carbon – 12 (^{12}C) which is assigned a mass of exactly 12 atomic mass units ($12u$). Therefore, we can define an Atomic Mass Unit (a.m.u or u) as exactly $1/12th$ the mass of a ^{12}C atom.

The atomic mass of an element is defined as the weighted average of the masses of the naturally occurring isotopes of that element. To estimate the atomic mass of an element, we need to know the atomic masses of its isotopes and the fractional natural abundances of these isotopes.

Example 1

Use the Table below to calculate the average atomic mass of carbon.

Isotope	a.m.u (u)	% Abundance
^{12}C	12.00000	98.892
^{13}C	13.00335	1.108

Solution

Step 1: Convert % abundances to fractional abundances:

$$98.892\% = \frac{98.892}{100} = 0.98892$$

Similarly, $1.108\% = 0.01108$

Step 2: Multiply the mass of each isotope with its fractional abundance then add up the two products. This gives the required average atomic mass of C:

$$\text{Average atomic mass} = (1200000 \times 0.98892) + (13.00335 \times 0.01108)$$
$$= 11.86704 + 0.144077118$$
$$= 12.01111712 \approx 12.0111U$$

Example 2

The table below gives some isotopic data of naturally occurring copper. Use the data to calculate the percent natural abundance of the two isotopes of copper if the average atomic mass of copper is 63.546U.

Isotope	a.m.u (u)	% Abundance
^{63}Cu	62.9298	?
^{65}Cu	64.9278	?

Solution

Step 1: Let the fractional abundance of $^{63}Cu = x$

Therefore, the fractional abundance of $^{65}Cu = 1 - x$

Step 2: Multiply the mass of each isotope with its fractional abundance. Add up the products and equate this to the given average atomic mass for copper:

$$[62.9298 \times x] + [64.9278 \times (1 - x)] = 63.546$$

Step 3: Solve for x and hence find the fractional and finally the percent abundances of the two isotopes:

$$62.9298x + 64.9278 - 64.9278x = 63.546$$
$$\therefore 62.9298x - 64.9278x = 63.546 - 64.9278$$
$$-1.998x = -1.3818$$
$$\therefore x = \frac{-1.3818}{-1.998} = 0.691591591$$

Also $1 - x = 1 - 0.691591591 = 0.308408408$

$\therefore$ % abundance for $^{63}Cu = 0.6916 \times 100\% = 69.16\%$, and % abundance for $^{65}Cu = 0.3084 \times 100\% = 30.8408\%$

Exercises

1. Some isotopic data on naturally occurring Neon are given below:

Isotope	a.m.u (u)	% Abundance
^{20}Ne	19.99244	90.51
^{21}Ne	20.99395	0.27
^{22}Ne	21.99138	9.22

Calculate the average atomic mass of Neon.

2. Indium has two naturally occurring isotopes and average atomic mass of 114.82 u. If one of the isotopes has a mass of 112.9043 u, calculate the mass (u) of the second isotope.

3. The table below shows some isotopic data on chlorine. Use the data to compute the percent natural abundance of chlorine.

Isotope	a.m.u (u)	% Abundance
^{35}Cl	34.9688	?
^{37}Cl	36.9659	?

Average atomic mass of Cl = 35.45 a.m.u.

References

1. Ndiokwere C. L., Ukhun M. E., 2006. First year University inorganic and physical chemistry. Mindex Publishing, Benin City.
2. Hill J. W., Petrucci R. H., McCreary T. W., Perry S. S. 2005. General chemistry, 4th edn. Pearson prentice Hall, New Jersey.
3. Chang R., 1991. Chemistry, 4th edn. McGraw- Hill Inc. New Jersey.

STOICHIOMETRY AND CHEMICAL EQUATIONS

P. O. Okolo

Stoichiometry deals with the quantities of chemical species involved in a chemical reaction. To perform stoichiometric calculations, we need the knowledge of a) mole concept and chemical equation b) law of conservation of mass, and c) the Periodic Table of elements. In practical terms, stoichiometry will address the following issues:

i. predicting quantities of reactants and products,
ii. interpreting the results of a chemical analysis,
iii. deciding the most feasible routes for chemical reactions,
iv. computing the energy (heat, light, electricity, etc) required for certain chemical reactions,
v. calculating empirical formula, molecular formula, percentage composition,
vi. computations involving limiting reagents, percentage yield, theoretical yield, and
vii. solution chemistry involving various concentration expression, like molarity (M); molality (m); percent weight in weight (% w/w), weight in volume (% w/v), volume in volume (% v/v); normality (N); and mole fraction (x).

Balancing Chemical Equation

A chemical equation is a shorthand representation of a chemical reaction. The following symbols are used in chemical equations:

(↑) *or* (*g*) =gas/steam: (l) = liquid; (s) = solid; (aq) = aqueous (water) solution; ht or Δ = applied heat; and ↓ or ppt = precipitate.

Example 1

Balance the equation $Fe_{(s)} + O_{2(g)} \longrightarrow Fe_2O_{3(s)}$

Solution

$*4Fe_{(s)} + *3O_{2(g)} \longrightarrow *2Fe_2O_{3(s)}$

* are the stoichiometric coefficients.

Example 2

Balance the equation $C_2H_{6(g)} + O_{2(g)} \longrightarrow CO_{2(g)} + H_2O_{(g)}$

Solution:

$2C_2H_{6(g)} + 7O_{2(g)} \longrightarrow 4CO_{2(g)} + 6H_2O(g)$

Exercises

Balance the following equations:

i. $H_{2(g)} + O_{2(g)} \longrightarrow H_2O_{(l)}$

ii. $SiCl_4 + H_2O \longrightarrow SiO_2 + HCl$

iii. $PCl_5 + H_2O \longrightarrow H_3PO_4 + HCl$

iv. $CaO + P_4O_{10} \longrightarrow Ca_3(PO_4)_2$

v. $CH_3CH_2CH_2CH(OH)CH_2OH + O_2 \longrightarrow CO_2 + H_2O$

vi. $MnO_4^- + H^+ + 5e^- \longrightarrow Mn_2^+ - H_2O$

Percentage Composition

Example

Calculate the percentage composition of nitrogen (N) in magnesi nitrate $Mg(NO_3)_2\, Mg = 24.31, N = 14.01, O = 16.00)$

Solution:

$Mg(NO_3)_2 \rightarrow Mg + 2N + 6O$

Imputing molar masses (mm), get

Mg + 2N + 6O

$24.31 + 2 \times 14.01 + 6 \times 16.00$

$= 24.31 + 28.02 + 96$

Total = 148.33g/mol

$$\text{Now, } \%N = \frac{\text{molar mass due to N}}{\text{molar mass of } Mg(NO_3)_2} \times 100\%$$

$$= \frac{28.02}{148.33} \times 100\% = 18.89\%N$$

-1/3

Exercises

Calculate the percentage composition of oxygen in each of the following compounds:

i. $C_3H_8O_3$ (ii) $(NH_4)_2SO_4$ (ii) $N(CH_2CH_2OH)_3$

Empirical Formula

Empirical formula of a compound is its simplest formula. For example, the simplest formula of hydrogen peroxide (H_2O_2) is HO.

Empirical Formula from Elemental Analysis

Example 1

Establish the empirical formula of phenol with the following composition 76.57%C, 6.49%H, and 17.00% O by mass ($C = 12; H = 1; O = 16$).

Solution

Follow the scheme below:

	C	H	O
% composition	76.57	6.49	17.00
Mole ratio	$\frac{76.57}{12}$	$\frac{6.49}{1}$	$\frac{17.00}{16}$

	6.3825	6.49	1.0625
Simplest ratio	$\frac{6.3825}{1.0625}$	$\frac{6.49}{1.0625}$	$\frac{1.0625}{1.0625}$
	6.0070	6.1082	1
	C_6	H_6	O_1
Desired empirical formula		$\mathbf{C_6H_6O}$	

Exercises

1. Establish the empirical formula of cyclohexanol with the following composition:

   71.95% *C*; 12.08% *H*; *and* 15.97% *O*;($C = 12.011$; $H = 1.0079$; $O = 15.99$

2. Determine the empirical formula of mebutamate having the following percentage composition:

   51.70%*C*,8.68%*H*,12.06%*N*, *and* 27.55%*O*. ($C = 12.011$; $H = 1.0079$; $N = 14.0067$).

3. A compound was found to contain 19.3%*C*,2.5%*H*, *and* 11.69%*N*. What is its empirical formula? ($C = 12.011, H = 1.0079, N = 14.0067$).

4. Ascorbic acid (vitamin C) has the following elemental composition by mass 40.91%*C*,4.58%*H*, *and* 54.51% *O*. Determine its simplest (empirical) formula.

Empirical Formula from Chemical Analysis

Examples 1

A 0.10 g sample of an alcohol, known to contain only C, H, and O wa combusted in oxygen to produce CO_2 and H_2O. These products wer collected separately and weighed. 0.1910g of CO_2 and 0.1172g of H_2O

were found. Establish the empirical formula of the alcohol. ($C = 12, H = 1, O = 16$)

Solution

To calculate the mass of C in g.

$CO_2 = 44g/mol$. This means that 44g CO_2 contain 12 g C.

$\therefore 0.1910gCO_2$ will contain

$$\frac{0.191\ g\ CO_2}{44g\ CO_2} \times 12g\ C = [0.0521g\ C]$$

To calculate the mass of H in g.

$H_2O = 18g/mol$, i.e. 18 gH_2O contain 2 g H

$\therefore 0.1172gH_2O$ will contain

$$\frac{0.1772g\ H_2O}{18g\ H_2O} \times 2gH = [0.0130g\ H]$$

To calculate the mass of O in g. The mass of O is obtained by difference.

$$0.10g\ sample - (0.0521g\ C + 0.0130g\ H) = [0.0349\ g\ O]$$

	C	H	O
% composition by mass	0.0521	0.0130	0.0349
Mole ratio	$\frac{0.0521}{12}$	$\frac{0.0130}{1}$	$\frac{0.0349}{16}$
	0.00434	0.0130	0.00218
Simplest ratio	$\frac{0.00434}{0.00218}$	$\frac{0.0130}{0.00218}$	$\frac{0.00218}{0.00218}$
	1.9908	6.1032	1.0000
	C_2	H_6	O_1
Required empirical formula	C_2H_6O		

Example 2

A sample of a gas is found to contain 2.34 g of N and 5.34 g of O. What is the empirical formula of the gas? $(N = 14, O = 16)$

Solution

	N	O
Mass ratio:	2.34g	5.34g
Mole ratio:	$\frac{2.34}{14}$	$\frac{5.34}{16}$
	0.167	0.334
Simplest ratio:	$\frac{0.167}{0.167}$	$\frac{0.334}{0.167}$
	1	2

∴ The required empirical formula is NO_2

Exercise

25 g sample of a compound that contains only C and H was burned in oxygen to give CO_2 and H_2O as products. These products were trapped separately and weighed. It was found that 3.007 g of CO_2 and 1.845 g H_2O were formed in this reaction. What is the empirical formula of the compound?

Molecular formula

The molecular formula of a compound indicates the full number of each atom in the molecule. For example

Compound	Molecular formula	Empirical formula
Hydrogen peroxide	N_2O_2	HO
Hydrazine	H_2N_4	NH_2
Glucose	$C_6H_{12}O_6$	CH_2O
Benzene	C_6H_6	CH

The relationship between molecular formula and empirical formula is

$$\begin{bmatrix} empirical \\ formula \end{bmatrix} n = \begin{matrix} molecular \\ formula \end{matrix}$$

Example

If the molecular formula of fruit sugar is 180.2 and its empirical formula is CH_2O, what is the molecular formula of fruit sugar? (C=12, H=1, O=16)

Solution

$(CH_2O)n = 180.2$

i.e. $30n = 180.2$

$$\therefore\ n = \frac{180.2}{30} = 6.007 \approx 6$$

The required molecular formula is $(CH_2O)_6 = C_6H_{12}O_6$

Exercises

1. Adenine has a molecular mass of 135.1 and consists of 44.45% C, 3.73%H and 52.82% N by mass. Determine its molecular formula
2. An unknown compound on elemental analysis, was found to contain 49.02%C, 2.74%H and 48.24% Cl by mass. Its molecular mass was determined as 147.0. Calculate the molecular formula of the compound.
3. The empirical formula of hydroquinone is C_3H_3O. If its molecular mass is 110. Establish its molecular formula.

4. Ethylene (28.0 g/mol), cyclohexane (84.0 g/mol), and 1- pentene (70.0 g/mol) all have the empirical formula of CH_2. Find the molecular formula of each species.
5. If the empirical formula of a liquid substance is NO_2 and its molecular mass is 92.0. What is its molecular formula?

Stoichiometry and Chemical Equation

Example

How many moles of CO_2 will be produced in the complete combustion of 4.30 moles of C_6H_6? The chemical equation for the reaction is given as:

$$C_6H_6 + O_2 \longrightarrow CO_2 + H_2O$$

Solution

Step 1: First, balance the equation:

$$2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + 6H_2O$$

Step 2:

Impute the necessary number of moles based on stoichiometry:

	$2C_6H_6 + 15O_2 \longrightarrow$	$12CO_2 + 6H_2O$
Stoich:	**2mol**	**12mol**
Given:	**4.30mol**	**?**

Step 3: Calculate the required quantity.

From stoichiometry,

$$2mol\ C_6H_6 \equiv 12mol\ CO_2$$

$$\therefore 4.60mol\ C_6H_6 = \frac{4.30mol\ C_6H_6}{2mol\ C_6H_6} \times 12mol\ CO_2 = 25.8mol\ CO_2$$

Exercises

1. How many moles of H_2O are produced from 0.40 mol of C_6H_6 according to the equation?

$$2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + 6H_2O$$

2. In the hydrolysis of phosphorous tetrachloride, what mass of HCl can be produced from 15.0g of PCl_3? The equation for the reaction is;

$$PCl_3 + 3H_2O \longrightarrow 3HCl + H_3PO_3$$

3. What is the mass of Al required to produce 4.0 g of hydrogen according to the equation?

$$2Al + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2$$

4. How many grams of CO_2 is formed when 73.9 g of propane is burnt

$$C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$$

5. How many grams of oxygen can be obtained from 2.50 g of potassium chlorate?

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

Stoichiometry, Limiting Reagent and Percent Yield

When reactants react to yield products, they do so in discrete stoichiometric quantities. For examples, let us assume that one mole of reactant A requires one mole of reactant B for complete reaction. Let us also assume that the reaction mixture contains four moles of A and five moles of B. What will happen is that the four moles of A will react with four moles of B and one mole of B will be left unreacted. It is also important to note that the quantity of the product will depend on A, because when all of A have reacted, the reaction will stop, even though an excess of B remains in the reaction mixture.

Here we term reactant A the limiting reagent because it limits the amount of the product. Reactant B is termed excess reagent.

Example 1

Identify the limiting and excess reagent if 12.0 moles of water are mixed with 5.0 moles of barium metal. The stoichiometric equation is given as:

$$Ba_{(s)} + 2H_2O_{(l)} \longrightarrow Ba(OH)_{2(aq)} + H_2\uparrow$$

Solution

$$Ba_{(s)} + 2H_2O_{(l)} \longrightarrow Ba(OH)_{2(aq)} + H_2\uparrow$$

Stoich:	1 mol	2 mol
Given:	5 mol	12 mol

To identify the limiting reagent.

From the stoichiometry,

1 mol Ba reacts with 2 mol H_2O

$\therefore$ 5 mol Ba will react with

$$\frac{5\ mol\ Ba}{1\ mol\ Ba} \times 2\ mol\ H_2O = 10\ mol\ H_2O$$

The above calculations show that the 5.0 mols of Ba will react exactly with 10 mols of H_2O, leaving an excess of 2 mols of water unreacted. Therefore, limiting reagent is Ba; excess reagent is H_2O.

Examples 2

Suppose 25.0 g of nitrogen gas and 5.0 g of hydrogen are mixed to form ammonia. Calculate the mass (g) of ammonia produced at the end of the reaction ($N = 14, H = 1$). The stoichiometry is

$$N_2(g) + H_2(g) \longrightarrow NH_3(g)\ \text{(unbalanced)}$$

Solution

Step 1: Determine the limiting reagent

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)\ \text{(balanced)}$$

Stoich:	28 mol	6 mol
Given:	25 mol	5 mol

From stoichiometry,

28g N_2 react with 6g H_2

$$25g\ N_2 \text{ will react with } \frac{25gN_2}{28gN_2} \times 6g\ H_2 = 5.357g\ H_2$$

$$\frac{25gN_2}{28gN_2} \times 6g\ H_2 = 5.357g\ H_2$$

From this calculation, the 25 g of N_2 will need 5.357 g of H_2 for complete reaction. Since we are given only 5 g of H_2 (which is less than the 5.357 g of H_2 needed), it is clear that H_2 is not enough and so becomes the limiting reagent.

The limiting reagent= H_2.

Step 2: To calculate the mass (g) of the product (ammonia).

Since the quantity of the product will depend on the quantity of the limiting reagent, we base the calculation on the limiting reagent:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

Stoich:	**6 g**	**34 g**
Given:	**5 g**	**?**

From the stoichiometry,

6g of H_2 yield 34g of NH_3

$$\therefore 5g\ H_2 \text{ will yield } \frac{5g\ H_2}{6g\ H_2} \times 34g\ NH_3 = 28.33g\ NH_3$$

Percent Yield

When we calculate the yield (product) of a reaction based on the balanced stoichiometric equation, this is termed the theoretical yield. But in practical terms, the real yield will always be less than the theoretical yield as a result of side reactions, impurities, losses during reaction, weighing

errors, etc. This real yield is termed the actual yield. The percent yield is given as:

$$\%\,Yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100\%$$

Example

Calculate the percent yield of magnesium nitride that can be made in the reaction of 40.00 g of magnesium and 17.00 g of nitrogen. Also calculate the amount (g) of the excess reagent. The actual yield is 59.01 mg ($Mg = 24.3, N = 14$).

The stoichiometry is

$$3Mg_{(s)} + N_{2(g)} \longrightarrow Mg_3N_{2(s)}$$

Solution

Step 1: Determine the limiting reagent

	$3Mg_{(s)} + N_{2(g)} \longrightarrow Mg_3N_{2(s)}$	
Stoich:	**27.9 g**	**28 g**
Given:	**40 g**	**17 g**

From stoichiometry,

$27.9g\ Mg \equiv 28g\ N_2$

$$\therefore 40mg \equiv \frac{40g\ Mg}{27.9g\ Mg} \times 28g\ N_2 \equiv 40.143\ N_2$$

Therefore N_2 is the limiting reagent.

Step 2: To calculate the % Yield

	$3Mg_{(s)} + N_{2(g)} \longrightarrow$	$Mg_3N_{2(s)}$
Stoich:	**28 g**	**100.9g**
Given:	**17 g**	**?**

From stoichiometry,

28g N_2 yielded 100.9 g Mg_3N_2,

$\therefore 17g N_2$ will yield

$$\frac{17gN_2}{28gN_2} \times 100.9gMg_3N_2 = 61.26g\ Mg_3N_2$$

Therefore: $theorectical\ yield = 61.26g\ Mg_3N_2$

But actual yield $= 59.01g\ Mg_3N_2$

$$\therefore \%\ yield = \left(\frac{59.01g}{61.26g}\right) \times 100\% = 96.32\%$$

Exercises

1. Which is the more effective antacid 0.50g of $NaHCO_3$ or 0.50g of $Mg(OH)_2$? (Na=23, H=1, C=12, O=16, Mg=24.3, Cl=35.5)

   Stoichiometry:

   $$Na_2HCO_{3(s)} + 2HCl_{(aq)} \longrightarrow NaCl(aq) + H_2O_{(l)} + CO_{2(g)}$$

   $$Mg(OH)_{2(s)} + 2HCl_{(aq)} \longrightarrow 2H_2O_{(l)} + MgCl_{2(aq)}$$

2. If 12.0 g HCl is mixed with 14.5 g FeS, how many grams of H_2S can be produced? (H=1, Cl=35.5, Fe = 56, S=32)

   Stoichiometry:

   $$FeS_{(s)} + 2HCl_{(aq)} \longrightarrow FeCl_{(aq)} + H_2S_{(aq)}$$

3. What mass of acetic acid is needed to produce 150 g ethyl acetate if the percent yield is 80%. Assume that ethanol is present in excess. (C=12, H=1, O=16).

   Stoichiometry:

   $$\underset{\text{Acetic acid}}{CH_3COOH_{(aq)}} + \underset{\text{ethanol}}{C_2H_5OH} \longrightarrow \underset{\text{ethyl acetate}}{CH_3COOCH_2CH_3} + H_2O$$

4. Calculate the theoretical yield of isopentyl acetate we can possibly get from 15.0 g isopentyl alcohol and 26.9g acetic acid $(C = 12, H = 1, O = 16)$. Stoichiometry:

   $$\underset{\text{Acetic acid}}{CH_3COOH} + \underset{\text{Isopenty alcohol}}{HOCH_2CH_2CH(CH_3)_2} \longrightarrow \underset{\text{Isopentyl acetate}}{CH_3COOCH_2CH_2CH(CH_3)_2} + H_2O$$

ii. When the % yield of the reaction is 80%, calculate the actual yield of isopentyl alcohol.

5. The explosive trinitrotoluene (TNT) is made from toluene, (C_7H_8) and nitric acid (HNO_3), based on the stoichiometry: $C_7H_8 + 3HNO_3 \rightarrow C_7H_5(NO_2)_3 + 3H_2O$. When 50.0 g of C_7H_8 and 100.0 g of HNO_3 are reacted together, 105.0 g of TNT were produced. Calculate, a. the theoretical yield, and b. the percent yield. $(C = 12, H = 1, N = 14, O = 16)$

Stoichiometry in Solution Chemistry

Most chemical reactions take place in solution. Here, the solute (analyte), solvent and solution are emphasized.

$$Solution = Solute + Solvent$$

Concentration of a solute in a solution may be expressed as molarity (M), molality (m), % (w/v), %(v/v), mole fraction (x), normality (N). When the analyte is a solid dispersed in a solid matrix its concentration is expressed in $(\%w/w = \%m/m)$.

Molarity (M)

The molarity, symbol M, is defined as the number of moles (mols) of solute per litre (or dm^3) of the solution (unit = mol/L)

$$Molarity\ (M) = \frac{number\ of\ moles\ of\ solute}{litre\ solution}$$

Example 1

Calculate the molarity (M) of a solution prepared by dissolving 13.5 g of pure solid NaOH in enough water to make 2500 mL of solution. $(Na = 23, O = 16, H = 1)$.

Solution

$$Amount\ of\ NaOH = 13.5g = \frac{13.5}{40} moles$$

Volume of final solution = 2500 mL = 2.5L

Using the relationship

$$Molarity\ (M) = \frac{Moles\ of\ solute}{L\ of\ solution}$$

$$= \frac{\left(\frac{13.5}{40}\right)}{2.5L} moles = \frac{0.3375\ moles}{2.5L}$$

$$= \frac{0.135mol}{L} = 0.135M$$

Example 2

Give the concentration of all the ions in each of the following aqueous solutions: (a)$0.50MK_2SO_4$ (b) $2.0MHCl$

Solution

a. K_2SO_4 is a very soluble salt, such that

$$\underset{(0.5M)}{K_2SO_4} \longrightarrow \underset{2(0.5M)}{2K^+} + \underset{0.5M}{SO_4}$$

Therefore, the solution will contain

$2 \times 0.5M\ K^+ = 1.0M\ K^+ and\ 0.5M\ SO_4^{2-}\ ion$

b. HCl is completely soluble in water, such that

$$\underset{2.0M}{HCl} \longrightarrow \underset{2.0M}{H^+} + \underset{2.0M}{Cl^-}$$

Therefore, 2.0 M HCl aqueous solution will contain 2.0 M H^+ and 2.0M Cl^-.

Example 3

Solution

$$Amount\ of\ NaOH = 13.5g = \frac{13.5}{40} moles$$

Volume of final solution = 2500 mL = 2.5L

Using the relationship

$$Molarity\ (M) = \frac{Moles\ of\ solute}{L\ of\ solution}$$

$$= \frac{\left(\frac{13.5}{40}\right)}{2.5L} moles = \frac{0.3375\ moles}{2.5L}$$

$$= \frac{0.135mol}{L} = 0.135M$$

Example 2

Give the concentration of all the ions in each of the following aqueous solutions: (a)$0.50MK_2SO_4$ (b) $2.0MHCl$

Solution

a. K_2SO_4 is a very soluble salt, such that

$$\underset{(0.5M)}{K_2SO_4} \longrightarrow \underset{2(0.5M)}{2K^+} + \underset{0.5M}{SO_4}$$

Therefore, the solution will contain

$2 \times 0.5M\ K^+ = 1.0M\ K^+ and\ 0.5M\ SO_4^{2-}\ ion$

b. HCl is completely soluble in water, such that

$$\underset{2.0M}{HCl} \longrightarrow \underset{2.0M}{H^+} + \underset{2.0M}{Cl^-}$$

Therefore, 2.0 M HCl aqueous solution will contain 2.0 M H^+ and 2.0M Cl^-.

Example 3

How many moles of Na^+ ions are there in 15 cm^3 of a 0.8 M NaCl aqueous solution? Assume that NaCl is completely soluble in the solution.

Solution

0.8 M NaCl means that 1.0 L (1000 cm^3) of the solution contains 0.8 moles of NaCl. Therefore, 15 cm^3 of this solution will contain

$$\frac{15cm^3}{1000cm^3} \times 0.8\,moles = 0.012\,mol\,NaCl$$

Since NaCl is very soluble in water, we will have:

$$\underset{0.012\ mol}{NaCl} \longrightarrow \underset{0.012\ mol}{Na^+} + \underset{0.012\ mol}{Cl^-}$$

Therefore, 15cm^3 of a 0.8 M NaCl aqueous solution will contain 0.012 moles of Na^+.

Example 4

What mass of HNO_3 (63 g/mol) is contained in 300 mL of 0.30 M HNO_3 solution?

Solution

0.30 M HNO_3 means that 1000 mL of this solution contain 0.30 moles of HNO_3.

Therefore, 300 mL will contain

$$\frac{300\,mL\,soln}{1000\,mL\,soln} \times 0.30\,mol\,HNO_3$$

$$= 0.09\,mol\,HNO_3$$

$$= (0.09 \times 63)g\,HNO_3$$

$$= 5.67g\,HNO_3$$

Percent Concentration

Example 1

Calculate the percent concentration of sodium chloride (NaCl) in a solution made up of 1.5 g NaCl and 45 g H_2O

Solution

Total mass of solution $= 1.5gNaCl + 45g \rightleftarrows H_2O = 46.5g$solution

$$\therefore \%(m/m)NaCl = \frac{mass of solute(NaCl)}{mass\ of solution(NaCl + H_2O)} \times 100\%$$

$$= \frac{1.5g}{46.5g} \times 100\% = 3.23\%\ (m/m)$$

Example 2

What is the mass percent of ethanol in a solution composed of 110.0 g of water and 2.0 g of ethanol?

Solution

Total mass of solution = 110.0 g water + 2.0 g ethanol = 112.0 g solution

$\therefore$ % (m/m) of ethanol

$$= \frac{mass\ of\ ethanol}{mass\ of\ solution} \times 100\% = \frac{2.0g}{112.0g} \times 100\% = 1.786\%\ (m/m)$$

Molality (m)

The molality, symbol m, of a solution is defined as the number of moles of solute per kg solvent:

$$Molality\ (m) = \frac{number\ of\ \ moles\ of\ solute}{kg\ of\ solvent}$$

Example 1

What is the molality of a solution prepared by dissolving 25.0 g of I_2 (254 g/mol) in 200 g of CCl_4?

Solution

$$\text{Moles of solute } (I_2) = \left(\frac{25}{254}\right) \text{mol } I_2 = 0.098 \text{ mol } I_2$$

$$\text{Kg of solvent } (CCl_4) = \left(\frac{200}{1000}\right) \text{kg } CCl_4 = 0.2 \text{Kg } CCl_4$$

$$Molality\ (m) = \frac{mols\ of\ solute}{kg\ solvent}$$

$$= \frac{0.098\ mol}{0.2\ kg} = 0.49 \frac{mol}{kg} = 0.49M$$

Example 2

What is the molality of 1.65 M H_2SO_4 (98g/mol)? Solution that has a density of 1.29 g/ mol.

Solution

Step1: Calculate the mol of H_2SO_4 and wt (kg) of H_2O.

Density = 1.29 g/mL means that

1 mL of this solution weights 1.29 g

$\therefore$ 1000 mL of the solution will weigh

$$\frac{1000\ mL}{1\ mL} \times 1.29g = 1{,}290g$$

But 1.65 M H_2SO_4 means that 1000 mL contain 1.65 mol H_2SO_4

$$= (1.65 \times 98)g\ H_2SO_4 = 161.7g\ H_2SO_4$$

The weight of H_2SO_4 = 161.7 g = 1.65 mol H_2SO_4

The weight of water = 1290 − 161.7 = 1128.3 g = 1.1283 kg

$$\therefore Molality\ (m) = \frac{mol\ solute}{kg\ solvent}$$

$$= \frac{1.65\ mol}{1.1283\ kg\ solvent} = 1.46\frac{mol}{kg} = 1.46\ M$$

Mole Fraction

Assume that a solution is composed of two components, A and B. Then the mole fraction of A (x_A) is given as: $x_A = \frac{n_A}{n_A + n_B}$

Where $n_A = moles of A$; $n_B = moles of B$.

Example

What is the mole fraction of glucose ($C_6H_{12}O_6$ = 180 g/mol) in a 30% (m/m) aqueous solution of glucose? (H_2O = 18 g/mol).

Solution

30% (m/m) means that 100 g solution contain 30 g glucose

$$\therefore Moles\ of\ glucose = \frac{30}{180} = 0.167\ mol$$

$$Moles\ of\ water = \frac{(100 - 30)}{18} = 3.89\ mol$$

$$\therefore Mole\ fraction\ of\ glucose\ (X_{glucose}) = \frac{0.167}{0.167 + 3.89} = 0.04$$

Standard Solution and Stock Solution

A standard solution is a solution whose concentration is exactly known. A stock solution is a highly concentrated standard solution from which dilute solutions can be made. To obtain a dilute solution from a stock solution, the following relationship is used.

$$M_1V_1 = M_2V_2$$

where M_1 = concentration of the stock solution, V_1 = volume of the stock solution required for dilution, M_2 = final concentration of the dilute solution, V_2 = final volume of the dilute solution.

Example

Describe how you would prepare 500 mL of 2.0 M acetic acid (CH_3COOH) from a 16.0 M stock solution of acetic acid.

Solution

Data: M_1 = 16.0 M, V_1=? M_2 = 2.0 M, V_2 = 500 mL

Using $M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{2.0\ M \times 500\ mL}{16.0} = 62.5\ mL$$

Pipette exactly 62.5 mL of the stock solution into a 500 mL volumetric flask and make up to the mark with distilled water. Then shake thoroughly to mix.

Exercises

1. What volume of 5.0 M HCl must be taken to prepare 0.70 L of 0.10 M HCl?
2. What mass of CO and H_2, respectively are required to form 5.0 kg of CH_3OH according to the stoichiometry

   $$CO_{(g)} + 2H_{2(g)} \longrightarrow CH_3OH_{(g)}$$

3. What mass of water must be added to 425 g of formaldehyde to prepare a 50% (w/w) solution of formaldehyde.
4. Oxygen can be generated in the laboratory by thermal decomposition of potassium chlorate ($KClO_3$):

$$KClO_{3(s)} \longrightarrow 2KCl_{(s)} + 3O_2\uparrow$$

a) How many grams of $KClO_3$ must react to produce 4.00 moles of O_2? b) How many grams of O_2 can be prepared by the complete decomposition of 9.25 g of $KClO_3$? c) Calculate the number of moles of $KClO_3$ that must be decomposed to prepare 12.0 g of O_2. d) How many grams of KCl are produced during a decomposition that also produced 11.0 g of O_2?

5. What is the mass of Al in 1.5 kg of each of the following compounds? (a) $AlCl_3$ (b) Al_2O_3 (c) $KAl(SO_4)_2.12H_2O$
6. Calculate the percentage composition of the stared species in the following compounds (a) LiBr* (b) *CH_4 (c) *K_2SO_4 (d) *$CuSO_4$
7. If 12.3 g of CCl_4 are made from the reaction of 20.0 g of CS_2, with 24.0 g of Cl_2, what is the percent yield of CCL?
8. Iron metal can be produced by the action of CO on Fe_2O_3 according to the stoichiometry:

$$Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow 2Fe_{(s)} + 3CO_{2(g)}$$

a) What is the maximum amount of iron which could be produced by the reaction of 130.0 g of Fe_2O_3 and 130.0 g of CO?

b) Identify the excess reagent and how much of it would be left over after producing the maximum amount of Fe?

9. A 26.0% (m/m) aqueous solution of KI has a density of 1.30 g/mL.

a) How many grams of KL are in 70.0 mL of the solution?

b) How many grams of solution contain 25.0 g of KI?

c) How many millilitres of the solution contain 20.0 g of KI?

10. Calculate the % (m/m) composition of nitric acid in 3.0 molal (m) HNO_3 solution.

11. How many moles of solute are contained in i) 13.05mL of 3.00 M $CaCl_2$, ii) 1,200 mL of 0.25 M $(NH_4)_2SO_4$, iii) 180 mL of 0.21 M HCl.
12. A concentrated HCl contains 37.0% (m/m) HCl and has a density of 1.19 g/mL. Calculate the concentration of this solution in a) Molarity (M), b) Molality (m), c) Mole fraction (X).

Reference

1. Ndiokwere CL, Ukhun ME, 2006. First year University inorganic and physical chemistry. Mindex Publishing, Benin City.
2. Hill JW, Petrucci RH, McCreary TW, Perry SS, 2005. General chemistry, 4th Edn. Pearson Prentice Hall, New Jersey.
3. Chang R, 1991. Chemistry, 4th edn. McGraw-Hill Inc. New Jersey.

ACIDS, BASES AND SALTS

P. O. Okolo & O. S. Lawal

Definitions of Acids and Bases

Arrhenius Definition of Acid

Arrhenius (1884) defined an acid as any chemical species which when dissolved in water undergoes dissociation to yield hydrogen ions (H^+) as the only positive ions. Examples are shown below:

$$HCl \longrightarrow H^+ + Cl^-$$
$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$
$$HNO_3 \longrightarrow H^+ + NO_3^-$$
$$CH_3COOH \longrightarrow H^+ + CH_3COO^-$$

Another name for hydrogen ion (H^+) is proton.

Since proton does not exist independently in aqueous solution, we have, for example:

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

Monoprotic acids yield one proton per molecule:

$$HCl \longrightarrow H^+ + Cl^-$$
$$HNO_3 \longrightarrow H^+ + NO_3^-$$

Polyprotic acids yield more than one proton per molecule:

$$H_2CO_3 \longrightarrow 2H^+ + CO_3^{2-}$$
$$H_3PO_4 \longrightarrow 3H^+ + PO_4^{3-}$$

Arrhenius Definition of Base

Arrhenius defined base as any chemical species which when dissolved in water dissociates to yield hydroxide ion (OH^-) as the only negative ion.

Examples include:

$$NaOH \longrightarrow K^+ + OH^-$$

$$NH_3 + H_2O \longrightarrow NH_4^+ + OH^-$$

$$N_2H_4 + H_2O \longrightarrow N_2H_5^+ + OH^-$$

Bronsted-Lowry Definition of Acids and Bases

Bronsted and Lowry defined an acid as any substance which donates proton and a base as any substance which accepts a proton. For example

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

Conjugate Acid/ Base System.

Consider the acid/base reaction;

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

The HCl is a Bronsted-Lowry acid because it can donate a proton to H_2O. The H_2O is a base because it can accept the proton from HCl.

Now, examine the reverse process of this reaction: the H_3O^+ is a potential Bronsted-Lowry acid because it can donate a proton to Cl^- to reform the original acid, while the Cl^- is a potential base because it can accept the proton from H_3O^+. The H_3O^+ is therefore termed a conjugate acid, while the Cl^- is the conjugate base. Other conjugate acid/ base systems include:

$$HCl + NH_3 \rightleftharpoons NH_4^+ + Cl^-$$

$$HNO_3 + F^- \rightleftharpoons HF + NO_3^-$$

$$NH_4^+ + S_2^- \rightleftharpoons HS^- + NH_3$$

Acid Base Conj. acid Conj. Base

Autoionization or Self-ionization

When a chemical species act as both Bronsted-Lowry acid and base, it is termed autoionization or self-ionization. Examples are:

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$$

$$NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

$$\underset{\text{Acid}}{CH_3COOH} + \underset{\text{Base}}{CH_3COOH} \rightleftharpoons \underset{\text{Conj. acid}}{CH_3COOH_2^+} + \underset{\text{Conj. Base}}{CH_3COO}$$

In general, an auto-ionization reaction involves the formation of a cation-anion pair from two neutral molecules of the same substance.

Lewis Definition of Acid and Base

Lewis defined an acid as any chemical species that can accept a pair of electrons, and a base as any species that can donate a pair of electrons to form a covalent bond. Examples include:

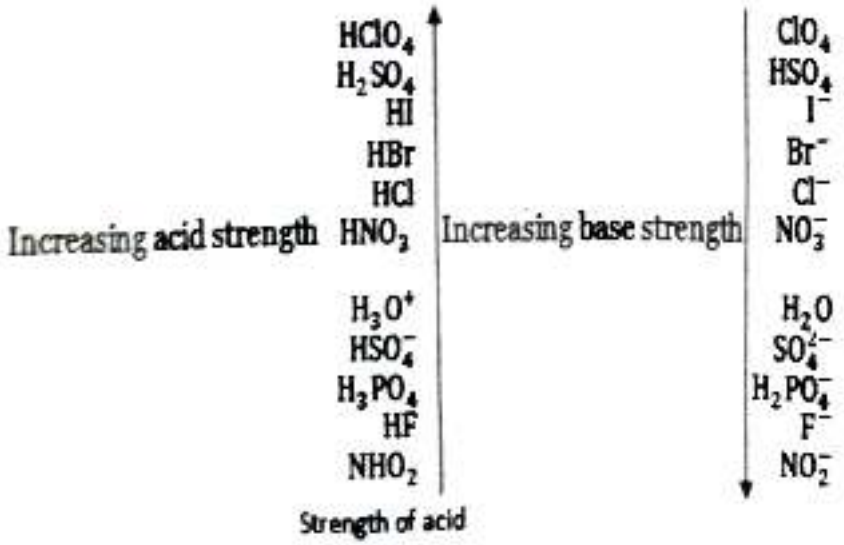
$$H^+ + OH^- \longrightarrow H_2O$$

$$\underset{\text{Lewis acid}}{BCl_3} + \underset{\text{Lewis base}}{NH_3} \longrightarrow Cl_3BNH_3$$

Lewis concept has a more universality than both Arrhenius and Bronsted because electrons are involved in all chemical reactions.

Strength of Acids and Bases

As acids become stronger, their conjugate bases become weaker. Based on this, acids have been arranged in order of their relative strength as depicted below:



Levelling and Differentiating Solvents

It is not possible to compare the strengths of all acids when the reference base is a strong one. For example, HCl, NHO_3 and $HClO_4$ all appear to be 100% ionized (dissociated) in water. With water as the reference base, each of these acids appears to be of equal strength. Their differences are removed or levelled out and this is termed levelling effect.

If, on the other hand, pure ethanoic acid (glacial acetic acid) is used as a solvent instead of water (ethanoic acid is a much weaker base than water) there is an appreciable difference between the extent of dissociation of the following:

$$HCl + HC_2H_3O_2 \longrightarrow H_2C_2H_3O_2^+ + Cl^-$$

$$HNO_3 + HC_2H_3O_2 \longrightarrow H_2C_2H_3O_2^+ + NO_3^-$$

$$HClO_4 + HC_2H_3O_2 \longrightarrow H_2C_2H_3O_2^+ + ClO_4^-$$

Here the ethanoic acid is a differentiating solvent.

Salts

A salt is the compound formed when any of the ionizable proton in an acid has been replaced by corresponding amount of metallic ions:

$$Na^+ + HCl \longrightarrow NaCl + H^+$$

$$Ba^{2+} + H_2SO_4 \longrightarrow BaSO_4 + 2H^+$$

Five types of salt are immediately identifiable, namely: normal salts, acid salts, basic salts, double salts, and complex salts.

Normal Salts

These are formed when all the replaceable hydrogen ions in a Bronsted-Lowry acid have been completely replaced by metallic ions.

$$2HCl_{(aq)} + Na_2CO_{3(aq)} \longrightarrow 2NaCl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$$

$$H_2SO_{4(aq)} + 2NaOH_{(aq)} \longrightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

Normal salts are neutral to litmus, though some normal salts such as sodium trioxocarbonate (iv) and aluminium chloride will undergo hydrolysis in water to give an acidic or alkaline solution.

Acid Salts

This is formed by partial replacement of ionizable protons in an acid by a metallic species. Thus, acids with two replaceable hydrogen ions will form only one acid salt:

$$H_2SO_4 + Na^+ \longrightarrow \underset{\text{Sodium sulphate}}{NaHSO_4} + \underset{\text{Hydrogen}}{H^+}$$

Acids with three replaceable hydrogen ions can form two different acid salts:

$$H_3PO_4 + K^+ \longrightarrow KH_2PO_4 + H^+$$

$$H_3PO_4 + 2K^+ \longrightarrow K_2HPO_4 + 2H^+$$

Acid salts are formed when there is an insufficient supply of metallic ions to saturate the place of the replaceable hydrogen ion in the acid.

$$H_2SO_4 + KOH \longrightarrow KHSO_4 + H_2O$$

The proton (H^+) in an acid salt can still be replaced by a metal ion, thus possessing the properties of a salt and that of an acid:

$NaHSO_4 \longrightarrow Na^+ + HSO_4^{2-}$;Then $HSO_4^- \longrightarrow H^+ + SO_4^{2-}$

Basic Salts

When a base reacts with an insufficient amount of an acid, a basic salt is formed. They can be viewed as normal salts in which part of the protons are replaced by oxygen or hydroxyl group:

$$\underset{}{SbCl_3} + \underset{\text{Excess}}{H_2O} \longrightarrow \underset{\text{Basic salt}}{SbOCl} + H_2$$

Basic salts have the properties of a base - they turn red litmus to blue and react with excess acid to from a normal salt and water.

Double Salts

Double salts are capable of ionizing into three different ionic species in aqueous solutions. They are formed by the combination of two simple salts in stoichiometric ratio but give reaction(s) characteristic of the constituent simple salts. Example of double salt is ammonium iron (ii) tetraoxosulphate(vi)hexahydrate, $(NH_4)SO_4.FeSO_4.6H_2O$.

Usually, two of the three types of ions in solution are cations while the other is an anion. In the above case, the ions are NH_4^+, Fe^{2+} and SO_4^{2-}.

Alums

Alums are double salts with the general formula $m^+m^{3+}(SO_4)_2.12H_2O$. Aluminum ion(Al^{3+}), and Fe^{3+}can form double salts. An example of alum is the potassium alum $KAl(SO_4)_2.12H_2O$, which may be applied to treat cotton fibres to render it more effective adsorbent for dyes.

Complex Salts

A complex salt contains a complex ion. A complex ion contains ligands such as $NH_4^+, NO_3^-, SO_4^{2-}, CN^-$. An example of a complex salt

is $K_4[Fe(CN)_6]$. When dissociated in water this forms a metal ion, k^+ and a stable complex ion, $[Fe(CN)_6]^{4-}$.

$$K_4[Fe(CN)_6] \longrightarrow 4K^+ \ [Fe(CN)_6]^{4-}$$

Hexacyanoferrate(ii) ion

Exercises

1. a. Define with suitable examples what is meant by an acid and a base according to
   i. Arrhenius concept,
   ii. Bronsted-Lowry concept, and
   iii. Lewis concept
   b. Explain how the reaction between Na_2O and SO_3 is a neutralization reaction using Lewis definition of acid and base.
2. a. Explain if ammonia can act as an acid (proton donor) in water
   b. The conjugate base of NH_3 is NH_2^-. Compare and explain the basic strengths of NH_2^- and OH^-.
3. What are the conjugate bases of (i) Na_2S (ii) HCN (iii) OH^-
4. Write the conjugate acids of (i) C_5H_5N (ii) CO_3^{2-} (iii) Br^-
5. Identify the acid-base conjugate pairs in the following:

$$N_2H_4 + NH_4^+ \rightleftharpoons N_2H_5^+ + NH_3$$
$$CH_3OH + H^+ \rightleftharpoons H_2 + CH_3O^-$$
$$C_2H_3O_2^- + NH_4^+ \rightleftharpoons HC_2H_3O_2 + NH_3$$

6. Identify the Lewis acids and bases in the following:

$$Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+$$
$$FeCl_3 + Cl^- \rightleftharpoons FeCl_4^-$$
$$(CH_3)_2O + BF_3 \rightleftharpoons (CH_3)_2OBF_3$$

7. What is the advantage of Lewis concept of acid and base over Bronsted-Lowry concept?

Calculations Involving Acids and Bases

pH Concept

The pH of a solution is the minus logarithm of the proton (H^+) ion concentration:

$pH = -log[H^+]$,
where $[H^+]$ = molar concentration of H^+ ion.

Similarly, $pOH = -log[OH^-]$

Example 1

Calculate

a) The pH of a solution 0.030 M in H^+ ion.

b) The pH of a solution $2.0 \times 10^{-2} M$ in OH^- ion.

c) If the pH of an aqueous system is 8.5, calculate the $[H^+]$ in this system.

Solution

a. Since $[H^+] = 0.03M$

$$pH = -log[H^+] = -log(0.03) = 1.5$$

b. Using the relation

$$pH + pOH = 14$$

But $pOH = -log[OH^-] = -log(2.0 \times 10^{-2}) = 1.7$

$$\therefore pH = 14 - pOH = 14 - 1.7 = 12.3$$

c. Since pH = 8.5

$$\therefore pH = -log[H^+] = 8.5$$

$$\therefore [H^+] = log^{-1}(-8.5) = 3.16 \times 10^{-9}$$

pH of Buffer Solutions

A buffer solution is a solution of a weak acid (or weak base) and its salt. Both components must be present. The buffer solution has the ability to resist changes in pH upon the addition of a small amount of either acid or base. In other words, a buffer is an acid-base conjugate pair, that is, a weak acid and its conjugate base or a weak base and its conjugate acid.

One of the simplest buffers is the acetic acid-sodium acetate system. A solution containing these two substances has the ability to neutralize either the added acid or added base as follows:

If a base is added to the buffer system, the OH^- ions will be neutralized by the acid in the buffer:

$$CH_3COOH_{(aq)} + OH^-_{(aq)} \longrightarrow CH_3COO^-_{(aq)} + H_2O_{(l)}$$

If an acid is added, the H^+ ions will be consumed by the conjugate base in the buffer:

$$CH_3COO^-_{(aq)} + H^+_{(aq)} \longrightarrow CH_3COOH_{(aq)}$$

Here, the acetate ions are provided by the complete dissociation of the sodium acetate.

$$CH_3COONa_{(s)} \longrightarrow CH_3COO^-_{(aq)} + Na^+_{(aq)}$$

In general, a buffer system can be represented as salt/acid or conjugate base/acid. Thus, the sodium acetate-acetic acid buffer system can be written as

$$CH_3COONa/CH_3COOH \text{ or } CH_3COO^-/CH_3COOH$$

pH of Acid Buffer

The pH of an acid buffer is calculated using the formula, called Henderson equation:

$$pH\ of\ acid = pKa + log\left[\frac{salt}{acid}\right]$$

Example

1. Calculate the pH of a buffer system containing 1.0 M CH_3COOH and 1.0 M CH_3COONa.
2. What is the pH of this buffer system after the addition of 0.10 moles of gaseous HCl to IL of the solution? Assume that the volume of the

solution does not change when the HCl is added. (K_a for acetic acid $= 1.8 \times 10^{-5}$).

Solution

1. $[CH_3COOH] = 1.0M$
   $[CH_3COO^-] = 1.0M$
   Using the Henderson equation, $pH = pK_a + log\frac{[salt]}{[CH_3COOH]}$

$$= -\log 1.8 \times 10^{-5} + \log\left(\frac{1.0}{1.0}\right) = 4.74$$

2. After addition of HCl, complete dissociation of the HCl occurs because it is a strong acid.

$$\underset{0.1\ mol}{HCl_{(aq)}} \longrightarrow \underset{0.1\ mol}{H^+_{(aq)}} + \underset{0.1\ mol}{Cl^-_{(aq)}}$$

But the CH_3COO^- already in solution will consume the H^+, which came from HCl:

$$\underset{0.1\ mol}{H^+} + \underset{0.1\ mol}{CH_3COO^-} \longrightarrow \underset{0.1\ mol}{CH_3COOH}$$

Therefore, at equilibrium,

$[CH_3COOH] = [1.0 + 0.1]M = 1.1M$

$[CH_3COO^-] = [1.0 - 0.1] = 0.9M$

Using the Henderson equation,

$$pH = -\log 1.08 \times 10^{-5} + \log\frac{0.9}{1.1} = 4.66$$

Note that the pH of the buffer did not change significantly after the addition of the 0.1 mol HCl.

Buffer Range

This is the limit to which a buffer can resist pH change, and there is also a range over which the buffer is most effective. This range is often called the buffer range, and this is defined as:

$$pH\ range = pKa \pm 1.00$$

Since the pKa for acetic acid is 4.74, the buffer range for the CH_3COO^-/CH_3COOH system is given by pH 3.74 to 5.74.

Example

Calculate the buffer range for the following:

HCO_3^-/H_2CO_3; $pKa = 6.38$

Solution

$$\text{pH range} = \text{pKa} \pm 1.00 = 6.38 \pm 1.00 = 5.38 \text{ to } 7.38$$

Exercises

1. A buffer solution was prepared by adding 8.3 g acetic acid and 16.0 g of sodium acetate together. Then 0.07 mL of 1.0 M HCl was added. Calculate the pH of the buffer solution (K_a for acetic acid = $1.8x10^{-5}$).
2. How many moles of sodium acetate must be added to 1.0 litre of 0.2 M CH_3COOH solution to make a buffer of pH 6? (Ka for $CH_3COOH = 1.8 \times 10^{-5}$).
3. What is the buffer range of the following?
   CO_3^{2-}/HCO_3^-; pKa = 10.32
4. Describe how you would prepare a "phosphate buffer" at a pH of about 7.40.

Salt Hydrolysis and Solubility Products

Salts of weak acids and weak bases will dissociate completely in water. When a weak acid or a weak base is dissolved in water, the amount dissociated (say x) is always small such that:

$[weak\ acid] - x \approx [weak\ acid]$ *and* $[weak\ base] - x \approx [weak\ base]$

The above is true if the dissociation constraints for the weak acid (k_a) and weak base (K_b) is less than or equals to $1.0x10^{-4}$, this is called fair

approximation and may be used to simplify some calculations involving hydrolysis.

The extent of dissociation of a weak acid (or weak base) in water is further discouraged if salt of the weak acid (or weak base) is introduced into the solution; this is due to common ion effect.
Hydrolysis is any reaction involving water molecule (s) as a reactant.

A salt of strong base-weak acid will undergo anionic hydrolysis in water, e.g.

$$\underset{\text{salt of strong}}{CH_3COONa} \longrightarrow \underset{\text{anion}}{CH_3COO^-} + \underset{\text{cation}}{Na^+}$$

base-weak acid

$$\text{Then } CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$$

Hydrolysis constant, $K_h = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]} = \frac{K_w}{K_a}$

Where, K_w = hydrolysis constant for water = 1.0×10^{-14}

Salt of weak base-strong acid will undergo cationic hydrolysis in water, e.g.

$$\underset{\text{Salt of weak}}{NH_4Cl} \longrightarrow \underset{\text{cation}}{NH_4^+} + \underset{\text{anion}}{Cl^-}$$

base-strong acid

$$\text{Then } NH_4^+ + H_2O \longrightarrow NH_3 + H_3O^+ \ (H_3O^+ = H^+)$$

$$K_h = \frac{[NH_3][H_3O^+]}{[NH_4^+]} = \frac{K_W}{K_b}$$

Also $K_W = K_aK_b$ – this is the relationship between the dissociation constant of a base and that of its conjugate acid.

When a strong acid dissociates in water, it does so completely, e.g.

$$\underset{0.3M}{HCl} \longrightarrow \underset{0.3M}{H^+} + \underset{0.3M}{Cl^-}$$

Similarly, for a strong base;

$$\underset{0.1\ M}{Ca(OH)_2} \longrightarrow \underset{0.1\ M}{Ca^{2+}} + \underset{2(0.1M)}{2OH^-}$$

Hydrolysis is not possible in the last two cases above.

Solubility Products (K_{sp})

The solubility product of a compound is the product of the molar concentrations of the constituent ions, each raised to the power of its stoichiometric coefficient in the equilibrium equations. The value of K_{sp} indicates the solubility of a compound - the smaller the K_{sp}, the less soluble the compound, e.g.

$$AgCl_{(s)} \longrightarrow Ag^+_{(aq)} + Cl^-_{(aq)}$$

At equilibrium, $Ksp = [Ag^+][Cl^-] = 1.6 \times 10^{-10}$

The Ion Product

For the AgCl, if ion product = $Q = [Ag^+]_o[Cl^-]_o$

The subscript "o" reminds us that these are initial concentrations before equilibrium.

The possible relationships between Q and K_{sp} are:

i. $Q < K_{sp}$

   i.e. $[Ag^+]_o[Cl^-]_o < 1.6\times10^{-10}$ .... unsaturated solution

ii. $Q = K_{sp}$

   i.e. $[Ag^+]_o[Cl^-]_o = 1.6 \times 10^{-10}$ .... saturated solution

iii. $Q > K_{sp}$

i.e. $[Ag^+]_o[Cl^-]_o > 6 \times 10^{-10}$ .... supersaturated solution;
AgCl will precipitate out until the product of the ionic concentration is equal to 1.6×10^{-10}

Molar Solubility

The molar solubility of a compound is the number of moles of the compound (solute) in one litre of a saturated solution ($molL^{-1}$), at 25°C.

Solubility

Solubility is the number of grams of a solute in one litre of a saturated solution (gL^{-1}), at 25°C.

Example 1

The molar solubility of silver sulphate is 1.5×10^{-2} $molL^{-1}$. Calculate the solubility product of the salt.

Solution

Step 1: Write the solubility equilibrium equation

$$Ag_2SO_{4(s)} \longrightarrow 2Ag^+_{(aq)} + SO_4^{2-}{}_{(aq)}$$

From the stoichiometry, 1 mol of Ag_2SO_4 dissociates into 2 mols of Ag^+ and 1 mol of SO_4^{2-} in solution. Therefore, when 1.5×10^{-2} mol Ag_2SO_4 is dissolved in 1L of solution, the concentrations are:

$$[Ag^+] = 2[1.5 \times 10^{-2}M] = 3.0 \times 10^{-2}M$$
$$[SO_4^{2-}] = 1.5 \times 10^{-2}M$$

Step 2: Now we can calculate the solubility product as follows:

$$K_{sp} = [Ag^+]^2[SO_4^{2-}]$$
$$= (3.0 \times 10^{-2})^2(1.5 \times 10^{-2}) = 1.4 \times 10^{-5}$$

Exercises

1. Experimental facts reveal that the solubility of calcium sulphate is $0.67gL^{-1}$ at a certain temperature. Estimate the value of K_{sp} for calcium sulphate ($CaSO_4 = 136.2g/mol$).
2. If the K_{sp} for CuI is 5.1×10^{-12} calculate the molar solubility of CuI.
3. Given K_{SP} for calcium phosphate (310.2g/mol) to be 1.2×10^{-26} calculate the solubility of calcium sulphate in gL^{-1}.

Predicting Precipitation Reactions

Example

Exactly $200cm^3$ *of* $0.0040M\ BaCl_2$ are added to exactly $600cm^3$ *of* $0.0080M\ K_2SO_4$. Will a precipitate form? (K_{sp} for $BaSO_4 = 1.1 \times 10^{-10}\ K_{sp}$)

Solution

Step 1: the ions in solution are:

Ba^{2+} Cl^-
K^+ SO_4^{2-}

The only precipitate that might form is $BaSO_4$ (KCl ordinarily does not form a precipitate).

$$Ba^{2+}_{(aq)} + SO_4^{2-}{}_{(aq)} \longrightarrow BaSO_{4(s)}$$

Step 2: The number of moles of Ba^{2+} present in the original $200cm^3$ of solution is:

$$200cm^3 \times 0.004M = 0.8m\ mol$$

(where $m = 1 \times 10^{-3}$) = $8.0 \times 10^{-4}\ mol\ Ba^{2+}$.

The total volume after combining the two solutions is $(200cm^3 + 600cm^3) = 800cm^3$. The concentration of Ba^{2+} in the $800cm^3$ volume before equilibrium is:

CHEMICAL BONDING AND INTERMOLECULAR FORCES

E.G. Olumayede

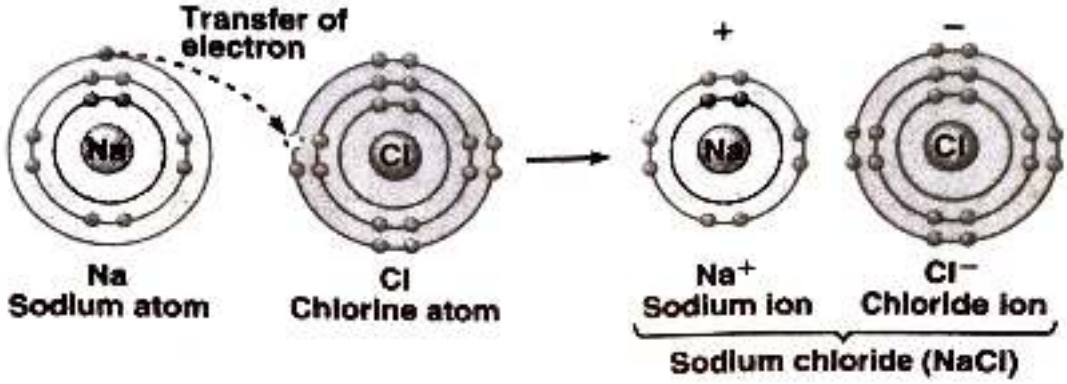
Atoms, ions and molecules come together to form compounds and the chemical bonds are formed, which hold together these bonded atoms or ions. It is the electrostatic force of attraction between negatively charged electrons and positively charged nuclei. Roussel and Lewis, G.N put forward the theory that atoms tend to react together, so as to attain stable configuration, which is the octet configuration (noble gases electron). This is referred to as OCTET RULE. The octet rule states that when an atom form a chemical bond by gaining, loosing or sharing of valence electrons, its electron configuration becomes the same as that of noble gas before or after it in the periodic table.

This theory goes for many atoms but not always true in some cases. The major criterion for chemical reaction is energetic of the reaction. There are three methods of attaining octet configuration viz i. electrovalent bonding, ii. Covalent bonding, and iii. Coordinate bonding.

Electrovalent (Ionic) Bonding

This is the electrostatics force of attraction between two opposite charged ions. The bonding involves the complete transfer of one or more electrons from one atom to another. Electrovalent bond can be formed between those atoms which readily lose their electrons (Group I and II) and those that readily accept electrons (Group VII and VI) elements.

For example, ionic bonding in sodium chloride is formed by transfer of an electron from sodium atom to chlorine atom, so as to attain Neon and Argon configuration, respectively.

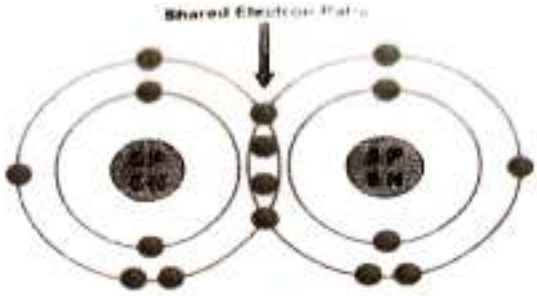


Properties of Electrovalent Compounds

1. Electrovalent compounds are made up of positive and negative ions arrange in crystal lattices. The crystal lattices can be regarded as macromolecules.
2. They usually have high boiling point and high melting point
3. Electrovalent (Ionic) compounds are usually soluble in polar solvents, that is, solvents with high dielectric constant for solvating the individual ions.
4. Ionic compounds react faster than covalent compounds. In ionic compounds, the reacting ions just collide with each other to form new compounds.

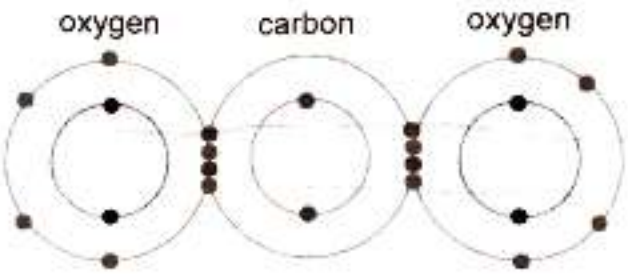
Covalent Bonding

Covalent bond exists when the electronegativity difference between two elements is small. It is formed from sharing of electrons between the atoms. For example, in the covalent bonding of oxygen molecule, an oxygen atom shares two electrons with another atom as shown in the diagram below:



Oxygen (O_2) Molecule

Similarly, in carbon dioxide molecule, carbon has 4 of 8 electrons in its outer shell and oxygen has 6 of eight electrons. By combining two oxygen atoms with one carbon atom, the atoms can share electrons such that each atom has a full outer shell.



carbon dioxide

Other examples include H_2, Cl_2, CCl_4.

Covalent bonds are brought about by the overlap of singly filled orbital. There are two types of covalent bonds. One is simply shown above while the other is coordinate covalent bond.

Coordinate covalent bonding occurs when a pair of electrons donated by one atom is shared between two atoms e.g. NH_4^+

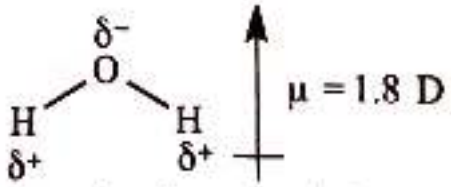
Properties of Covalent Compounds

1. Covalent compounds are made up of discrete molecules and the only force existing between the individual molecules is the weak Van der Waal's forces.
2. Covalent compounds are usually gaseous, liquid and soft solid, with low melting and boiling points.

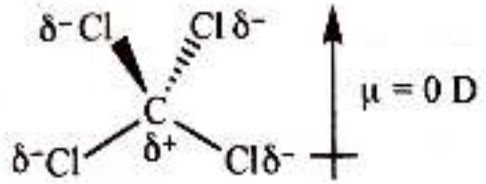
3. Covalent compounds are usually soluble in non-polar solvent, that is, solvent with low dielectric constant
4. In covalent compounds, old bonds are broken to form new bonds and this requires high energy hence they react slowly.

Dipole

When two atoms of the same element are joined together to form covalent compound, there is no dipole of the resulting molecules. However, when two different atoms form a covalent bond, there is usually a dipole. For instance, in water (H_2O), O shared four electrons provided by itself with two hydrogens. Here, O has greater affinity for electrons than hydrogen, so the molecule has dipole moment and it is a polar solvent.



Meanwhile, in CCl4 molecule, the central C atom is relatively positively charged and four Cl atoms are negatively charged. Here, the four Cl atoms possess greater affinity than carbon, so the molecule has no dipole moment and is a non-polar solvent.



The dipole moment is defined as the product of the distance separating equal charges of opposite sign and the magnitude of the charge.

Metallic Bonding

This is the bonding type which holds the atoms of element together in solid metals. The metallic cations are arranged in definite geometric

pattern (crystal lattice) and the valence electrons are free to move about throughout the crystal. The presence of these free electrons account for the high thermal conductivity or high electric conductivity of metals.

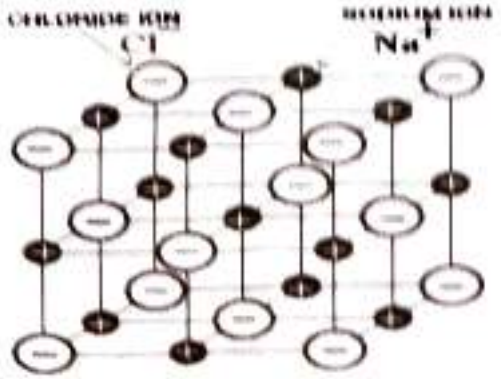


Figure 13: Lattice Structure of NaCl

Intermolecular Bonding

Between all particles, but with particular reference to covalently bonded molecules, there always exists some very weak electrical attractive forces known as **intermolecular forces** or **intermolecular bonding**. These constantly acting attractive forces or intermolecular bonds are very much weaker than full covalent or ionic chemical bonds. Some of these forces include:

1. Hydrogen Bonding

When a hydrogen atom is bonded to a very electronegative atom, including fluorine, oxygen, or nitrogen, a very polar bond is formed. The electronegative atom obtains a negative partial charge and the hydrogen obtains a positive partial charge. These partial charges are similar to what happens in every polar molecule. However, because of the big difference in electro negativities between these two atoms and the amount of positive charge exposed by the hydrogen, the dipole is much more dramatic. These molecules will be attracted to other molecules which also have partial charges. This attraction for other molecules which also have

a hydrogen bonded to a fluorine, nitrogen, or oxygen atom is called **hydrogen bond**. For example, in water and methanol.

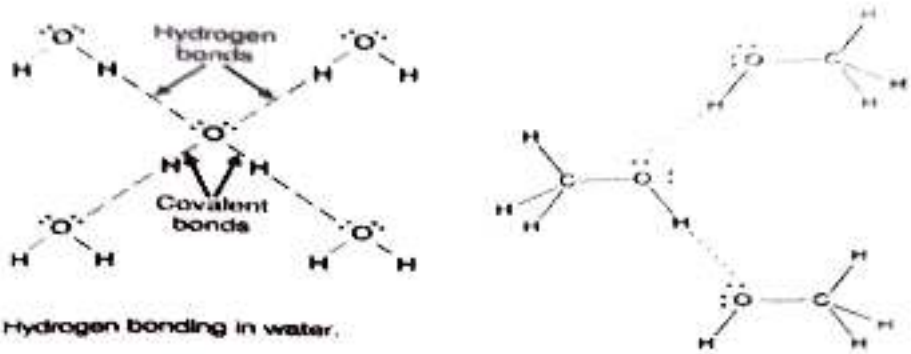


Figure 14: Hydrogen Bonding Scheme

2. Van Der Waals Forces

This is a relatively weak electrical force that attract neutral molecules to one another in gases, in liquids and solidified gases, especially organic liquids and solids. Solids that are held together by van der Waals forces have lower melting points and are softer than those held together by the stronger ionic, covalent and metallic bonds. Van der Waals forces are due to one of these factors; (1) the molecules possession of permanent electric dipoles – in some materials, although they are electrically neutral, may have fixed distortion in the distribution of electric charge in the very structure, hence, one side of the molecule is always positive and the opposite side negative. The tendency of such permanent dipoles to align with each other results in a net attractive force. (ii) Polarization induced by nearby polar or nonpolar molecules, that is, due to attractive force from the interaction of a neighbouring dipole. (iii) The force of attraction exists between the molecules e.g., in the noble gas or the organic liquid such as argon and benzene respectively.

CHEMICAL EQUILIBRIUM AND THERMODYNAMICS

O. S. Ayanda

Chemical Equilibrium

Chemical equilibrium is the state where the concentrations of all reactants and products remain constant with time. However, on the molecular level, equilibrium is not static, but is a highly dynamic situation.

Therefore, equilibrium systems are:

- dynamic (in constant motion)
- reversible
- can be approached from either direction

Reversible Reaction

A reversible reaction is one which can be made to go in either direction by changing the condition of the experiments.

$$A + B \rightleftharpoons C + D$$

where A and B are the reactants and C and D are the products.

Examples are:

1. Action of steam on red hot iron

   $$3Fe_{(s)} + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_{2(g)}$$

2. Action of heat on ammonium chloride

   $$NH_4Cl \rightleftharpoons NH_3 + HCl$$

Equilibrium in Chemical Reactions

When a reversible reaction proceeds, the concentration of the reactants decreases. Sooner or later, they level up and become constant. Since the

rate of the forward reaction now equals the rate of the backward reaction, a state is established in which the concentration no longer changes. This state is known as the state of chemical **equilibrium.**

Equilibrium state of a reaction is therefore defined as a stage where the concentrations of the reactants and products cease to change with time in a reversible reaction or a point where the rate of the forward reaction is equal to the rate of the backward reaction.

Properties of a System at Equilibrium

1. Equilibrium is dynamic
2. Rate of reaction is equal in opposite directions
3. The concentration of each of the reactants and product is constant
4. $\Delta G = 0$
5. The system resist disturbance.

In a reversible reaction, reaction is represented by

$$A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} C + D$$

Rate of forward reaction $R_1 = k_1[A][B]$

Rate of backward reaction $R_2 = k_2[C][D]$

At equilibrium, the rates are equal; $R_1 = R_2$

Therefore, $k_1[A][B] = k_2[C][D]$

$$\frac{k_1}{k_2} = \frac{[C][D]}{[A][B]} = K$$

k is the equilibrium constant.

$$\text{In } aA + bB \underset{k_2}{\overset{k_1}{\rightleftharpoons}} cC + dD$$

$$R_1 = k_1[A]^a[B]^b$$
$$R_2 = k_2[C]^c[D]^d$$

At equilibrium, $R_1 = R_2$

Therefore, $k_1[A]^a[B]^b = k_2[C]^c[D]^d$

$$\frac{k_1}{k_2} = \frac{[C]^c[D]^d}{[A]^a[B]^b} = K$$

Note that equilibrium does not have units in some instances.

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

$$K = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

To find the unit using the concentration: $K = \frac{[mol/L][mol/L]}{[mol/L]}$, the mol/L crosses out and we left with mol/L. The unit of k is therefore mol/L.

Question 1

For each of the following equilibria, write the expression for the equilibrium constant K_c and state their units.

1. $2NO_2 \rightleftharpoons N_2O_4$
2. $N_2 + 3H_2 \rightleftharpoons 2NH_3$
3. $CH_3CH_2CO_2H_{(l)} + CH_3CH_2OH_{(l)} \rightleftharpoons CH_3CH_2CO_2CH_2CH_{3(l)} + H_2O_{(l)}$

Homogeneous and Heterogeneous Equilibrium

Homogeneous systems are systems in which all the substances involved are in the same phase. Therefore, homogeneous solutions are characterized by having only one phase. Example is the synthesis of ammonia.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

In contrast, heterogeneous systems are systems in which the components are in different phases. Thus, heterogeneous solutions are composed of two phases. The formation of an aqueous solution of lead (II) iodide is a heterogeneous mixture dealing with molecules in both the solid and aqueous states.

$$PbI_{2(s)} \rightleftharpoons Pb^{2+}_{(aq)} + 2I^{-}_{(aq)}$$

In calculations involving a heterogeneous equilibrium constant, pure solids and pure liquids will always be excluded. In a mathematical perspective, concentration of pure solids and pure liquids equal one, which does not affect the overall k value.

Example:

$$Ca(OH)_{2(s)} \rightleftharpoons Ca_2^{+}{}_{(aq)} + 2OH^{-}_{(aq)}$$

Its equilibrium constant k will not include $Ca(OH)_2$. Hence,

$$k = [Ca^{2+}][OH^{-}]^2$$

Factors Affecting the Position of Equilibrium

These effects may be predicted by Le Chatelier's principles or the law of mobile equilibrium. This states that if a system at equilibrium is disturbed, the system tends to shift its equilibrium position to counter the effect of the disturbance.

The main factors that affect the position of equilibrium are:

1. Temperature
2. Pressure
3. Concentration

Effect of Changing Concentration on Equilibrium

Changing the concentration of a substance will shift the equilibrium to the side that would reduce that change in concentration. The chemical

system will attempt to partially oppose the change affected to the original state of equilibrium. In turn, the rate of reaction, extent and yield of products will be altered corresponding to the impact on the system.

Considering the reaction:

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

If the concentration of PCl_5 in the system is increased, more of Cl_2 will be used up and the amount of PCl_5 will increase. Moreover, the subtraction of a species would cause the reaction to fill the gap and favour the side where the species was reduced. The equilibrium constant, however, remains the same.

Effect of Pressure on Equilibrium

Changes in pressure are attributable to changes in volume. The equilibrium concentrations of the products and reactants do not directly depend on the pressure subjected to the system. However, a change in pressure due to a change in volume of the system will shift the equilibrium.

Considering the reaction:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$
$$\text{4 moles} \rightleftharpoons \text{2 moles}$$

From the reaction, 4 moles on the left-hand side form 2 moles on the right-hand side. if there is an increase in pressure, the equilibrium shift to where there is less molecules and vice-versa. The formation of NH_3 from N_2 and H_2 decreases the total number of molecules in the system by 50%, therefore, reducing the total pressure exerted by the system. Experiments show that increase in the pressure drives the reaction to the right. On the other hand, lowering the pressure on the system favours the decomposition of NH_3 to N_2 and H_2

There is no effect on a reaction where the number of moles of gas is the same on each side of the chemical equation.

i.e. $N_2 + O_2 \rightleftharpoons 2NO$

2 moles $\rightleftharpoons$ 2 moles

Effect of Temperature on Equilibrium

All chemical reactions involve evolution or absorption of heat. When the reaction is exothermic (ΔH is negative, puts energy out), heat is included as a product, and, when the reaction is endothermic (ΔH is positive, takes energy in), heat is included as a reactant. The reaction of N_2 and I_2 to give HI is exothermic.

$$H_2 + I_2 \rightleftharpoons HI + \text{heat}$$

Therefore, if the temperature was increased, the heat content of the system would increase, so the system would consume some of that heat by shifting the equilibrium to the left, thereby producing less HI (increase in temperature favours the decomposition back to H_2 and I_2). More HI would be produced if the reaction was run at a lower temperature.

Effect of a Catalyst

A catalyst has no effect on the position of a system at equilibrium. It helps equilibrium to be attained in a shorter time.

Thermodynamics

Thermochemistry is the branch of chemistry concerned with the heat effects that accompany chemical reactions. It is a subfield of a larger discipline called thermodynamics.

What is Thermodynamics?

Thermo means heat

Dynamics means motion or change

Chemical thermodynamics is the branch of chemistry which studies the energy transformation that accompanies chemical or physical changes.

Some basic terms are:

A system is the part of the universe chosen for study, and it can be as large as all the oceans on Earth or as small as the contents of a beaker.

The surroundings are that part of the universe outside the system with which the system interacts.

An open system freely exchanges energy and matter with its surroundings. The beaker of hot coffee transfers energy to the surroundings - it loses heat as it cools. Matter is also transferred in the form of water vapor.

A **closed system** can exchange energy, but not matter, with its surroundings. The flask of hot coffee transfers energy (heat) to the surroundings as it cools. Because the flask is stoppered, no water vapor escapes and no matter is transferred.

An **isolated system** does not interact with its surroundings. Hot coffee in an insulated container approximates an isolated system. No water vapor escapes, and, for a time at least, little heat is transferred to the surroundings.

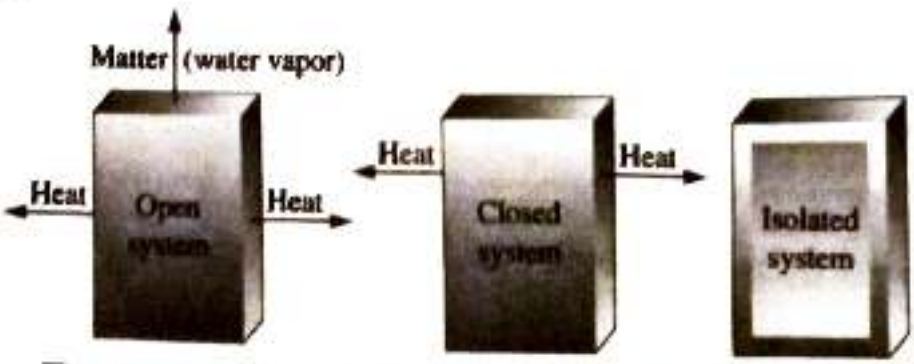


Figure 15: Open, Closed and Isolated Systems

Energy is the capacity to do work. **Work** is done when a force acts through a distance. The energy of a moving object is called **kinetic energy**. The kinetic energy of an object is based on its mass m and velocity u through the equation:

$$Kinetic\ energy = \frac{1}{2}mv^2$$

$$Work = Force \times Distance$$

The unit of work, heat and energy is the joule, but work and heat are not forms of energy but *processes* by which the energy of a system is changed.

Potential Energy is the energy resulting from condition, position, or composition; it is an energy associated with forces of attraction or repulsion between objects.

The kinetic energy associated with random molecular motion is called **thermal energy**. Thermal energy is proportional to the temperature of a system. The more vigorous the motion of the molecules in the system, the hotter the sample and the greater is its thermal energy.

First Law of Thermodynamics

For a closed system, the change in the internal energy of the system is equal to the sum of the energy in the form of heat exchanged between system and surroundings (q) and the work done on the system (w). The First Law postulates the concept of internal energy, E, and states, while energy may change forms, it is not created or destroyed.

Mathematically, the first law of thermodynamics can be written as:

$$\Delta E = q + w$$

ΔE is the change in internal energy of the system, q is the heat added and w is the work done by the system on its surrounding.

Question 2

If 600 Kg of heat is added to a system and 450kJ of work is done. What is the change in the internal energy of the system?

An isolated system is unable to exchange either heat or work with its surroundings, so that $\Delta E_{isolated\ system} = 0$, and hence the energy of an isolated system is constant and energy is conserved.

Any energy entering the system carries a positive sign. Thus, if heat is absorbed by the system $q > 0$, If work is done on the system $w > 0$. Moreover, any energy leaving the system carries a negative sign. Thus, if heat is given off by the system $q < 0$, If work is done *by* the system $w < 0$.

In general, the internal energy of a system changes as a result of energy entering or leaving the system as heat and/or work. If, on balance, more energy enters the system than leaves, ΔE is positive. If more energy leaves than enters, ΔE is negative.

Enthalpy and Enthalpy Change

Enthalpy, *H*, is the sum of the internal energy and the pressure - volume product of a system:

$$H = E + PV$$

where P is the pressure, E is internal energy and V is the volume of the system

Since, the **enthalpy change** $(\Delta H) = H_1 - H_2$

$$\Delta H = (E_2 + P_2V_2) - (E_1 + P_1V_1)$$

Expand brackets and solve to get:

$$\Delta H = \Delta E + (P_2V_2 - P_1V_1)$$

If pressure (P) is constant i.e. $P1 = P2$. Hence, $\Delta H = \Delta E + P\Delta V$

When ΔH for a chemical reaction is positive, the reaction is endothermic and if ΔH is negative, the reaction is exothermic.

Endothermic and Exothermic Reaction

An **exothermic reaction** is one that produces a temperature increase in an isolated system or, in a non-isolated system, gives off heat to the surroundings. For an exothermic reaction, the heat of reaction is a negative quantity. While in an **endothermic reaction,** the corresponding situation is a temperature decrease in an isolated system or a gain of heat from the surroundings by a non-isolated system. In this case, the heat of reaction is a positive quantity.

Second Law of Thermodynamics:

The law states that all physical and chemical changes occur such that at least some energy disperses: the total concentrated or organized energy of the universe decreases, the total diffuse or disorganized energy of the universe increases. The Second Law has to do with moving from concentrated energy to dispersed energy in the form of random thermal motion. It postulates a new thermodynamic variable *S*, the **Entropy**, a measure of the dissipated energy within a system at each temperature that is unavailable to do work. Entropy is the quantity which measures the degree of disorderliness of a system. Increasing the temperature increases the random motion and thus increases the entropy. The entropy change of a system is denoted by ΔS. Note that all physical and chemical changes occur such that the total entropy of the universe increases i.e. $S_{universe} = increases$ or $\Delta S_{universe} > 0$. The entropy of an isolated system never spontaneously decreases i.e. $\Delta S_{isolated} \geq 0$.

Gibb's Free Energy

Free energy is a quantity indicating the extent to which a reaction between substances occurs. The Gibb's free energy is defined as:

$$G = H - TS$$

where H is the enthalpy, S is the entropy and T is the absolute temperature.

ΔG is the free energy change;

$$\Delta G = \Delta H - T\Delta S$$

$\Delta G = \Delta H - T\Delta S$ tells the direction of spontaneous change at constant temperature and pressure.

1. If ΔG is less than zero, the reaction is spontaneous in the forward direction.
2. If ΔG is greater than zero the reaction is not spontaneous and must be driven by an external source of energy. It is actually spontaneous in the reverse direction.
3. If $\Delta G = 0$, the reaction is at equilibrium. There is no driving force for change either forward or reverse.

Question 3

The reaction represented by the equation:

$$C_{(s)} + H_2O_{(l)} \longrightarrow CO_{(g)} + H_{2(g)}$$

was carried out at 27°C. If the enthalpy change was 4500 J and the entropy change was 12 J. What was the free energy change ΔG?

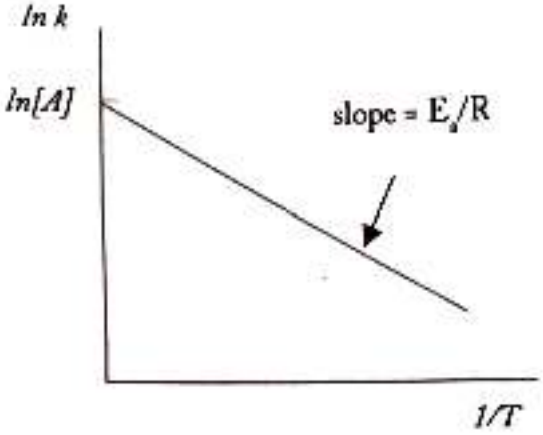
Activation Energy (E_a)

The activated energy of a reaction is an energy barrier that must be overcome before a reaction can occur. It is also the difference between the energy of the reactant molecules and that of the activation complex. At the top of the energy barrier, the molecules are at the transition stage (activated complex) before going to form the product of the reaction.

$$k = Ae^{\frac{-E_a}{RT}}$$

The equation is linearized to:

$$lnk = lnA - \frac{E_a}{R} \cdot \frac{1}{T}$$



A plot of $ln\ k$ versus $\frac{1}{T}$ gives a straight-line graph with the $slope = \frac{-E_a}{R}$ and intercept equals $ln\ A$.

Energy Profile Diagram

In the energy profile diagram shown in Figure 2, the number of molecules that possess the activation energy will determine the rate of the reaction. Thus, this part of the reaction (activation complex) is the rate determining step.

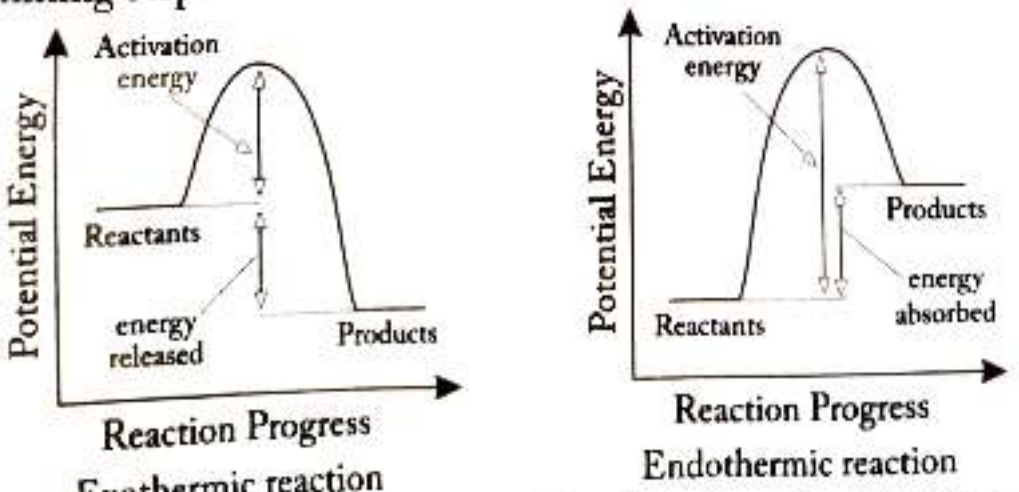


Figure 16: Energy Profile Diagram of an Endothermic and Exothermic Reaction

Types of Enthalpy Reaction

Enthalpy of Formation

This is defined as the quantity of heat evolved or absorbed when one mole of the compound is formed from its elements. It is expressed as ΔH_f.

Enthalpy of Combustion

It is *the quantity of heat evolved when one mole of the substance is completely oxidized.*

Enthalpy of Solution

The quantity of heat evolved or absorbed when one mole of a solute is dissolved completely in large excess of water, so that further dilution of solution does not produce any heat change.

Enthalpy of Neutralization

This is *the quantity of heat evolved when one equivalent (or equivalent mass) of an acid is completely neutralized by one equivalent (or equivalent mass) of a base in dilute solution.*

Enthalpy of Dilution

The quantity of heat evolved or absorbed when solution containing one mole of a solution is diluted from one concentration to another.

Enthalpy of Precipitation

The quantity of heat evolved in the precipitation of one mole of a sparingly soluble substance on mixing dilute solutions of suitable electrolytes.

Enthalpy of Hydration

The quantity of heat evolved or absorbed when one mole of an anhydrous or a partially hydrated salt is combined with the required number of moles of water to form a specific hydrated substance.

Enthalpy of Fusion

It is the change in enthalpy during conversion of 1 mole of a substance from solid to liquid state at its melting point (mostly endothermic).

Enthalpy of Vaporization

It is the change in enthalpy when 1 mole of a substance is converted from liquid to gaseous state at its boiling point.

Enthalpy of Sublimation

It is the change in enthalpy when 1 mole of a substance is converted directly from solid to vapour (gaseous) state.

INTRODUCTORY CHEMICAL KINETICS AND NUCLEAR CHEMISTRY

O. S. Ayanda

Chemical Kinetics

Chemical kinetics is the branch of chemistry that deals with reaction rate. It is the study of the rate at which a chemical process occurs. Besides information about the speed at which reactions occur, kinetics also sheds light on the reaction mechanism (how the reaction occurs). Chemical kinetics are dependent on various factors such as concentration, temperature, catalyst etc.

A study into the kinetics of a chemical reaction is usually carried out with one or both of two main goals in mind:

1. Analysis of the sequence of elementary steps giving rise to the overall reaction. i.e. the reaction mechanism.
2. Determination of the absolute rate of the reaction and/or its individual elementary steps.

Rate Law

Rate law is the equation describing the relationship between the reaction rate and concentration of a reactant or reactants.

Rates are negative for reactants and positive quantities for products.

For the reaction:

$$A \longrightarrow P$$

$$\text{rate} = -\frac{\Delta[A]}{\Delta t} = \frac{\Delta[P]}{\Delta t}$$

For a general reaction;

$$aA + bB \longrightarrow cC + dD$$

$$rate = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

$$R \propto [A]^m[B]^n$$

$$R = k[A]^m[B]^n$$

where k is the rate constant, m is the order of reaction with respect to [A] and n is the order of reaction with respect to [B]. Therefore, m + n is the overall order of the reaction. Note that m ≠ a and n ≠ b.

m and n indicate the sensitivity of the rate to concentration changes of each reactant. An exponent of 0 means the reaction is zero order in that reactant; meaning the rate does not depend on the concentration of that reactant. An exponent of 1 means the rate is directly proportional to the concentration of that reactant i.e. if concentration is doubled, rate doubles, reaction is first order in that reactant. An exponent of 2 means the rate is proportional to the square of concentration of that reactant i.e. if concentration is double, rate is quadrupled; reaction is second order in that reactant.

Kinetic Classification of Chemical Reaction

We classified reaction based on:

1. **Molecularity**

a. Unimolecular reaction

Include some molecular decomposition and intramolecular rearrangement.

$$I_2 \longrightarrow 2I$$

For simple unimolecular reaction;

$$Rate = kC$$

where C is the concentration of initial substance. Unimolecular reactions are often first-order reactions.

b. Bimolecular reaction

Are reactions resulting from collision of 2 molecules of the same or different species.

$$H_2 + I_2 \longrightarrow 2HI$$

For simple bimolecular reaction;

$$\text{Rate} = kC_1C_2,$$

where C_1 and C_2 are the concentration of the reactants. The rate of reaction depends on the product of the concentrations of both species involved, which makes bimolecular reaction second-order reactions.

c. Termolecular reaction

Include reactions that requires a collision between 3 molecules of the same or different species for the reaction to take place.

$$2NO + H_2 \longrightarrow N_2O + H_2O$$

For simple termolecular reaction;

$$\text{Rate} = kC_1C_2C_3.$$

All termolecular reactions are third-order.

Step	Molecularity	Law
$A \longrightarrow$ Products	unimolecular	rate = k[A]
$2A \longrightarrow$ Products $A + B \longrightarrow$ Products	bimolecular	rate = $k[A]^2$ rate = k[A][B]
$3A \longrightarrow$ Products $2A + B \longrightarrow$ Products $A + B + C \longrightarrow$ Products	termolecular	rate = $k[A]^3$ rate = $k[A]^2[B]$ rate = [A][B][C]

The molecularity of a reaction is determined by the number of molecule(s) that simultaneously take part in the reaction. Theoretically higher molecularity is possible but in practice the simultaneous collision of 3 molecules is improbable and are rarely encountered. Practically, no reactions of higher molecularity are known, when large numbers of molecules participate in reactions, they are assumed to proceed in stages

2. Order of Reaction

a. Zero order reaction
b. First-order reaction
c. Second-order reaction
d. Third-order reaction.

Order of reaction is the sum of the exponent of the concentration in the equation for the dependence of the rate on the concentration of the reactant.

i.e. For reaction A → P;

$$rate = -\frac{d[A]}{dt} = k[A]^x$$

x is the order of the reaction.

For **First-Order Reaction** (i.e. $x = 1$):

$$rate = -\frac{d[A]}{dt} = k[A]^1$$

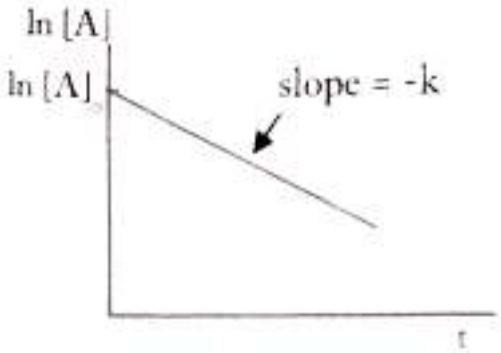
Therefore, $-\frac{d[A]}{[A]} = Kdt$

If $[A]$ is the initial concentration at initial time t and $[A]$ is the concentration after t unit of time,

Then: $\int_{A_0}^{A_t} \frac{d[A]}{[A]} = -k \int_{t-0}^{t} dt$

$$\ln A_{[A]_0}^{[A]_t} = -k(t)_0^t$$

$\ln[A]_t - \ln[A]_0 = -kt$-------- First-order kinetic equation



The first-order reaction kinetic equation could be arranged and written as:

$$\ln \frac{[A]_0}{[A]_t} = kt;$$

$$\ln \frac{[A]_t}{[A]_0} = -kt;$$

$$\frac{[A]_t}{[A]_0} = e^{-kt}; or$$

$$[A]_t = [A]_0 e^{-kt}$$

Half-life for First-order Reaction:

This is the time it takes for half of the original concentration to be used up.

$$\text{At } t_{1/2}: [A]_t = \frac{[A]_0}{2}$$

First-order equation = $\ln \frac{[A]_0}{[A]_t} = kt$

Substitute $\frac{[A]_0}{2}$ for $[A]_t$ in the first-order equation.

$$\ln \frac{[A]_0}{\frac{[A]_0}{2}} = kt_{1/2} \approx \ln\left([A]_0 \times \frac{2}{[A]_0}\right) = kt_{1/2}$$

$$\ln 2 = kt_{1/2}$$

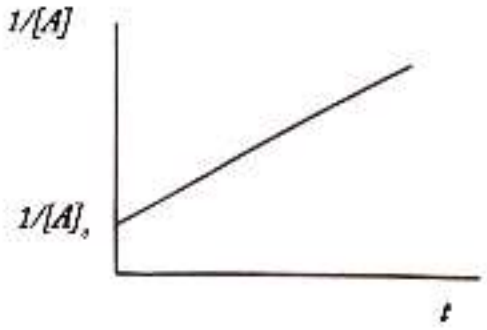
$$\text{Therefore; } t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

A summary of the kinetic and half-life equations for a second-order, third-order and zero-order reaction are as follows:

Second-order Kinetic (i.e. $x = 2$):

$$rate = -\frac{d[A]}{dt} = k[A]^2$$

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$



$$t_{1/2} = \frac{1}{[A]_0 k}$$

Third-order Kinetic (i.e. $x = 3$):

$$rate = -\frac{d[A]}{dt} = k[A]^3$$

$$\frac{1}{2}\left[\frac{1}{[A]_t^2} - \frac{1}{[A]_0^2}\right] = kt$$

$$t_{1/2} = \frac{3}{2k[A]_0^2}$$

Zero order kinetic (i.e. $x = 0$):

$$[A]_0 - [A]_t = kt$$

The **General Kinetic Equation** for all the order of reaction is:

$$\frac{1}{n-1}\left[\frac{1}{[A]_t^{n-1}} - \frac{1}{[A]_0^{n-1}}\right] = \text{kt}$$

$$t_{1/2} = \frac{2^{n-1} - 1}{[A]_0^{n-1}(n-1)k}$$

TRT

Question 1

For the reaction $A \rightarrow P$, the data obtained were presented in the Table below:

[A] M	Rate V (m/hr)
0.170	0.050
0.340	0.100
0.680	0.200

i. What is the order of the reaction?

ii. Calculate the rate constant

Nuclear Chemistry

In 1896, a French physicist named Henri Becquerel discovered that uranium-containing crystals emitted rays that could expose and fog photographic plates. He found that these "rays" originated from changes within the atomic nuclei of the U atoms. He proposed that the uranium atoms were unstable. They emitted particles and/or energy to become more stable. These emissions he called *uranic rays*. In France, Marie

Sklodowska Curie carried out her doctoral work to determine if other substances emitted uranic rays. She discovered two elements, which she later named polonium (after her native Poland) and radium, which emitted high levels of radioactivity. Marie Curie changed the name of uranic rays to **radioactivity** (or **radioactive decay**). She shared the Nobel Prize in physics with Becquerel and her husband, Pierre Curie, for discovering radioactivity. **Radioactivity** is the spontaneous emission of particles and/or electromagnetic radiation by unstable nuclei, also known as radionuclides. Radiation emitted includes alpha particles, beta particles and gamma rays.

Radiation and Nuclear Reactions

Nuclear reactions involve changes in the particles in an atom's nucleus and thus cause a change in the atom itself. All elements heavier than bismuth (Bi) (and some lighter) exhibit natural radioactivity and thus can decay into lighter elements. Unlike normal chemical reactions that form molecules, nuclear reactions result into the transmutation of one element into different isotope or a different element altogether. Note that the number of protons in an atom defines the element, so a change in the protons results in a change in the atom. There are 3 common types of radiation and nuclear changes:

1. Alpha (α) Radiation

This is the emission of an alpha particle from an atom's nucleus. An alpha particle contains two protons and two neutrons (and is similar to the helium nucleus ${}^{4}_{2}He$). When an atom emits an alpha particle, the atom's atomic mass will decrease by 4 units (because two protons and two neutrons are lost) and the atomic number (Z) will decrease by two units. The element is said to transmute into another element that is two Z unit smaller. An example is uranium decay into the element thorium (Th) by emitting an alpha particle, as depicted in the following equation:

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

2. **Beta (β) Radiation**

This is the transmutation of a neutron into a proton and an electron (followed by the emission of the electron from the atom's nucleus: $^{0}_{-1}e$). When an atom emits a beta particle, the atom's mass will not change (since there is no change in the total number of nuclear particles), however the atomic number will increase by one (because the neutron transmuted into additional proton). Example is the decay of the isotope of carbon named carbon-14 into nitrogen.

$$^{14}_{6}C \longrightarrow {}^{0}_{-1}e + {}^{14}_{7}N$$

3. **Gamma (γ) Radiation**

This involves the emission of electromagnetic energy (similar to light energy) from an atom's nucleus. No particles are emitted during gamma radiation, and thus gamma radiation does not itself cause the transmutation of atoms, however, gamma radiation is often emitted during and simultaneously to alpha and beta radioactive decay. Gamma radiation does not change the mass or atomic number and is represented as $^{0}_{0}y$. X-rays emitted during the beta decay of cobalt-60 are common example of gamma radiation.

Type of change	Symbol	Change in protons (atomic number, Z)	Change in neutrons	Change in mass number, A
Alpha emission	$\propto$ or $^{4}_{2}He$	-2	-2	4
Beta emission	$\beta, \beta^{-}, {}^{0}_{-1}e$	+1	-1	0
Positron emission	$\beta^{+}, {}^{0}_{+1}e$, or ${}^{0}_{1}e$	-1	+1	0
Electron capture	E, C	-1	+1	0

Gamma emission	γ or ${}^{0}_{0}\gamma$	0	0	0

4. Positron Emission

A positron is a particle that has the same mass as an electron but opposite charge. The positron is represented as ${}^{0}_{1}e$. Carbon-11 is an example of a particle that undergoes positron emission.

$$^{11}_{6}C \longrightarrow {}^{11}_{5}B + {}^{0}_{1}e$$

5. Electron Capture

Capture by the nucleus on an inner shell electron of the electron cloud.

$$^{81}_{37}Rb + {}^{0}_{-1}e \text{ (orbital electron)} \longrightarrow {}^{81}_{36}Kr$$

In summary

Alpha emission $^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$

Beta emission $^{A}_{Z}X \longrightarrow {}_{Z+1}{}^{A}Y + {}_{-1}{}^{0}e$

Positron emission $^{A}_{Z}X \longrightarrow {}_{Z-1}{}^{A}Y + {}_{+1}{}^{0}e$

Electron capture $^{0}_{-1}e + {}^{A}_{Z}X \longrightarrow {}_{Z-1}{}^{A}Y$

Balancing Nuclear Reactions

The symbol ${}^{A}_{Z}X$ is used to represent a nuclide (single isotope), where "X" represents the element's atomic symbol, "A" represents the mass number (sum of protons and neutrons), and "Z" represents the atomic number (number of protons in the nucleus). When balancing nuclear reaction equations, both atomic numbers and mass numbers are conserved.

To find the missing nuclide in the following equation:

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{90}_{38}Sr + - + 3{}^{1}_{0}n$$

The sum of the mass numbers is 236 on the left, the sum of the atomic numbers is 92. On the right, the sum of the mass numbers is 93 and the sum of the atom numbers is 38. To balance the equation and obey the laws of conservation of mass and charge, the missing nuclide must have a mass number of 143 with an atomic number of 54. Thus, the missing nuclide is $^{143}_{54}Xe$.

Question 2

Write a balance equation for the following nuclear reactions:

$$^{254}_{98}\text{Cf} \longrightarrow {}^{a}_{b}\text{Pd} + {}^{132}_{52}\text{Te} + 4\,{}^{1}_{0}\text{n}$$

$$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}\alpha \rightarrow {}^{234}_{90}Th + {}^{c}_{d}\gamma$$

Half-Life

Half-life ($t_{1/2}$) is frequently used to compare and/or describe the decay rates for various radioactive nuclides. Half-life is the time required for 50% of the radioactive nuclides to decay.

For example, the radioactive element bismuth ^{210}Bican undergo alpha decay to form the element thallium ^{206}Tl with a reaction half-life $t_{1/2}$ = 5 days. If we start with 100 g of ^{210}Bi in a sealed lead container, there will be 50 g of ^{210}Bi and 50 g of ^{206}Tl after 5 days in the jar. After another 5 days (10 days from the start), one half of the remaining bismuth will decay and will be left with 25 g bismuth and 75 g of thallium. Thus, the reaction processes in half of whatever is left of the radioactive elements decaying every half-life period.

$$t_{1/2} = \frac{0.692}{\lambda}$$

where λ is the decay constant.

Decay constant is the instantaneous rate of decay per unit atom of a substance.

$$Decay\ constant\ (\lambda) = \frac{No.of\ atoms\ disintegrating\ per\ seconds}{No.of\ atoms\ in\ the\ source\ at\ that\ instant}$$

$$\lambda = \frac{0.692}{t_{1/2}}$$

Half-life is a measure of the stability of the nucleus. The decay is unaffected by temperature, pressure or whether the isotope is a solid, liquid or gas.

Question 3

If the half-life of a sample of a radioactive material is 60 days, what fraction of the original radioactive nuclei will remain after 120 days.

Natural and Artificial Radioactivity

Natural radioactivity: This is the spontaneous disintegration of unstable nuclei of radioactive elements bringing about changes in the nuclei of the atoms.

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

Artificial radioactivity: This is an induced nuclear reaction which occurs when nuclei are bombarded by high velocity particles such as neutrons, protons and alpha particles.

$$^{14}_{7}N + {}^{4}_{2}He \longrightarrow {}^{17}_{8}O + {}^{1}_{1}H$$

Types of Nuclear Reactions

Nuclear fission: It involves the splitting of heavier, unstable nuclides, into multiple, smaller, more stable nuclides and one or more neutrons.

Since the smaller nuclides are so much more stable, the process is highly exothermic. $^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{144}_{56}Ba + {}^{90}_{36}Ba + 3{}^{1}_{0}n +$ **Energy**

Nuclear fusion (thermonuclear reaction): It involves the combination of small nuclei into larger, more stable nuclei. The process is difficult to initiate because it requires very high temperatures, but the overall process is highly exothermic, produces minimal radioactive waste, and uses readily available fuel (hydrogen).

$$^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{3}_{2}H + {}^{1}_{0}He$$

Nuclear Transmutations

Nuclear reactions that are induced are called nuclear transmutations. It occurs when nuclei are bombarded with nucleons (protons, neutrons) or other nuclei to produce atoms of different element. Example is the conversion of nitrogen-14 into oxygen-17 by bombarding with alpha particles from radium.

$$^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$$

Difference between nuclear fusion and fission

1. Nuclear fusion releases more energy than nuclear fission
2. Fission requires initial high temperature while fission does not

Application of Radioactivity

In Medicine;

1. Cobalt-60 has been used to destroy cancer cells (radiotherapy)
2. A radio-isotope of iodine-131 ($^{131}_{53}I$) can be used to study the effectiveness of thyroid gland in the production of thyroxin. It can also be used to treat cancer.

3. Gamma rays from radioactive elements can be used to sterilize syringes and other medical apparatus.
4. Barium-131 is used to detect bone tumours.

In Agriculture;

1. Gamma rays are used to sterilize packages, foods thus preventing their spoilage
2. Dangerous pests on cattle can be destroy by sterilizing effect of gamma rays
3. Radioactive phosphorus-32 can be incorporated into a fertilizer enabling us to know the way a plant uses the phosphorus.

Question 4

a. Americium-243 is an alpha emitter used in smoke detectors. Write the nuclear equation for its alpha emission.
b. Iron-59, which is used to diagnose anaemia, has a half-life of 45 days. What fraction of it is left in 90 days?
c. Write short note on the biological effects of radiation.

INTRODUCTION TO COLLIGATIVE PROPERTIES OF LIQUIDS

P.O. Okolo & S.O. Afolabi

Colligative properties of a solution are those physical properties which depend solely on the population (concentration) of the solute in the solution, these physical properties are **vapour pressure, boiling point, freezing point** and **osmotic pressure**. The nature of the solute (ionic, molecular mass or size) does not affect the colligative properties of the solution. The difference between the colligative properties of a pure solvent and its solution depend only on the number of non-volatile solutes in the solution.

Knowledge of colligative properties of a solution may be used to calculate:

(i) The concentration of solute particles
(ii) The molecular weight of the solute and
(iii) The purity of solute samples

These are important in chemistry, pharmacy, medicine, industry, car radiators and management of road-surface snow, etc.

Vapour Pressure Depression

The law governing the vapour pressure of solution was established by F.M. Raoult (1887) as follows:

$$P_{soln} = x_{solv} \cdot P^0_{solv},$$

where P_{soln} is the vapour pressure arising from the solution, x_{solv} is the mole fraction of the solvent and P^0_{solv} is the vapour pressure of the pure solvent.

This implies that non-volatile solute lowers the vapour pressure. Raoult's law obeys strictly an ideal solution though it might work for non-ideal solution that are reasonably dilute ($x_{solv} > 0.97$).

In an ideal solution, the forces of attraction/repulsion between the solute molecules in the solution would be the same when the solutes molecules are not in the solution. The thermodynamic explanation of Raoult's law will be explained in a more advanced topic. It is also important to note that change in the vapour pressure (ΔP) between a pure solvent and a resulting solution is given by:

$\Delta P = x_{solute} \cdot P^0_{solv}$; where x_{solute} is the mole fraction of the solute.

Example 1

Calculate the vapour pressure of a perfect solution of 95.0g of glycerine ($C_3H_8O_3, 92.1g/mol$) in 190.0g ethanol ($C_2H_5OH, 46.0\,g/mol$) at 40°C. Given: $P^0_{ethanol}$ at 40°C is 135.3 torr. The glycerine is non-volatile at this temperature.

Solution

$$\text{Data: mole of glycerin} = \frac{95}{92.1} = 1.03 \text{ mol}$$

$$\text{Mole of ethanol} = \frac{190}{46} = 4.13 \text{ mol}$$

Mole fraction of solvent, $x_{solv} = \frac{4.13}{(4.13+1.03)}$

$= 4.13/5.16 = 0.8$

$P^0_{solv} = 135.3\ torr$

Now, using the relationship.

$$P_{soln} = x_{solv} P^0_{solv}$$

$$= 0.8 \times 135.3\ torr = 108.24\ torr$$

Note: The vapour pressure of the pure solvent (ethanol) has been depressed from 135.5 torr to 108.24 torr by the presence of the non-volatile impurity/solute (glycerol)

The difference in vapour pressure (ΔP)is:

$\Delta P = P^0_{solv} - P_{soln} = (135.3 - 108.24)$ torr

$= 27.06 torr$

ΔP can also be calculated straight from the formula

$\Delta P = x_{solute} P^0_{solv}$

$since\ x_{solute} = \dfrac{1.03}{(1.03 + 4.13)} = 0.2$

$\therefore \Delta P = 0.2 \times 135.3\ torr = 27.06\ torr$

Freezing Point Depression

The freezing point of a liquid is the temperature at which both the liquid phase and the solid phase have the same vapour pressure. This is also the temperature at which the first appearance of solid forms out of solution. A solution is known to freeze at a lower temperature than the pure solvent and the difference in these freezing points is directly proportional to the molal concentration of the non-volatile solute. Think of how this knowledge could be used in managing road – surface snow in cold regions of the world.

Mathematically, the extent of freezing point depression is given by:

$$\Delta T_f = T_f(soln) - T_f(solv) = -k_f m,$$

where ΔT_f is the difference in freezing point between the solution and the solvent, $T_f(soln)$ is the freezing point of the solution, $T_f(solv)$ is

the freezing point of the solvent, and k_f is the proportionality constant, molal freezing point depression constant when the non-volatile solute in the solution is 1.0 m (m = molality), this is also called the cryoscopic constant. The minus sign simply means that the freeing point of the resulting solution was lowered.

Example 2

Calculate the freezing point of an aqueous solution of sucrose whose composition is 25.0g $C_{12}H_{22}O_{11}$ (342.3) in 100.0g H_2O. Given: freezing point of $H_2O = 0.0^0C, k_f$(water)= 1.86^0cm^{-1}.

Solution

No of moles of sucrose = 25/342.3 = 0.073mol

Molality of sucrose = $\frac{number\ of\ moles\ of\ solute}{kg\ solvent}$

$$\frac{0.073\ mol\ sucrose}{(100/1000)kg\ H_2O}$$

$$0.73molkg^{-1} = 0.73m$$

Now, using the relationship below to calculate the freezing point depression

$$\Delta T_f = -k_f m, get$$

$$\Delta T_f = -1.86^0Cm^{-1} \times 0.73m$$

$$= -1.36^0C$$

Hence, the freezing point of pure solvent was depressed by 1.36^0C

∴ the resulting freezing point of the solution is freezing point of pure solvent minus the depression

$= 0.0^0C - 1.36^0C = -1.36^0C$

Boiling Point Elevation

The boiling point of a liquid is the temperature at which boiling just commences. A liquid boil when the vapour pressure is equal to the external pressure around it. Since the presence of a non-volatile solute lowers the vapour pressure, it is only natural that a higher temperature is needed to raise the vapour pressure of the solution to the point where it boils. For examples one of any non-volatile impurity (solute) in 1 kilogram of water elevates the boiling point by $0.512\ ^{0}C$.

Mathematically, the difference in the boiling point of a pure solvent and its solution (when non-volatile solute is dissolved) is given below.

$$\Delta T_b = T_b(soln) - T_b(solv) = k_b m$$

where ΔT_b is the change in the boiling point between that of pure solvent and the solution, $T_b(soln)$ is the boiling point of the solution, $T_b(solv)$ is the boiling point of the solvent, m is molality (mol kg^{-1}), k_b is the molal boiling point elevation constant i.e. the boiling point when the nonvolatile impurity (solute) concentration is 1.0 m, k_b is also called ebullioscopic constant. Think how this knowledge can be used to solve the problem of freezing in car radiators.

Example

Calculate the change in the boiling point of water when 1.0g of glycerine, $C_3H_8O_3(92.1)$ is dissolved in 4.7g of H_2O. $k_b for\ water = 0.512^0 cm^{-1}$.

Solution

Molality of glycerol(m) = $\frac{1.0}{92.1} = 0.011\ mol$

Molal conc. of glycerol = $\frac{0.011 mol}{(47.9/1000) kg}$

$= 0.011 mol/0.0479 = 0.2296 mol/kg = 0.2296 m$

Using $\Delta T_b = K_b m$, get

$$\Delta T_b = 0.512^0 C m^{-1} \times 0.2296 m$$
$$= 0.117578^0 C \approx 0.118^0 C$$

Osmotic Pressure

Osmosis is the passage of solvent molecules from region of low concentration to that of high concentration through a semipermeable membrane. Osmotic pressure is the pressure needed to be applied to the region of high concentration to balance (equilibrate) the movement of solvent molecules across the semipermeable membrane. Semipermeable membranes include parchment paper, animal skins, etc. Osmosis is a biological process witnessed by most living organisms, such as in pig's bladder, cell membranes, lining of the digestive tract, walls of blood vessels; water and some nutrients travel within plant-cells by means of osmosis.

Some solution of macromolecules, such as proteins and polymers can afford us the opportunities to estimate their osmotic pressure because of their high molar masses, low molarity and reasonably low osmic pressures. Osmotic pressure measurement may be used to calculate molar masses.

Mathematically, the expression for calculating osmotic pressure of dilute solutions of non-volatile solutes is given by vant Hoff's equation:

$$\pi = \frac{n}{v} RT,$$

where π is the osmotic pressure, n is the moles of solute, R is the ideal gas constant, T is the temperature in Kelvin and v is the volume of solution in litres.

Since $\frac{n}{v}$ =molarity (M), we can write:

$$\pi = \text{MRT}$$

Example

Calculate the molar mass of haemoglobin solution containing 20.0 g of haemoglobin in 1 L aqueous solution. The osmotic pressure of this solution is 5.92 torr at 22°C. Given R= 0.08206L atm/mol.K; 1 atm = 760 torr

Solution

$T = (22 + 273)k = 295k$

$$5.92torr = 5.92torr \times \frac{1atm}{760torr} = 7.8 \times 10^{-3}\ atm$$

from $\pi = MRT$

$$M = \frac{\pi}{RT} = \frac{7.8 \times 10^{-3} atm}{(0.08206atm\ k^{-1}mol^{-1}) \times 295k}$$

$$= 3.2 \times 10^{-4} mol\ L^{-1} = 3.2 \times 10^{-4} M$$

$$Since\ \left(\frac{20g\ haemoglobin}{molar\ mass\ haemoglobin}\right) mol/L = 3.2 \times 10^{-4} mol/L$$

$$\therefore \frac{20g\ haemoglobin}{molar\ mass\ haemoglobin} = 3.2 \times 10^{-4}$$

$$\therefore\ molar\ mass\ haemoglobin = \frac{20g}{3.2 \times 10^{-4}} = 62{,}500g/mol$$

Exercises

1. Assuming an ideal situation, calculate the vapour pressure of a solution composed of equal mole fraction of benzene C_6H_6 and toluene C_7H_8 at 25^0C. Given: vapour pressure of pure benzene and pure toluene = 95.1 and 28.4mm Hg, respectively.

2. What is the vapour pressure at 23^0C of a solution prepared by dissolving 10.0 g of wax (av. formula, $C_{12}H_{46}$) in 40.0 g of CCl_4? Given: vapour pressure for CCl_4 at $23^0C = 100\ torr$
3. Calculate the vapour pressure at 20^0C of a solution composed of 0.25mol sucrose($C_{12}H_{22}O_{11}$) and 75.0g urea $\left[CO(NH_2)_2 \frac{60.06g}{mol}\right]$ in 1kg of H_2O $(18.02g/mol)$ Give : $P^0_{water} = 17.5mm\text{Hg}$ at 20^0C

Solution

No of moles of sucrose =0.250mol. $C_{12}H_{22}O_{11}$ given

No of mole of urea = 75g urea/ 60.06g/mol = 1.25mol $CO(NH_2)_2$

No of mole of water = 1000g/18.02g/mol = 55.5mol H_2O

Total no of mole of solute = = $0.25 + 1.25 = 1.50mol\ solute$

Mole fraction of water, X_{water}

$$= \frac{55.5mol H_2O}{1.50mol\ solute + 55.5mol\ H_2O} = 0.974$$

Finally, use $P_{H_2O} = x_{H_2O}P^2_{H_2O} = 0.974 \times 17.5mm\ Hg = 17.0mmHg$

4. Calculate the freezing point of a solution composed of 92.1g of I_2 in 800.0g chloroform, $CHCl_3(253.8)$. Given: K_f for chloroform, $CCl_3 = 4.68\ ^0C/m$, $T^0$for chloroform = -63.5
5. A solution composed by dissolving 1.5g hemocyanin in 250ml of H_2O. This solution has an osmotic pressure of 0.00342 atm at 277K.
   (a) Calculate the molar mass of hemocyanin and
   (b) the freezing point of resulting solution.
   Given R= 0.08206Latm $mol^{-1}K^{-1}$; $K_f = 1.86^0Cm^{-1}$.

REDUCTION-OXIDATION REACTIONS

S.O. Afolabi & P.O. Okolo

Reduction-oxidation (redox) reaction is one in which electron(s) are transferred from one reactant to another. In a redox reaction, one reactant is reduced and the other is oxidized, simultaneously. Redox reactions include combustion of fossil fuels, actions of bleaching agent and the reaction between carbon and heptaoxodichromate(VI) solution in a determination of soil organic carbon.

Various Forms of Redox Reactions

Reduction and oxidation are complementary, that is, one completes the other, and takes place in various forms in chemistry; thus, they are expressed generally in five ways as defined with the following examples.

1. Removal and addition of oxygen

Reduction is the removal of oxygen from a reacting species:

$$CuO_{(s)} + H_{2(g)} \longrightarrow Cu_{(s)} + H_2O_{(l)}$$

CuO is reduced to Cu, while H_2 is oxidizing to H_2O. The CuO is the oxidizing agent (oxidant) because it caused oxidation of H_2, while H_2 is the reducing agent (reductant) because it caused reduction of CuO.

Oxidation is the addition of oxygen to a species:

$$Mg_{(s)} + O_{2(g)} \longrightarrow MgO_{(s)}$$

Here, Mg is oxidized to MgO; the Mg is a reductant while O_2 is the oxidant.

2. Addition and removal of hydrogen to a chemical species

Reduction is the addition of hydrogen to a reactant:

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)}$$

Here, O_2 is reduced to H_2O, O_2 is the oxidant while H_2 is the reductant.

Oxidation is the removal of hydrogen from a reactant:

$$H_2S_{(g)} + 2SO_{2(g)} \longrightarrow 3S_{(s)} + H_2O_{(l)}$$

H_2S is oxidized to S. H_2S is the reductant while SO_2 is the oxidant.

3. **Decrease and increase in the number of electronegative atoms surrounding an element in the cause of a chemical reaction**

Reduction is the decrease in the number of electronegative atoms:

$$2FeCl_{3(s)} \longrightarrow 2FeCl_{2(s)} + Cl_{2(g)}$$

Here, $FeCl_3$ is reduced to $FeCl_2$.

Oxidation is increase in the number of electronegative atoms:

$$PbCl_{2(g)} + Cl_{2(g)} \longrightarrow PbCl_{4(s)}$$

Here, $PbCl_2$ is oxidized to $PbCl_4$.

4. **Gain and loss of electrons**

Reduction is the gain of electron(s) by a reacting species:

$$Cl_2 + 2e \longrightarrow 2Cl^-$$

Here, Cl_2 is reduced. Note that this is a half-reaction.

Oxidation is the loss of electron(s) by a chemical species:

$$Na \longrightarrow Na^+ + e$$

The Na is oxidized in this half-reaction

5. **Change in oxidation number (oxidation state)**

Reduction is decrease in oxidation sate.

$$^{(0)}C_{(s)} + {}^{(o)}O_{2(g)} \longrightarrow {}^{(+4)}C^{(-2)}O_{2(g)}$$

Here, O_2 is reduced from oxidation state of zero to -2, while C is oxidized from zero to +4. The C is the reductant and O_2 is the oxidant.

Oxidation is the increase in oxidation state:

$$^{(0)}Na_{(s)} + {}^{(0)}O_{2(g)} \longrightarrow {}^{(+1)}Na_2{}^{(-2)}O_{(s)}$$

Here, Na is oxidized from oxidation state of zero to +1, while O_2 is reduced from zero to -2. Na is the reductant while O_2 is the oxidant.

Reducing and Oxidizing Agents

Generally, reducing agents are mostly electron donors such as metals and some non-metals like C and H, and certain compounds like CO, $LiAlH_4$ and $NaBH_4$. Oxidizing agents are electron acceptors such as non-metals (O_2, F_2, Cl_2 etc.). Their oxidizing power decreases down a group, thus, lower members of a group cannot displace higher ones from a solution of their salts:

$$2F_{2(g)} + 2KCl_{(aq)} \longrightarrow 2KF_2 + Cl_{2(g)}$$

$$Cl_{2(g)} + KF_{2(aq)} \longrightarrow \text{No reaction}$$

Certain compounds like acidified potassium tetraoxomanganate(VII) ($KMnO_4$), acidified potassium heptaoxodichromate(VI) ($K_2Cr_2O_7$), concentrated hexaoxosulphate(VI) acid (H_2SO_4) and concentrated trioxonitrate(V) acid (HNO_3) are notable oxidizing agents.

Oxidizing and Reduction as Relative Tendencies

Reduction and oxidation may be relative tendencies, in that some substances can act either as reducing agents or oxidizing agents depending on the ability of the other reactant to accept or donate electron. For example, in

$$2H_2S_{(g)} + SO_{2(g)} \longrightarrow 3S_{(s)} + 2H_2O_{(l)}$$

SO_2 is an oxidizing agent because H_2S is a better reducing agent.

But in $O_{2(g)} + 2SO_2 \longrightarrow 2SO_{3(g)}$

SO_2 is a reducing agent because O_2 is a better oxidizing agent.

Examples

Identify the oxidant and reductant in each of the following redox reactions:

a. $2H_2S_{(g)} + SO_{2(g)} \longrightarrow 3S_{(s)} + H_2O_{(l)}$

b. $2SO_{2(g)} + O_{2(g)} \longrightarrow 2SO_{3(g)}$

c. $2Mg_{(s)} + O2_{(g)} \longrightarrow 2MgO_{(s)}$

Solution

a. SO_2 is the oxidant because it oxidizes H_2S (oxidation number of S increases from -2 in H_2S to zero in S).H_2S is the reductant because it reduces the oxidation number of S from +4 in SO_2 to zero in S.

b. Oxygen (O_2) is the oxidant because it oxidizes SO_2 (S = +4) to SO_3 (S = +6). SO_2 is the reductant because it reduces O_2 (O = zero) to SO_2 (O = -2).

c. O_2 is the oxidant because it oxidizes Mg (Mg = zero) to MgO (Mg = +2). Mg is the reductant because it reduces O_2 (O = zero) to MgO (O = -2).

Some Tests for Reducing Agents

a. Some reducing agents decolourize aqueous solution of $KMnO_4$ (purple to colourless)

$$\underset{\text{Purple}}{MnO_4^-{}_{(aq)}} + 8H^+_{(aq)} + 5e \longrightarrow \underset{\text{Colourless}}{Mn_2^+{}_{(aq)}} + 4H_2O_{(l)}$$

b. Certain reducing agent turn acidified aqueous $K_2Cr_2O_7$ from orange to green

$$\underset{\text{orange}}{Cr_2O_7^{2-}{}_{(aq)}} + 14H^+ + 6e \longrightarrow \underset{\text{green}}{2Cr^{3-}{}_{(aq)}} + 7H_2O_{(l)}$$

c. Some other reducing agents change the brown solution of aqueous Fe^{3+} salt to green Fe^{2+}

$$\underset{\text{Brown}}{Fe^{3+}{}_{(aq)}} + e \longrightarrow \underset{\text{Green}}{Fe^{2+}{}_{(aq)}}$$

Some Tests for oxidizing agents

a. Some oxidizing agents liberates iodine(I_2) from aqueous solution of KI:

$$2I^-_{(aq)} \longrightarrow I_{2(aq)} + 2e$$

The liberated I_2 will blacken starch solution/paper.

b. Certain oxidizing agents will liberate a whitish-yellow deposit of sulphur(S) from hydrogen sulphide(H_2S).

$$2H_2S_{(g)} + SO_{3(g)} \longrightarrow 3S_{(s)} + 2H_2O_{(l)}$$

c. Some other oxidizing agents will liberate chlorine (Cl_2) from concentrated hydrochloric acid (HCl). This reaction is employed in the preparation of chlorine gas (Cl_2).

d. Some oxidizing agents, when added to a freshly prepared solution of $Fe^{2+}{}_{(aq)}$ ions acidified with dilute $H_2SO_{4(aq)}$ and, on warming, change from a pale green colour (Fe^{2+}) to a pale-yellow brown colour (Fe^{3+}).

$$\underset{\text{Pale-green}}{Fe^{2+}{}_{(aq)}} \longrightarrow \underset{\text{Pale-yellow-brown}}{Fe^{3+}{}_{(aq)}} + e$$

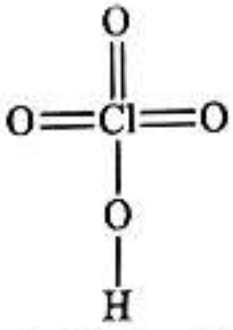
Rules for Assigning Oxidation Number/State to Elements

As a first step in balancing redox equations, we need to be able to assign oxidation numbers to every element in all the compounds involved in the redox reaction. The following rules are applicable:

1. When the elements are in free (uncombine) state, each element has an oxidation number of zero. Thus, each atom in H_2, Br_2, K, U, Xe, O_2, P_4, S_8, etc. has the same oxidation number, zero.
2. For monoatomic ions, the oxidation number is equal to the charge on the ion. Thus Na^+ has an oxidation number of +1; Ba^{2+}, +2; Fe^{2+}, +2; Al^{3+}, +3; I^-, -1; O^{2-}, -2; etc. All alkali metal ions have an oxidation number of +1, and all alkaline earth metal ions have an oxidation number of +2 in their compounds. Aluminium always has an oxidation numbers of +3 in all its simple compounds.
3. The oxidation number of oxygen in most compounds (e.g. MgO and H_2O) is -2, but in hydrogen peroxide (H_2O_2) and peroxide ions(O_2^{-2}), its oxidation state is -1.
4. The oxidation state of hydrogen is +1, except when it is bonded to metals in binary compounds (metal hydrides). For example, LiH, NaH, and CaH_2, its oxidation number is -1.
5. Fluorine has an oxidation number of -1 in all its compounds. Other halogens (Cl, Br, and I) have negative oxidation numbers when they occur as halide ions in their compounds. When they combine with oxygen, for example in oxides, oxoacids and oxoanion, they have positive oxidation states.

The oxoacids contain H, O, and one other element which occupies a central position:

E.g.,



Perchloric acid

Other oxoacids include $HBrO_4$ (perboric acid), $HClO_3$ (chloric acid), HClO (hypochloric acid), and HIO_4 (periodic acid).

6. In neutral molecules, the sum of the oxidation numbers of all the atoms must be zero. In a polyatomic ion, the sum of oxidation numbers of all the elements in the ion must be equal to the net charge of the ion.

For example, in the ammonium ion, NH_4^+, the oxidation number of N is -3 and that of H is +1. Thus, the sum of the oxidation number is -3+4(+1) = +1, which is equal to the net charge of the ion.

Example

Assign oxidation numbers to all the elements in the following compounds and ions: (a) Li_2O (b) PF_3 (c) HNO_3 (d) $Cr_2O_7^{2-}$

Solution

a) Using rule 2, we see that Li has an oxidation number of +1 (Li^+) and oxygen an oxidation number of -2 (O^{2-}).

b) Using rule 5, we note that fluorine has an oxidation number of -1, so that P must have an oxidation number of +3.

c) This is the formula for trioxonitrate (V)acid, which yields one H^+ ion and a NO_3^- ion in aqueous solution. From rule 4, we know that H has an oxidation number of +1. Thus, the other group, the nitrate ion, must have a net oxidation number of -1. Since oxygen has an oxidation

number of -2, the oxidation number of nitrogen, say x, is given by -1 = x+3(-2), therefore, x = +5

d) From rule 6, we see that the sum of the oxidation number in $Cr_2O_7^{2-}$ must be -2. The oxidation number of O is -2. The problem is to determine the oxidation number of Cr, say y. The sum of the oxidation number in the ion is -2=2y+7(-2), therefore, y = +6

Types of Redox Reaction

The following are different types of redox reactions.

1. Combustion Reaction

Reactions involving elements as reactants are largely redox reactions. Thus, all combustion reactions, which involve elemental oxygen, as well as many reactions involving elements themselves are redox in nature:

$$^{(0)}S_{(s)} + {}^{(0)}O_{2(g)} \longrightarrow {}^{(+4)}S^{(-2)}O_{2(g)}$$

$$^{(0)}Mg_{(s)} + {}^{(0)}N_{2(g)} \longrightarrow {}^{(+4)}Mg_3{}^{(-3)}N_2$$

In the above redox equations, the number in parenthesis above each element donates the oxidation number. This format allows for easy identification of element that are oxidized (an increase in oxidation number) or reduced (a decrease in oxidation number).

2. Decomposition Reaction

Some redox reactions are decomposition reactions. These produce one or more free elements.

$$2^{(+2)}Hg^{(-2)}O_{(s)} \longrightarrow 2^{(0)}Hg_{(l)} + {}^{(0)}O_{2(g)}$$

$$2^{(+1)}K^{(+5)}Cl^{(-2)}O_3 \longrightarrow 2^{(+1)}K^{(-1)}Cl_{(s)} + 3^{(0)}O_{2(g)}$$

$$2^{(+1)}Na^{(-1)}H_{(s)} \longrightarrow 2^{(0)}Na_{(s)} + {}^{(0)}H_{2(g)}$$

3. Displacement Reaction

In displacement reaction, an ion (or atom) in a compound is replaced by another ion (or atom) of another element. It is represented by:

$$A + BC \longrightarrow AC + B$$

Most displacement reactions can be conveniently divided into three major categories: hydrogen displacement, metal displacement, and halogen displacement

An easy way to predict whether a displacement reaction in both hydrogen and metal displacement cases will actually occur can be gleaned from the **Activity (Electrochemical) Series**. According to this series, any metal above hydrogen will reduce (displace) it from water or from acid; metals below hydrogen will not react with either water or an acid. In general, any metal listed in the series will displace the metals below it from a compound. For example, Zn is above Ag, so Zn will displace Ag^+ ions from, say, Ag_2SO_4 solution.

Metal displacement reactions find many applications in metallurgical processes in which pure metals are obtained from their compounds in ores. For example, vanadium is obtained by treating vanadium(V)oxide with metallic calcium.

$$^{(+5)}V_2O_{5(s)} + 5^{(o)}Ca_{(l)} \longrightarrow 2^{(0)}V_{(l)} + 5^{(+2)}CaO_{(s)}$$

Similarly,

$$^{(+4)}TiCl_{4(g)} + 2^{(o)}Mg_{(l)} \longrightarrow {}^{(0)}Ti_{(s)} + 2^{(2+)}MgCl_{2(s)}$$

In each case, the metal that acts as the reducing agent is above the metal that is reduced. Ca is above V while Mg is above Ti in the series.

4. Hydrogen Displacement Reaction

Group 1A metals and some Group IIA metals (Ca, Sr and Ba) are the most reactive of all the metals; they displace hydrogen from cold water as follows;

$$2^{(0)}Na_{(s)} + 2^{(+1)}H_2O_{(l)} \longrightarrow 2^{(+1)}NaO^{(+1)}H_{(aq)} + {}^{(0)}H_{2(g)}$$

$$^{(0)}Ca_{(s)} + 2^{(+1)}H_2O_{(l)} \longrightarrow {}^{(+2)}Ca(OH)_2 + {}^{(0)}H_{2(g)}$$

Only the oxidation numbers of elements that are oxidized or reduced are shown.

Less reactive metals, such as Al and Fe, react with steam to give hydrogen gas;

$$2^{(0)}Al_{(S)} + 3^{(+1)}H_2O_{(g)} \longrightarrow {}^{(+3)}Al_2O_{3(S)} + 3^{(+1)}H_{2(g)}$$

$$2^{(0)}Fe_{(s)} + 3^{(+1)}H_2O_{(l)} \longrightarrow {}^{(+3)}Fe_2O_{3(S)} + {}^{(0)}3H_{2(g)}$$

Some other metals are capable of displacing hydrogen from acids:

$$^{(0)}Cd_{(s)} + {}^{(+1)}HCl_{(aq)} \longrightarrow {}^{(-2)}CdCl_{2(aq)} + {}^{(0)}H_{2(g)}$$

$$^{(0)}Sn_{(s)} + 2^{(+1)}HCl_{(aq)} \longrightarrow {}^{(+2)}SnCl_{2(aq)} + H_{2(g)}$$

5. Metal Displacement Reaction

A metal in a compound may be displaced by another metal which was uncombined. For example, when uncombined zinc is introduced to a solution containing copper tetraoxosulphate (VI) ($CuSO_4$), the Zn displaces Cu^{2+} from the solution;

$$^{(0)}Zn_{(S)} + {}^{(+2)}CuSO_{4(aq)} \longrightarrow {}^{(+2)}ZnSO_{4(aq)} + {}^{(0)}Cu_{(s)}$$

6. Halogen Displacement Reaction

The action of the halogen as oxidizing agents diminishes as we go down the group, i.e., F to I in Group 7. That is, F_2 can displace Cl^-, Br^- and I^-

ions in solution. Fluorine reacts so vigorously with water that it is not advisable to carry out these reactions in water. Similarly, Cl_2 can reduce Br^- and I^- to Br_2 and I_2 in aqueous solution;

$$^{(0)}Cl_{2(g)} + 2K^{(-1)}Br_{(aq)} \longrightarrow 2K^{(-1)}Cl_{(aq)} + {}^{(0)}Br_{2(l)}$$

$$^{(0)}Cl_{2(g)} + 2Na^{(-1)}I_{(aq)} \longrightarrow 2Na^{(-1)}Cl_{(aq)} + {}^{(0)}I_{2(s)}$$

Molecular bromine, in turn, can displace iodide ion in aqueous solution.

$$^{(0)}Br_{2(l)} + 2^{(-1)}I^-_{(aq)} \longrightarrow 2^{(-1)}Br^-_{(aq)} + {}^{(0)}I_{2(s)}$$

7. Disproportionation Reaction

In disproportionation reaction, an element in one oxidation number is both oxidized and reduced:

$$2H_2{}^{(-1)}O_{2(aq)} \longrightarrow 2H_2{}^{(-2)}O_{(l)} + {}^{(0)}O_{2(g)}$$

$$^{(0)}Cl_{2(g)} + NaOH_{(aq)} \longrightarrow Na^{(+1)}ClO_{(aq)} + Na^{(-1)}Cl_{(aq)} + H_2O_{(l)}$$

Resolving a Redox Reaction into Half-Reactions

Example

For the redox reaction; $2Sr + O_2 \longrightarrow 2SrO$

i) Resolve it into half reactions

ii) Identify the oxidation and reduction ½-reactions

iii) Identify the oxidizing and reducing agent

Solution

$$2Sr + O_2 \longrightarrow 2SrO$$

Assign oxidation state to each of the element;

$$2^{(0)}Sr + {}^{(0)}O_2 \longrightarrow 2^{(+2)}Sr^{(-2)}O$$

Following the change in oxidation state, separate the redox reaction into half-reactions:

$$2Sr \longrightarrow 2Sr_2 + 4e \text{ (oxidation ½ - reaction)}$$

$$O_2 + 4e \longrightarrow 2O_2^- \text{ (reduction ½ - reaction)}$$

O_2 is the oxidizing agent (oxidant), Sr is the reducing agent (reductant)

Problems

1. For each of the following redox reactions:

$$2Ca + O_2 \longrightarrow 2CaO$$

$$2Na + S \longrightarrow Na_2S$$

$$2Li + H_2 \longrightarrow 2LiH$$

$$3Mg + N_2 \longrightarrow Mg_3N_2$$

$$Zn + I_2 \longrightarrow ZnI_2$$

$$2C + O_2 \longrightarrow 2CO$$

   a) Resolve each into its half-reactions
   b) Identify the oxidation and reduction ½ reactions
   c) Identify the oxidant and the reductant

2. Arrange the following species in order of increasing oxidation number of the sulphur atom.
   (a) H_2S (b) S_3 (c) H_2SO_4 (d) S^{2-} (e) HS^{-1} (f) SO_2 (g) SO_3

3. Give the oxidation states of underlined atoms in the following molecules and ions.
   a) $\underline{Cs}O$ (b) $\underline{Ca}I_2$ (c) $\underline{Al}_2O_3$ (d) $H_3\underline{As}O_3$ (e) $\underline{Ti}O_2$ (f) $\underline{Mo}O_4^{2-}$
   (g) $\underline{Pt}Cl_4^{2-}$ (h) SnF_2 (j) $\underline{Cl}F_3$

4. Give the oxidation number for underlined atoms.
   (a) $H\underline{Cl}O_4$ (b) $H\underline{Br}O_4$ (c) $H\underline{I}O_4$ (d) $H\underline{Cl}O_3$ (e) $H\underline{Cl}O_2$ (f) $H\underline{Cl}O$

Balancing Redox Equations

In the laboratory we often encounter more complex redox reactions that invariably involve oxoanions such as chromate (CrO_4^{2-}), dichromate

(CrO_7^{2-}), permanganate (MnO_4^-), nitrate (NO_3^-), and sulphate (SO_4^{2-}). There are some special techniques for balancing redox reactions, which deal with electron transfer processes. One of such techniques is the **ion-electron method**.

The Ion-Electron Method

In this approach, the given overall redox reaction is resolved into two half-reactions, one for oxidation and the other for reduction. The equations for the two half-reactions are balanced separately and then recombined to give the overall (now balanced) equation. The detailed processes take six steps.

Example 1

The following redox equation is employed in determining the soil organic content, the result of which could be used to advise farmers as regards the viability of a farm land to support agriculture:

$$Cr_2O_7^{2-}(aq) + C \longrightarrow Cr^{3+} + CO_2 \text{ (in acidic medium)}$$

Balance the redox equation using the ion-electron method.

Solution

Step 1: Write the skeletal equation (as given) and assign oxidation numbers to each atom:

$$^{+6}Cr_2\,^{-2}O_7^{2-} + 0C \longrightarrow {}^{+3}Cr^{3+} + {}^{+4}C\,^{-2}O_2 \text{ (acidic medium)}$$

Step 2: Tracing the change in oxidation state, separate the skeletal equation into two ½ - reactions:

$$Cr_2O_7^{2-} \longrightarrow Cr^{3+} \text{ (reduction ½ - reaction)}$$

$$C \longrightarrow CO_2 \text{ (oxidation ½ - reaction)}$$

Note: We identified the first ½ -reaction as reduction ½-reaction because the oxidation state of Cr decreased from +6 to +3 in step 1. Similarly, the oxidation ½-reaction is established because C increased in oxidation state from zero to +4.

Step 3: Balance each ½-reaction

* "atomically", using H_2O for oxygen and H^+ for hydrogen
* ionically, using electrons

$$Cr_2O_7^{2-} + 14H^+ + 6e \longrightarrow 2Cr^{3+} + 7H_2O$$

$$C + 2H_2O \longrightarrow CO_2 + 4H^+ + 4e$$

Step 4: Equalize the electrons

$$2\,(Cr_2O_7^{2-} + 14H^+ + 6e \longrightarrow 2Cr^{3+} + 7H_2O)$$

$$3\,(C + 2H_2O \longrightarrow CO_2 + 4H^+ + 4e)$$

$$2Cr_2O_7^{2-} + 28H^+ + 12e \longrightarrow 4Cr^{3+} + 14H_2O$$

$$3C + 6H_2O \longrightarrow 3CO_2 + 12H^+ + 12e$$

Step 5: Add up the two ½-reactions and simplify.

$$2Cr_2O_7^{2-} + 28H^+ + 3C + 6H_2O + 12e \longrightarrow 4Cr^{3+} + 14H_2O + 3CO_2 + 12H^+ + 12e$$

Step 6: neatly write out the final balanced redox equation and check if everything is alright:

$$2Cr_2O_7^{2-} + 16H^+ + 3C \longrightarrow 4Cr^{3+} + 8H_2O + 3CO_2$$

Example 2

Balance the following redox equation (in basic medium) using the ion-electron method.

$$MnO_4^- + I^- \longrightarrow MnO_2 + I_2$$

Step 1: Write the skeletal equation for the reaction and assign oxidation numbers

$$^{+7}MnO_4^- + {}^{-1}I^- \longrightarrow {}^{+4}MnO_2 + {}^{0}I_2$$

Step 2: Separate into two ½-reactions

$$I^- \longrightarrow I_2 \text{ (oxidation ½-reaction)}$$

$$MnO_4^- \longrightarrow MnO_2 \text{(reduction ½-reaction)}$$

Step 3: Balance each ½-reaction "atomically" and ionically.

To balance the I atoms in the oxidation ½-reaction, we write

$$2I^- \longrightarrow I_2$$

To balance the O atoms in the reduction ½-reaction, we add two H_2O molecules on the right

$$MnO_4^- \longrightarrow MnO_2 + 2H_2O$$

To balance the H atoms, we add four H^+ ions on the left

$$MnO_4^- + 4H^+ \longrightarrow MnO_2 + 2H_2O$$

Since the reaction occurs in a basic medium and there are four H^+ ions, we add four OH^- ions to both sides of the equation:

$$MnO_4^- + 4H^+ + 4OH^- \longrightarrow MnO_2 + 2H_2O + 4OH^-$$

Combining the H^+ and OH^- ions to form H_2O, we write

$$MnO_4^- + 2H_2O \longrightarrow MnO_2 + 4OH^-$$

Next, we balance the charges of the two ½-reactions as follows

$$2I^- \longrightarrow I_2 + 2e$$

$$MnO_4^- + 2H_2O + 3e \longrightarrow MnO_2 + 4OH^-$$

Step 4: Equalize the electrons, get

$$3(2I^- \longrightarrow I_2 + 2e)$$

$$2(MnO_4^- + 2H_2O + 3e \longrightarrow MnO_2 + 4OH^-)$$

$$6I^- \longrightarrow 3I_2 + 6e$$

$$2MnO_4^- + 4H_2O + 6e \longrightarrow 2MnO_2 + 8OH^-)$$

Step 5: Add up and simplify

$$6I^- + 2MnO_4^- + 4H_2O + \cancel{6e} \longrightarrow 3I_2 + 2MnO_2 + 8OH^- + \cancel{6e}$$

Step 6: Check

$$6I^- + 2MnO_4^- + 4H_2O \longrightarrow 3I_2 + 2MnO_2 + 8OH^-$$

Problems

Use the ion-electron method to balance the following redox equation

a) $ClO_3^- + Cl^- \longrightarrow Cl_2 + ClO_2$ (acidic medium)

b) $Mn_2^+ + H_2O_2 \longrightarrow MnO_2$ (in basic/alkaline medium)

c) $Cr_2O_7^{2-} + C_2O_4^{2-} \longrightarrow Cr^{3+} + CO_2$ (acidic medium)

d) $S_2O_3^{2-} + I_2 \longrightarrow I^- + S_4O_6^{2-}$ (acidic medium)

e) $Bi(OH)_3 + SnO_3^{2-} \longrightarrow SnO_3^{2-} + Bi$ (basic medium)

$CN^- + MnO_4^- \longrightarrow CNO^- + MnO_2$ (basic medium)

References

1. Ndiokwere, C.L. and Ukhun, M.E. (Coordinating editors) (2006). First Year University Inorganic and Physical Chemistry. Mindex Publishing Company Ltd. Benin
2. Olumayede, E.G. and Abitogun, A.S. (2010). General and Physical Chemistry. Alabi-Eyo and Co. Ltd. Akure.
3. Avinde, F.O. and Asubiojo, F.O.I. (2007) Lamlard's SSCE and UTME Chemistry. Lamlard Publication Ltd. Ado-Ekiti.
4. Chang, R. (1991). Chemistry, 4th Edition, McGraw-Hill Inc. New Jersey.

PRACTICE QUESTIONS AND ANSWERS

Compiled by all Authors

Physical Quantities and Error Treatments: Structure of the Atom; States of Matter

1. What symbol is used to express the factor, 10^{-6}?
   A) M B) m C) μ D) n
   Answer: C

2. The factor 0.001 corresponds to which prefix?
   A) deka B) deci C) centi D) milli
   Answer: D

3. How many significant figures are there in the answer to the following problem? $23.1 \div 0.3848 \times 11 = ?$
   A) one B) two C) three D) four
   Answer: B

4. Round off 0.003075 06 to four significant figures.
   A) 0.031 B) 3.076 C) 3100 D) 3.075×10^5
   Answer: D

5. Convert 0.008723 to standard scientific notation.
   A) 8.723×10^{-3} B) 8.723×10^{-6} C) 8.723×10^{3} D) 8723×10^{6}
   Answer: A

6. Convert 7.0 μm to meter.
   A) 7.0×10^{9} m B) 7.0×10^{-9} m C) 7.0×10^{3} m D) 7.0×10^{6} m
   Answer: B

7. The fundamental SI unit of mass is the _____.
   A) kilogram B) Kelvin C) Meter D) mL
   Answer: A

8. According to history, the concept that all matter is composed of atoms was first proposed by
   A) the Greek philosopher Democritus, but not widely accepted until modern times. B) Dalton, but not widely accepted until the work of Mendeleev. C) Dalton, but not widely accepted until the work of Einstein. D) Dalton, and widely accepted within a few decades.
   Answer: A

9. Mendeleev arranged the elements according to
   A) atomic number and atomic weight. B) atomic weight and chemical reactivity. C) electron configuration and atomic weight. D) physical state and relative abundance.
   Answer: B

10. Elements in a periodic group have similar
    A) chemical properties. B) densities. C) masses. D) physical properties.
    Answer: A

11. Which group of elements reacts violently with water?
    A) halogens B) noble gases C) alkali metals D) alkaline earth metals
    Answer: C

12. The existence of electrons in atoms of all elements was demonstrated by
    A) Millikan's oil drop experiment. B) Rutherford's gold foil experiment. C) Thomson's cathode ray tube experiment. D) None of these
    Answer: C

13. Most of the alpha particles directed at a thin gold foil in Rutherford's experiment
    A) bounced directly back from the foil. B) passed directly through the foil undeflected. C) passed through the foil but were deflected

A) number of collisions with the container wall increases. B) number of moles of the gas increases. C) temperature of the gas decreases. D) volume of the container decreases.
Answer: C

29. Which statement about real gases is true?
A) The volume of the gas particles is zero. B) The mass of the gas particles is zero. C) Forces of attraction and repulsion exist between gas particles at close range. D) The behavior of real gases can be exactly predicted using the ideal gas law.
Answer: C

30. Which one of the following is not used to describe the condition of a gas?
A) number of moles B) electronic configuration C) temperature D) pressure
Answer: B

The Mole Concept and Avogadro's Number; Determination of Relative and Average Atomic Masses; Stoichiometry and Chemical Equations

1. Molarity is defined as
A) moles of solute per liter of solution. B) moles of solute per liter of solvent. C) moles of solvent per liter of solution. D) moles of solvent per liter of solvent.
Answer: A

2. Chemical equations are balanced in order to obey the law of
A) definite proportions. B) mass action. C) mass conservation. D) multiple proportions.
Answer: C

3. Which contains Avogadro's number of formula units?
A) 36.5 g of Cl B) 36.5 g of Cl_2 C) 36.5 g of HCl D) All of the above
Answer: C

4. What is the molar mass of aspartic acid, $C_4O_4H_7N$?
A) 43 g/mol B) 70 g/mol C) 133 g/mol D) 197 g/mol
Answer: C

5. How many moles are in 1.50 g of ethylamine, $CH_3CH_2NH_2$?
A) 0.0222 mol B) 0.0332 mol C) 90.16 mol D) 45.08 mol
Answer: B

6. Ozone is unstable, decomposing to oxygen, as shown in the balanced equation $2O_3 \rightarrow 3O_2$. In this reaction, how many grams of oxygen can be formed from the decomposition of 96 grams of ozone?
A) 96 g B) 16 g C) 32 g D) 6 g
Answer: A

7. What is the oxidation number of the chromium atom in $K_2Cr_2O_4$?
A) -2 B) +3 C) +6 D) +7
Answer: B

8. How many grams of KNO_3 are needed to make 250. mL of a solution that is 0.135 M?
A) 1.71 g B) 0.341 g C) 3.41 g D) 6.82 g
Answer: C

9. How many milliliters of a 6.0 M HNO_3 solution are needed to make 0.25 L of a 3.5 M HNO_3 solution?
A) 686 mL B) 428 mL C) 146 mL D) 129 mL
Answer: C

10. Write a balanced net ionic equation for the reaction of H_2SO_4 with $Ba(OH)_2$:
A) $H_2SO_{4(aq)} + Ba(OH)_{2(aq)} \longrightarrow BaSO_{4(s)} + 2H_2O_{(l)}$

B) $2H^+_{(aq)} + SO_4^{2-}_{(aq)} + Ba^{2+}_{(aq)} + 2OH^-_{(aq)} \longrightarrow BaSO_{4(s)} + 2H_2O_{(l)}$

C) $H^+_{(aq)} - OH^-_{(aq)} \longrightarrow H_2O_{(l)}$

D) $Ba^{2+}_{(aq)} + SO_4^{2-}_{(aq)} \longrightarrow BaSO_{4(s)}$

Answer: B

11. How many liters of oxygen are needed to exactly react with 55.6 g of methane at STP?

$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} - 2H_2O_{(l)}$

A) 19.5 L B) 39.0 L C) 156 L D) 77.8 L

Answer: C

12. Molality is defined as moles of solute per

A) kilogram of solvent. B) liter of solvent. C) mole of solvent. D) total moles present.

Answer: A

13. A 3.17 *m* solution of $CaCl_2$ in water has a density of 1.24 g/mL. What is the molarity?

A) 2.56 M $CaCl_2$ B) 2.91 M $CaCl_2$ C) 3.50 M $CaCl_2$ D) 3.93 M $CaCl_2$

Answer: B

14. What mass of $HNO_3(63g/mol)$ is contained in 300 mL of 0.30 M HNO_3 solution?

A) 30.0 g B) 0.30 g C) 6.30 g D) 5.67 g

Answer: D

15. Calculate the percentage composition of nitrogen (N) in magnesium nitrate $Mg(NO_3)_2 (Mg = 24.31, N = 14.01, O = 16.00)$

A) 18.89% B) 1.889% C) 0.1889% D) 0.01889%

Answer: A

1. A saturated solution is defined as

A) a concentrated solution. B) a solution that is in equilibrium with pure solute. C) a solution that is in equilibrium with undissolved solute. D) a solution that is not in equilibrium with both pure solvent and undissolved solute.

Answer: C

2. Which of the following compounds is not an Arrhenius acid?

A) HF B) KOH C) HClO D) H_2SO_3

Answer: B

3. When dissolved in water, NaOH behaves as

A) an acid that forms Na^+ and OH^- ions. B) an acid that forms NaO^- and H^+ ions. C) a base that forms Na^+ and OH^- ions. D) a base that forms NaO^- and H^+ ions.

Answer: C

4. What ion is provided when Arrhenius bases dissolve in water?

A) H^+ B) Na^+ C) Cl^- D) OH^-

Answer: D

5. How many H^+ ions can the acid, H_2CO_3, donate per molecule?

A) 0 B) 1 C) 2 D) 3

Answer: C

6. Write a net ionic equation for the neutralization reaction of $HCN_{(aq)}$ with $NaOH_{(aq)}$.

A) $HCN_{(aq)} + NaOH_{(aq)} \longrightarrow NaCN_{(aq)} + H_2O_{(l)}$

B) $HCN_{(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \longrightarrow Na^+_{(aq)} + CN^-_{(aq)} - H_2O_{(l)}$

C) $HCN_{(aq)} + OH^-_{(aq)} \longrightarrow CN^-_{(aq)} + H_2O_{(l)}$

D) $H^+_{(aq)} - OH^-_{(aq)} \longrightarrow H_2O_{(l)}$

Answer: C

7. The reaction $2HNO_{3(aq)} + Ba(OH)_{2(aq)} \longrightarrow Ba(NO_3)_{2(aq)} + 2H_2O_{(l)}$ is best classified as;

A) acid-base neutralization reaction. B) oxidation-reduction reaction. C) precipitation reaction. D) single replacement reaction.

Answer: A

8. Calculate the hydroxide ion concentration in an aqueous solution that contains 3.50×10^{-3} M in hydronium ion.

A) 2.86×10^{-3} M B) 2.86×10^{-1} M C) 2.86×10^{12} M D) 3.50×10^{-12} M

Answer: C

9. If the ionization constant of water, K_w, at 40°C is 2.92×10^{-14}, then what is the hydronium ion concentration for a neutral solution?

A) $[H_3O^+] < 1.00 \times 10^{-7}$ M B) $[H_3O^+] > 1.71 \times 10^{-7}$ M C) $[H_3O^+] = 1.71 \times 10^{-7}$ M D) $[H_3O^+] < 1.71 \times 10^{-7}$ M

Answer: C

10. Calculate the pH for an aqueous solution of acetic acid that contains 2.15×10^{-5} M hydronium ion.

A) 4.65×10^{-10} B) 2.15×10^{-5} C) 2.67 D) 11.33

Answer: C

11. Calculate the pH for an aqueous solution of pyridine that contains 2.15×10^{-4} M hydroxide ion.

A) 4.65×10^{-11} B) 2.15×10^{-4} C) 3.67 D) 10.33

Answer: D

12. Which of the following solutions has the highest concentration of hydroxide ions $[OH^-]$?

A) pH = 3.21 B) pH = 7.83 C) pH = 10.93 D) pH = 12.04

Answer: D

13. Which of the following can be classified as a weak base?

A) CH_3NH_2 B) NH_2OH C) Both CH_3NH_2 and NH_2OH D) Neither CH_3NH_2 nor NH_2OH

Answer: C

14. Which of the following species cannot act as a Lewis base?

A) O^{2-} B) CH_4 C) H_2O D) NH_3

Answer: B

15. Covalent bonding is a

A) gain of electrons. B) loss of electrons. C) transfer of electrons. D) sharing of electrons.

Answer: D

16. What type of bonding is found in the compound PCl_3?

A) covalent bonding B) hydrogen bonding C) ionic b D) metallic bonding

Answer: A

17. Which bond should have the highest bond dissociation en

A) N—N B) N=N C) N≡N D)

Answer: C

18. The greater the electronegativity difference between t atoms, the

A) greater the bond order. B) greater the covalent cha bond. C) greater the ionic character of the bond. D) t the bond.

Answer: C

19. Which of the following compounds exhibits hydroge

A) CH_3F B) CH_3OCH_3 C) $(CH_3)_3N$ D) C

Answer: D

20. Which of the following is not a property of electrovalent compounds?

A) They are made up of positive and negative ions arrange in crystal lattices. B) They usually have high boiling point. C) They usually have low high melting point. D) They are usually soluble in polar solvents.

Answer: C

Chemical Equilibrium and Thermodynamics; Introductory Chemical Kinetics

1. Which is a measure of the sum of the kinetic and potential energies of each particle in the system?

A) E, the internal energy B) G, the Gibbs free energy C) H, the enthalpy D) T, the temperature

Answer: A

2. Which of the following can be interpreted as a measure of randomness?

A) enthalpy B) entropy C) free energy D) temperature

Answer: B

3. For a particular process that is carried out at constant pressure, q = 200 kJ and w = −85 kJ. Therefore,

A) ΔE = 115 kJ and ΔH = 200 kJ. B) ΔE = 200 kJ and ΔH = 115 kJ. C) ΔE = 200 kJ and ΔH = 285 kJ. D) ΔE = 285 kJ and ΔH = 200 kJ.

Answer: A

4. Under thermodynamic standard state conditions, the element oxygen occurs as

A) $O_{(g)}$ B) $O_{2(g)}$ C) $O_{(s)}$ D) $O_{2(s)}$

Answer: B

5. Which combination always results in a reaction being spontaneous?

A) ΔH is negative and ΔS is negative. B) ΔH is negative and ΔS is positive. C) ΔH is positive and ΔS is negative. D) ΔH is positive and ΔS is positive.

Answer: B

6. What is the thermodynamic criterion for equilibrium for a reaction at constant temperature and pressure (PV work only)?

A) $\Delta S = 0$ B) $\Delta G = 0$ C) $\Delta S > 0$ D) $\Delta G < 0$

Answer: B

7. The energy of motion is

A) electrical energy B) dam energy C) reserve energy. D) kinetic energy.

Answer: D

8. Which of the following is not a type of energy or energy transfer?

A) thermal energy B) heat C) height D) work

Answer: C

9. What is the minimum energy barrier that must be overcome for a chemical reaction to occur?

A) activation energy B) net energy C) potential energy D) rate limiting energy

Answer: A

10. A zeroth order reaction is one whose

A) rate is zero B) rate is independent of reactant concentration. C) rate can be found where $[A]_t = -kt + \ln [A]_o$ D) rate is dependent on none of the reactants.

Answer: B

11. According to kinetic molecular theory, which of the following will decrease the rate of reaction?

A) increase the temperature B) increase the concentration

C) increase the size of the molecule D) increase the size of the reaction vessel

Answer: C

12. What is the overall reaction order for the reaction that has the rate law: Rate = $k[H_2][NO]^2$?

A) zero order B) first order C) second order D) third order

Answer: D

13. Using the method of initial rates for the reaction A → B, if the initial concentration of A is doubled and the rate of reaction quadruples, what is the order of reaction with respect to A?

A) zeroth B) first C) second D) fourth

Answer: C

14. The rate constant, k, for a first-order reaction is equal to 4.2×10^{-4} s^{-1}. What is the half-life for the reaction?

A) 2.9×10^{-4} s B) 1.2×10^{3} s C) 1.7×10^{3} s D) 2.4×10^{4} s

Answer: C

15. The elementary reaction

$2HF \longrightarrow H_2 + F_2$, is an example of a _____ reaction.

A) unimolecular B) bimolecular C) termolecular D) tetramolecular

Answer: B

16. Which general rate law below corresponds to an elementary bimolecular reaction?

A) Rate = $k[B]$ B) Rate = $k[A][B][C][D]$ C) Rate = $k[A][B]^2$ D) Rate = $k[A]^2$

Answer: D

17. A catalyst increases the rate of a reaction by providing a different reaction pathway that

A) lowers only the activation energy. B) raises only the energy of the products. C) lowers only the energy of the reactants and products. D) All of these are affected by the presence of a catalyst.

Answer: A

18. A reaction reaches dynamic equilibrium at a given temperature when A) the amount of products exceeds the amount of reactants. B) k_{rev} equals k_{fwd}. C) opposing reactions cease and the system is static. D) the relative amounts of reactants and products are constant and $rate_{fwd} = rate_{rev}$.

Answer: D

19. Which equilibrium below is homogeneous?

A) $CaSO_{4(s)} \rightleftharpoons Ca_2^{+}{}_{(aq)} + SO_4^{2-}{}_{(aq)}$

B) $2H_2O_{2(l)} \rightleftharpoons 2H_2O_{(l)} - O_{2(g)}$

C) $NH_4NO_{3(s)} \rightleftharpoons N_2O_{(g)} + 2H_2O_{(g)}$

D) $2CO_{(g)} + O_{2(g)} \rightleftharpoons 2CO_{2(g)}$

Answer: D

20. Which of the following is true?

A) As a reaction at constant temperature and pressure goes to equilibrium, $|\Delta G|$ decreases. B) The larger ΔG°, the faster the reaction. C) The standard state for solutes is the pure solute at 1 atm. D) When a reaction reaches equilibrium, $\Delta G = 0$.

Answer: A

Nuclear Chemistry; Colligative Properties; Reduction-Oxidation Reactions

1. Which statement about nuclear reactions is true?
   A) New elements are never produced in a nuclear reaction. B) Nuclear reactions involve valence electrons. C) The rate of a nuclear reaction is affected by catalysts. D) Tremendous amounts of energy are involved in nuclear reactions.
   Answer: D
2. The number of nucleons in an atom or ion is the same as the
   A) atomic number. B) charge on the atom or ion. C) mass number. D) none of these
   Answer: C
3. The number of nucleons in a $^{236}_{92}U^{2+}$ nucleus is
   A) 92 B) 144. C) 236. D) 328.
   Answer: C
4. The number of neutrons in $^{62}_{26}Fe^{2+}$ is
   A) 26. B) 36. C) 60. D) 62.
   Answer: B
5. A beta particle is
   A) $^{0}_{-1}e$. B) $^{-1}_{0}e$. C) $^{1}_{1}p$. D) $^{4}_{2}He$.
   Answer: A
6. Gamma radiation can be described as
   A) a helium nucleus. B) a negatively charged free electron. C) high energy electromagnetic radiation. D) a positively charged free electron.
   Answer: C
7. The binding energy is defined as

A) the amount of energy absorbed when electrons are added to an ion. B) the amount of energy absorbed when protons and neutrons form a nucleus. C) the amount of energy released when electrons are removed from the atom. D) the amount of energy required to break apart a nucleus into individual protons and neutrons.
Answer: D

8. Elements with _____ atomic mass are best possible candidates for a fusion reaction.
   A) very low B) moderate C) moderate to heavy D) very heavy
   Answer: A
9. Elements with _____ atomic mass are best possible candidates for a fission reaction.
   A) very low B) moderate C) moderate to heavy D) very heavy
   Answer: D
10. What is the missing product of the nuclear reaction?
    $^{209}_{83}Bi + {}^{1}_{1}p \longrightarrow ? + {}^{210}_{85}Po$
    A) α B) $\beta+$ C) $\beta-$ D) n
    Answer: C
11. Radiation is detected by its _____ properties.
    A) gravitational B) ionizing C) kinetic D) thermal
    Answer: B
12. Which unit of radiation describes the amount of energy absorbed by a kilogram of tissue exposed to a radiation source?
    A) becquerel B) curie C) rad D) rem
    Answer: C
13. The effects of ionizing radiation depend on
    A) length of exposure to radiation. B) location of source (internal or external). C) type and energy of radiation. D) All of the above
    Answer: D

14. In most liquid solutions, the component present in the larger amount is called the

A) dispersed medium. B) emulsifying agent. C) solute. D) solvent.

Answer: D

15. One reason ionic compounds do not dissolve well in nonpolar solvents is that

A) ion-dipole interactions are too large for effective solvation to occur. B) ion-solvent interactions are not strong enough to solvate the ions in solution. C) not all cations and anions have the same magnitude of charge and therefore do not form neutral ion pairs. D) there are no forces of attraction between ions and nonpolar molecules.

Answer: B

16. When two similar liquids mix to form a solution, the entropy of solution (ΔS_{soln}) is expected to be

A) negative. B) zero. C) positive. D) negative at low temperatures but positive at high temperatures.

Answer: C

17. What is the mole fraction of ethanol in a solution made by dissolving 25.0 g of ethanol, C_2H_5OH, in 53.6 g of water?

A) 0.154 B) 0.846 C) 0.214 D) 0.272

Answer: A

18. Which of the following is not an application of colligative properties?

A) adding silver to mercury to lower the vapor pressure of mercury B) desalinating sea water by reverse osmosis C) melting snow by application of salt D) reduced boiling points of pure liquids at increased altitudes

Answer: D

19. A KCl solution is prepared by dissolving 60.0 g KCl in 250.0 g of water at 25°C. What is the vapor pressure of the solution if the vapor pressure of water at 25°C is 23.76 mm Hg?

A) 20.5 mm Hg B) 21.3 mm Hg C) 22.5 mm Hg D) 25.5 mm Hg

Answer: B

20. What is the expected freezing point of a 0.50 *m* solution of Na_2CO_3 in water? K_f for water is 1.86°C/*m*.

A) -0.93°C B) -1.9°C C) -2.8°C D) -6.5°C

Answer: C

21. Two aqueous solutions, A and B, are separated by a semipermeable membrane. The osmotic pressure of solution A immediately begins to decrease. Which of the following statements is true?

A) Solvent molecules are moving from solution B into solution A. B) The initial osmotic pressure of solution B is greater than that of solution A. C) The solvent molecules are moving from the solution of higher osmotic pressure to that of lower osmotic pressure. D) Both B and C are true statements.

Answer: A

22. How will the osmotic pressure of an aqueous solution change as evaporation occurs?

A) The osmotic pressure will increase. B) The osmotic pressure will not change. C) The osmotic pressure will decrease. D) The osmotic pressure will increase or decrease until it equals the vapor pressure of water.

Answer: A

23. An aqueous solution has a normal boiling point of 105.0°C. What is the freezing point of this solution? For water K_b is 0.51°C/*m* and K_f = 1.86°C/*m*.

A) 14.7°C B) -14.7°C C) 18.2°C D) -18.2°C

Answer: D

24. HBr, HCl, $HClO_4$, KBr, and NaCl are all classified as

A) acids. B) nonelectrolytes. C) strong electrolytes. D) weak electrolytes.

Answer: C

25. In the following reaction, Cl_2 is

$2Rb_{(s)} + Cl_{2(g)} \longrightarrow 2RbCl_{(s)}$

A) the reducing agent. B) the oxidizing agent. C) oxidized. D) the electron donor.

Answer: B

26. Which species functions as the oxidizing agent in the following reduction-oxidation reaction?

$5Fe^{2+}_{(aq)} + MnO_4^-{}_{(aq)} + 8H^+_{(aq)} \longrightarrow Mn^{2+}_{(aq)} + 5Fe^{3+}_{(aq)} + 4H_2O_{(l)}$

A) $Fe^{2+}_{(aq)}$ B) $H^+_{(aq)}$ C) $Mn^{2+}_{(aq)}$ D) $MnO_4^-{}_{(aq)}$

Answer: D

27. Determine the number of water molecules necessary to balance the following chemical equation.

$Cr_2O_7^{2-}_{(aq)} + Cl^-_{(aq)} + H^+_{(aq)} \longrightarrow Cr^{3+}_{(aq)} + Cl_{2(g)} + ____ H_2O_{(l)}$

A) 3 B) 5 C) 7 D) 14

Answer: C

28. What is the oxidation half reaction in the following chemical reaction?

$Cr_2O_7^{2-}_{(aq)} + 6Cl^-_{(aq)} + 14H^+_{(aq)} \longrightarrow 2Cr^{3+}_{(aq)} + 3Cl_{2(aq)} + 7H_2O_{(l)}$

A) $Cr_2O_7^{2-}_{(aq)} + 14H^+_{(aq)} - 6e^- \longrightarrow 2Cr^{3+}_{(aq)} + 7H_2O_{(l)}$

B) $Cr_2O_7^{2-}_{(aq)} + 14H^+_{(aq)} \longrightarrow 2Cr^{3+}_{(aq)} + 7H_2O_{(l)} + 6e^-$

C) $2Cl^-_{(aq)} \longrightarrow Cl_{2(aq)} + 2e^-$

D) $Cl_{2(aq)} + 2e^- \longrightarrow 2Cl^-_{(aq)}$

Answer: C

29. Which cell involves a non-spontaneous redox reaction?

A) concentration cell B) electrolytic cell C) fuel cell D) galvanic cell

Answer: B

30. During an electrochemical reaction, electrons move through the external circuit toward the _____ and positive ions in the cell move toward the _____.

A) anode, anode B) anode, cathode C) cathode, anode D) cathode, cathode

Answer: D

31. What is the shorthand notation that represents the following galvanic cell reaction?

$Fe_{(s)} + Cu(NO_3)_{2(aq)} \longrightarrow Fe(NO_3)_{2(aq)} + Cu_{(s)}$

A) $Fe(s) \mid Fe^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

B) $Cu(s) \mid Cu^{2+}(aq) \parallel Fe^{2+}(aq) \mid Fe(s)$

C) $Fe(s) \mid NO_3^-(aq) \parallel NO_3^-(aq) \mid Cu(s)$

D) $Cu(s) \mid Cu(NO_3)_2(aq) \parallel Fe(NO_3)_2(aq) \mid Fe(s)$

Answer: A

32. Given that $E° = +0.897$ V, calculate E at 25°C for

$Pb(s) \mid Pb^{2+}(0.0400\ M) \parallel Fe^{3+}(0.200\ M), Fe^{2+}(0.0100\ M) \mid Pt(s)$

A) +0.779 V B) +0.935 V C) +1.015 V D) +1.134 V

Answer: C

33. How many grams of chromium metal are plated out when a constant current of 8.00 A is passed through an aqueous solution containing Cr^{3+} ions for 40.0 minutes?

A) 3.45 g B) 6.15 g C) 10.3 g D) 31.0 g

Answer: A

34. How many grams of nickel metal are plated out when a constant

current of 15.0 A is passed through aqueous $NiCl_2$ for 80.0 minutes?

A) 14.7 g B) 21.9 g C) 43.8 g D) 48.4 g

Answer: B

35. Given the half-cell potentials below, calculate the cell potential.

$$Ag^+_{(aq)} + e^- \longrightarrow Ag_{(s)} \qquad 0.80\ V$$

$$Cu_2{}^+_{(aq)} + 2e^- \longrightarrow Cu_{(s)} \qquad 0.34\ V$$

A) 0.46 V B) 1.14 V C) 1.26 V D) 1.94 V

Answer: A